RELATÓRIU FINÁL CAPSTONE PROJECT

Sistema Deteksaun Anomalia Rede Eletrisidade Real Time uza metodu Random Forest (Capstone Project)

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ABSTRACT

Rede distribuisaun eletrisidade iha Timor-Leste hasoru anomalia oin-oin: voltajen tun iha área rurál, voltajen sa'e liu iha kalan, frekuénsia la estável bainhira jeradór ida para, kliente ne'ebé uza liu kontratu, na'ok eletrisidade no meter ne'ebé aat. Monitorizasaun agora dala barak sei reativu no seidauk iha sistema ne'ebé bele analiza readings smart meter ho volume boot iha tempu real. Peskiza ida ne'e dezenvolve **Sistema Deteksaun Anomalia Rede Eletrisidade Real Time uza metodu Random Forest** (PowerPredict TL), sistema Big Data ne'ebé integra bot smart meter, FastAPI, Apache Kafka, Apache Spark, data lake Parquet, MySQL no website Streamlit, no uza algoritmu **Random Forest** hodi klasifika kada reading ba Normal ka tipu anomalia neen: Voltajen Tun, Voltajen Sa'e Liu, Frekuénsia La Estável, Karga Liu, Na'ok Eletrisidade no Kontador Aat.

Dataset kria husi seed dadus reál (konsumu UCI ElectricityLoadDiagrams no temperature Open-Meteo) no eskala ba 5 GB, 10 GB no 20 GB (records 319.1 millaun, 35.8 GB CSV); jeradór hatama events anomalia no rai ground truth ba kada reading (3.80% anomalia). Random Forest ne'ebé treina iha dataset 20 GB no test iha meters ne'ebé model nunka haree atinje precision 0.942, recall 0.968, F1-score 0.954 no PR-AUC 0.990. Iha test set hanesan, Random Forest accurate liu Decision Tree (F1 0.925), Logistic Regression (F1 0.896) no Isolation Forest (F1 0.144), tan ne'e Random Forest mak métodu machine learning ida de'it iha sistema. Spark MLlib Random Forest ne'ebé treina iha readings 2.3 millaun no test iha readings 37.0 millaun atinje F1 0.867. Iha stream real-time, readings 3.14 millaun prosesa ho latency média 2.8 segundu no F1 0.962. Scalability testing hatudu katak iha komputadór RAM 32 GB, Pandas la konsege prosesa dataset 5 GB, 10 GB no 20 GB (out of memory), enkuantu Apache Spark prosesa 20 GB iha 189 segundu ho memory 13.0 GB. Website iha login Admin no User, butaun Refresh no Night mode, no versaun publiku (konvidadu, read-only) ne'ebé ema hotu bele loke husi internet ho HTTPS liuhusi Cloudflare Tunnel iha link permanente GitHub Pages; pakote Docker Compose prontu ona ba deployment 24/7. Sistema ne'e bele ajuda EDTL deteta problema rede no suspeita na'ok eletrisidade sedu.

Keywords: Big Data, Anomaly Detection, Apache Kafka, Apache Spark, Random Forest, smart meter, real-time, Timor-Leste.

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KAPITULU I
INTRODUSAUN

1.1. Antesedente

Eletrisidade mak fundasaun ba dezenvolvimentu ekonómiku no sosiál iha nasaun ida. Timor-Leste halo progresu boot: tuir World Development Indicators husi World Bank, asesu ba eletrisidade atinje 94.3% husi populasaun iha tinan 2024 (World Bank, 2025). Maibé jerasaun eletrisidade kuaze tomak (99.6% iha tinan 2021) sei depende ba kombustível mina, no konsumu kada ema sei ki'ik, maizumenus 364 kWh kada tinan (Our World in Data, 2025). Rede nasionál ne'ebé EDTL, E.P. jere liga sentrál elétrika Hera no Betano ho municipality sira liuhusi liña distribuisaun naruk, enkuantu Oecusse no Atauro iha sistema rasik.

Iha rede hanesan ne'e, **anomalia** akontese beibeik. Anomalia mak padraun iha dadus ne'ebé la tuir komportamentu normál (Chandola et al., 2009). Iha rede eletrisidade, anomalia iha grupu rua. Primeiru, anomalia **kualidade enerjia**: Voltajen Tun (voltage sag) iha oras pico kalan, Voltajen Sa'e Liu (over-voltage) bainhira karga ki'ik, no Frekuénsia La Estável bainhira jeradór ida para (generator trip). Daruak, anomalia iha **kliente no meter**: Karga Liu (kliente uza liu kVA kontratu), Na'ok Eletrisidade (bypass ka manipulasaun meter) no Kontador Aat (sensor korrente sala). Na'ok eletrisidade parte husi *non-technical losses* (NTL); iha nasaun balun NTL bele to'o 40% husi eletrisidade ne'ebé distribui (Glauner et al., 2017).

Smart meter ka *Advanced Metering Infrastructure* (AMI) produz reading kada minutu ka kada oras. Ho meters rihun ba rihun, readings sira sai millaun records kada loron — dadus ho karakterístika Big Data: volume, velocity no variety (Laney, 2001; Wang et al., 2019). Pandas ka database tradisionál ne'ebé prosesa dadus iha memória komputadór ida de'it la to'o ona; presiza Apache Kafka ba data ingestion stream (Kreps et al., 2011) no Apache Spark ba prosesamentu distribuídu (Zaharia et al., 2016). Aprezentasaun individuál Reinaldo N. M. Alves Costa nian hatudu ona katak Pandas lalais liu ba dadus ki'ik, maibé Spark sai lalais liu bainhira dadus boot (Costa, 2026a). Capstone ida ne'e aumenta eskala ba dataset 5 GB, 10 GB no 20 GB.

Machine learning permite deteta anomalia automatikamente. Algoritmu **Random Forest** (Breiman, 2001) koñesidu nu'udar klasifikadór ne'ebé forte no estável: Fernández-Delgado et al. (2014) avalia klasifikadór 179 iha dataset 121 no hetan katak família Random Forest mak

di'ak liu. Peskiza ida ne'e kontinua métodu Random Forest husi peskiza anterior Reinaldo N. M. Alves Costa nian kona-ba jestaun bee (WaterPredict) (Costa, 2026b), maibé aplika ba **deteksaun anomalia** iha rede eletrisidade: kada reading smart meter klasifika ba Normal ka tipu anomalia neen. Ho arkitetura Lambda — *speed layer* ba real-time no *batch layer* ba dataset to'o 20 GB — PowerPredict TL bele fó alerta sedu ba EDTL no lista suspeita na'ok eletrisidade ba ekipa inspesaun, liuhusi website ne'ebé proteje ho login.

1.2. Formula Problema (Problem Statement)

Problema prinsipál iha peskiza ida ne'e mak:

1. **Deteksaun Reativu** – Voltajen Tun, Frekuénsia La Estável no Karga Liu hatene de'it depois kliente keixa ka ekipamentu aat ona.
2. **Perda Non-Técnika** – Na'ok Eletrisidade no Kontador Aat la hatene too inspesaun manuál, tanba seidauk iha analítika ba readings meter.
3. **Volume no Velocity Dadus** – Readings smart meter to'o atus millaun records; iha komputadór RAM 32 GB, Pandas la konsege prosesa dataset 5 GB (memory to'o 20.3 GB antes para).
4. **Falta Sistema Inteligente** – Seidauk iha sistema ne'ebé uza machine learning (Random Forest) hodi klasifika tipu anomalia iha readings smart meter Timor-Leste iha tempu real, ho asesu seguru ba operadór no admin.

Husi problema sira ne'e, peskiza ida ne'e hakarak hatán:

Oinsá dezempeñu algoritmu Random Forest, ne'ebé integra iha arkitetura Big Data (FastAPI–Kafka–Spark), hodi deteta no klasifika anomalia iha readings smart meter iha tempu real, kompara ho model seluk?

1.3. Pergunta Peskiza (Research Question)

1. Oinsá dezempeñu Random Forest hodi deteta no klasifika anomalia bazeia ba Precision, Recall, F1-score, PR-AUC no ROC-AUC iha test set no iha stream real-time?
2. Tanba sa Random Forest fit (adekuadu) ho dataset smart meter ne'ebé iha ground truth label no klase la balansu?

3. Oinsá arkitetura Big Data (FastAPI, Kafka, Spark, data lake no MySQL) bele prosesa dataset 5 GB, 10 GB, 20 GB no stream real-time, no iha volume ida ne'ebé mak Pandas la konsege ona?
4. Feature saida mak fó influénsia boot liu (Feature Importance) ba deteksaun kada tipu anomalia?
5. Oinsá dezenvolve website ne'ebé fasil uza no seguru (login Admin / User) ba monitorizasaun anomalia, alerta no simulasaun?

1.4. Objetivu Peskiza (Research Objective)

1.4.1. Objetivu Jerál (General Objective)

Dezenvolve no avalia sistema Big Data PowerPredict TL ne'ebé uza algoritmu Random Forest hodi deteta anomalia iha readings smart meter rede eletrisidade Timor-Leste iha tempu real.

1.4.2. Objetivu Espesífiku (Specific Objectives)

1. Kria dataset Big Data (5 GB, 10 GB, 20 GB) husi seed dadus reál ho events anomalia neen no ground truth label.
2. Implementa pipeline Big Data kompletu: bot → FastAPI → Apache Kafka → Spark Streaming → MySQL, no batch layer Spark ETL → data lake Parquet.
3. Implementa Random Forest (scikit-learn) no Random Forest distribuídu (Spark MLlib) ba deteksaun anomalia.
4. Hatudu ho evidénsia tanba sa Random Forest fit ho dataset, inklui meter-level split, komparasaun ho model seluk no cross-validation.
5. Identifika Feature Importance ba kada tipu anomalia.
6. Avalia performance: latency, throughput, Data Quality no scalability Pandas kompara ho Spark.
7. Dezenvolve website Streamlit ho login Admin / User no pájina 9 ba monitorizasaun, events anomalia, deteksaun manuál, simulasaun no jestaun uzuáriu.
8. Publika website ba internet ho seguru (HTTPS, modu konvidadu read-only, link permanente) no prepara pakote deployment 24/7 (Docker) hodi ema hotu bele haree sistema live.

1.5. Benefísiu Peskiza (Research Benefit)

1.5.1. Benefísiu Teóriku (Theoretical Benefit)

- Kontribui ba literatura Big Data no anomaly detection iha setór enerjia Timor-Leste.
- Fornese evidénsia empíriku katak Random Forest fit ho dadus smart meter ne'ebé iha ground truth label no klase la balansu.
- Fornese evidénsia bainhira tenke muda husi Pandas ba Apache Spark (iha ne'e: husi dataset 5 GB ba leten).

1.5.2. Benefísiu Prátiku (Practical Benefit)

- EDTL hetan alerta automátiku ba Voltajen Tun, Voltajen Sa'e Liu no Frekuénsia La Estável antes kliente keixa.
- Ekipa inspesaun hetan lista suspeita Na'ok Eletrisidade no Kontador Aat, hodi hamenus perda non-técnika.
- Kliente ho Karga Liu bele hetan avizu antes fuse ka transformadór aat.
- Operadór (User) no administradór (Admin) iha asesu tuir sira-nia knaar; funsaun sensitivu (simulasaun, retrain model, jestaun konta) Admin de'it.
- DIT no estudante sira hetan protótipu Big Data kompletu ba aprendizajen.

1.6. Eskopu no Limitasaun Projetu (Project Scope)

1. **Dadus Sintétiku ho Seed Reál** – Meters kria husi seed reál (UCI husi Portugál no temperature Open-Meteo); la'ós dadus kliente EDTL, tanba dadus ne'e privadu. Events anomalia hatama husi jeradór.
2. **Stream Simuladu** – Dadus real-time mai husi bot ho meters simuladu 281, la'ós sensor fíziku.
3. **Komputadór Ida** – Spark hala'o iha modu local (core 16–32) no Kafka ho broker ida; RAM 32 GB mak limite ne'ebé halo Pandas la konsege dataset 5 GB ba leten.
4. **Structured Streaming** – Iha Windows, checkpoint Structured Streaming presiza Hadoop winutils; peskiza ne'e uza Spark micro-batch ho offset iha MySQL no watermark manuál.
5. **Métodu Ida** – Sistema uza Random Forest de'it (hili husi komparasaun iha test set hanesan); la inklui deep learning.

6. **Seguransa Website** – Login Admin / User uza password hash PBKDF2 no cookie sesaun iha rede lokál (http://localhost:8505). Versaun publiku iha internet hatudu modu konvidadu read-only de'it, ho HTTPS husi Cloudflare Tunnel no la iha login; seidauk iha autentikasaun faktór rua.
7. **Disponibilidade Website Publiku** – Website publiku online de'it bainhira komputadór grupu ativu no start_all.bat hala'o (Cloudflare quick tunnel gratuitu, enderesu muda kada dala, link permanente la muda). Pakote deployment 24/7 ba Oracle Cloud Always Free prontu maibé seidauk hala'o tanba presiza konta cloud.

KAPITULU II
REVIZAUN LITERATURA

2.1. Definisaun Revizaun Literatura

Revizaun literatura mak prosesu lee, avalia no rasionaliza teoria no publikasaun sientífiku ne'ebé relevante ho tópiku peskiza, hodi hatene saida mak ema halo ona no pozisiona peskiza ida ne'e iha gap koñesimentu. Iha estudu ida ne'e, revizaun literatura foka ba:

- Konseitu Big Data, data streaming, Apache Kafka, Apache Spark, FastAPI no Streamlit.
- Anomaly detection no tipu anomalia iha rede eletrisidade.
- Algoritmu machine learning Random Forest (no Decision Tree nu'udar nia komponente).
- Métrika evaluasaun ba dadus la balansu (Precision, Recall, F1-score, PR-AUC, ROC-AUC).
- Estudu empíriku kona-ba analítika smart meter no deteksaun na'ok eletrisidade.

2.1.1. Objetivu Revizaun Literatura

1. **Identifika gap peskiza** – deteksaun anomalia ne'ebé sei reativu no manuál.
2. **Justifika algoritmu** – tanba sa Random Forest loos ba anomalia.
3. **Justifika teknolojia Big Data** – tanba sa uza Kafka, Spark no data lake.
4. **Fornese baze sientífiku** ba tipu anomalia no métrika evaluasaun.

2.1.2. Baze Teórika (Theoretical Basis)

Seksaun ida ne'e esplika teoria sira ne'ebé sai baze ba sistema PowerPredict TL: Big Data, ingestaun no prosesamentu dadus, arkitetura Lambda, anomaly detection, algoritmu machine learning, métrika evaluasaun no tools.

2.1.3. Big Data

Big Data mak dadus ne'ebé nia volume, velocity ka variety liu kapasidade komputadór ida no tools tradisionál, entaun servisu tenke fahe ba prosesu ka komputadór barak (Laney, 2001). Tabela tuir mai hatudu "V" lima iha PowerPredict TL:

V	Signifikadu	Iha PowerPredict TL
Volume	Totál dadus	319.1 millaun records, 35.8 GB CSV (5 GB · 10 GB · 20 GB); readings live 3.14 millaun iha MySQL
Velocity	Velosidade dadus tama	Meters 281 haruka reading kada 2 segundu (~140 mensajen/s) liuhusi FastAPI ba Kafka
Variety	Formatu oin-oin	CSV, JSON (REST API + Kafka), Parquet, MySQL table
Veracity	Kualidade dadus	Missing value, duplicate no invalid value deteta no hamoos husi Spark
Value	Valór ba utilizadór	Alerta anomalia, lista suspeita Na'ok Eletrisidade, dashboard ho login ba EDTL

Tabela 2.1. Karakterístika 5V Big Data iha PowerPredict TL

2.1.4. Data Streaming no Apache Kafka

Data streaming mak prosesa dadus kontinuamente bainhira dadus to'o. Apache Kafka mak plataforma *distributed messaging* ne'ebé dezenvolve iha LinkedIn ba log ho volume boot (Kreps et al., 2011). *Producer* haruka mensajen ba *topic*; topic fahe ba *partition* hodi bele prosesa paralelu; kada mensajen iha *offset* ne'ebé permite *consumer* kontinua husi fatin ne'ebé nia para. Kafka 3.9 iha peskiza ne'e hala'o iha modu **KRaft** (la presiza ZooKeeper).

2.1.5. Apache Spark

Apache Spark mak engine unifikadu ba prosesamentu Big Data ne'ebé fahe dadus ba *partition* no prosesa paralelu iha core ka node barak (Zaharia et al., 2016). Spark fornese DataFrame no Spark SQL, **Spark MLlib** ba machine learning distribuídu (Meng et al., 2016), no **Structured Streaming** ho modelu micro-batch, window no **watermark** ba dadus ne'ebé tarde (Armbrust et al., 2018). Pandas *scale up* (komputadór ida, dadus hotu tenke tama iha RAM), enkuantu Spark *scale out* (dadus fahe ba partition no core barak; partition ne'ebé la tama iha RAM bele *spill* ba disk).

2.1.6. Arkitetura Lambda

Arkitetura Lambda (Marz & Warren, 2015) kombina **batch layer** ne'ebé prosesa dadus istóriku tomak no **speed layer** ne'ebé prosesa dadus foun iha tempu real. Rezultadu kamada rua ne'e kombina iha **serving layer** (iha ne'e MySQL) ne'ebé dashboard lee. Iha PowerPredict TL, batch layer treina model Random Forest no speed layer uza model ne'e ba kada reading.

2.1.7. Data Lake no Formatu Parquet

Data lake mak repozitóriu ne'ebé rai dadus iha formatu orijinál (raw) no dadus ne'ebé prosesa ona (curated). Data lake PowerPredict TL iha zona tolu: *raw zone* (CSV), *curated zone*

(Parquet husi Spark ETL, partition tuir fulan) no *stream zone* (Parquet husi readings real-time). Parquet mak formatu columnar ne'ebé kompresa no lalais ba analítika. Tanba dataset sira boot (CSV ~37 GB), data lake rai iha disk D: (konfigurasaun `PP_DATA_DIR` iha `.env`).

2.1.8. FastAPI, Streamlit no Autentikasaun

FastAPI mak web framework Python ba REST API ne'ebé uza *type hints* no Pydantic hodi valida dadus automatikamente no kria dokumentasaun OpenAPI (/docs). Iha sistema ne'e FastAPI sai *gateway* entre bot smart meter no Kafka. **Streamlit** mak framework Python ne'ebé transforma script analítika ba website interativu la presiza HTML/JavaScript; tan ne'e implementasaun website fasil no lalais. Estrutura dashboard tuir projetu capstone referénsia (alerta → deskritiva → tendénsia → estatístika → machine learning → real-time).

Website ne'ebé hatudu dadus kliente no funsaun hanesan simulasaun ka retrain model presiza **autentikasaun** (sé mak uza) no **autorizasaun** (saida mak nia bele halo). Password la rai nu'udar testu, maibé nu'udar *hash* ho salt no funsaun derivasaun ne'ebé neineik hanesan PBKDF2 (Moriarty et al., 2017), hodi password la fasil atu adivinha maske database lakon. *Role-based access control* fahe asesu tuir knaar: User (operadór) haree no analiza, Admin jere sistema.

Atu website bele asesu husi internet la ho IP públiku no la loke port iha router, Cloudflare Tunnel uza programa cloudflared ne'ebé halo koneksaun sai (outbound) husi komputadór ba Cloudflare edge; Cloudflare fó enderesu HTTPS no haruka pedidu vizitante liuhusi koneksaun ne'e (Cloudflare, 2026). GitHub Pages hatudu website estátiku gratuitu husi repozitóriu GitHub, no iha ne'e uza hanesan link permanente (GitHub, 2026). Ba operasaun 24/7, Docker Compose define servisu barak (Kafka, database, API, Spark, website) iha file ida no hala'o sira hanesan container iha servidór ida (Docker Inc., 2026).

2.1.9. Data Analytics

Data analytics mak prosesu analiza dadus hodi hetan informasaun ne'ebé ajuda halo desizaun. Manual Capstone Project (seksaun 5.5) husu nivel analítika tolu no nivel avansadu ida:

- **Descriptive Analytics** – saida mak akontese: totál, média, persentajen no distribuisaun (ezemplu readings, meters no Anomaly Rate kada municipality).

- **Trend Analysis** – oinsá dadus muda tuir tempu: oras, fulan ka janela minutu (ezemplu anomalia kada minutu iha stream).
- **Statistical Analysis** – distribuisaun no relasaun entre variavel: média, percentile (p05, p50, p95) tuir grupu.
- **Advanced Analytics** – Classification, Anomaly Detection no Prediction ho machine learning (iha ne'e Random Forest).

Iha Big Data, agregasaun sira ne'e hala'o distribuídu ho Spark (groupBy, window, percentile_approx) tanba dadus la tama iha memória komputadór ida (Zaharia et al., 2016).

2.1.10. Anomaly Detection

Chandola et al. (2009) define anomalia nu'udar padraun iha dadus ne'ebé la tuir nosaun komportamentu normál, no fahe ba tipu tolu: *point anomaly* (reading ida la normál, ezemplu voltajen 185 V), *contextual anomaly* (reading normál iha kontestu ida maibé la normál iha kontestu seluk, ezemplu konsumu 0.1 kW normál iha madrugada maibé suspeitu iha 20:00) no *collective anomaly* (sekuénsia readings, ezemplu na'ok durante semana). Métodu deteksaun iha rua: **supervised** — model aprende husi ezemplu ho label (normal no tipu anomalia) — no **unsupervised** — model buka pontu izoladu la ho label. Supervised presiza label maibé bele fó **tipu** anomalia; unsupervised la presiza label maibé dala barak hetan alarme falsu no la hatene tipu.

2.1.11. Tipu Anomalia iha Rede Eletrisidade

Limite normál voltajen no frekuénsia tuir padraun EN 50160 no IEC 60038 (voltajen 220 V ±10%, frekuénsia 50 Hz). Tipu anomalia neen ne'ebé sistema deteta mak:

Tipu anomalia	Saida mak akontese	Sinál iha dadus	Se mak tenke halo
Voltajen Tun	Voltajen feeder tun 14–35 V (avaria, transformadór sobrekarga, liña naruk)	Voltage deviation < −6% ... −10%	Operadór rede / subestasaun
Voltajen Sa'e Liu	Voltajen sa'e > +10% iha kalan bainhira karga ki'ik	Voltage deviation > +10%	Operadór rede
Frekuénsia La Estável	Jeradór trip ka load rejection iha sentrál	Frequency fora husi 49.5–50.5 Hz	Sentrál elétrika
Karga Liu	Kliente uza liu kVA kontratu	Load ratio > 1.0	Servisu kliente
Na'ok Eletrisidade	Bypass meter: rejista de'it 5–55% husi konsumu loos	Baseline ratio ki'ik	Ekipa inspesaun
Kontador Aat	Sensor korrente sala durante loron 1–7	Consistency ratio P/(V·I·PF) ≠ 1	Manutensaun meter

Tabela 2.2. Tipu anomalia neen ne'ebé PowerPredict TL deteta

2.1.12. Decision Tree

Decision Tree fahe dadus ho pergunta "se/entaun" (ezemplu: voltage deviation < −6%?) to'o folla ho klase ida (Quinlan, 1986; Breiman et al., 1984). Kada split hili ho Entropy (Information Gain) ka Gini Impurity. Fasil interpreta, maibé tree ida bele *overfit* no sensitivu ba noise.

2.1.13. Random Forest (RF)

Random Forest mak algoritmu *ensemble learning* ne'ebé konstrói decision tree barak no kombina rezultadu liuhusi votasaun (Breiman, 2001).

- **Mekanizmu:** kada tree treina ho bootstrap sample (*bagging*); iha kada node, tree haree de'it feature aleatóriu balun (*random feature selection*) no hili split ho Gini Impurity ki'ik liu.
- **Vantajen ba anomalia:** kaptura relasaun la-lineár no interasaun entre feature (ezemplu konsumu ki'ik *iha oras pico*), robustu ba noise sensor, simu feature numériku no kategóriku la presiza normaliza, fó probabilidade (votu) ne'ebé bele uza nu'udar Anomaly Score, no fó Feature Importance.
- **Class weight:** iha dadus la balansu, peza klase ida (iha ne'e Normal) bele aumenta hodi model la fó alarme falsu barak — balansu entre Precision no Recall.
- **Dezvantajen:** presiza label (ground truth), model boot liu (tree 150) no susar liu atu interpreta kompara ho tree ida.

2.1.14. Tanba sa Random Forest?

Peskiza ida ne'e uza métodu machine learning ida de'it: **Random Forest**. Razaun mak: (1) dadus iha ground truth label, entaun supervised learning bele fó **tipu** anomalia; (2) anomalia depende ba feature barak hamutuk (ezemplu power ki'ik *kompara ho baseline* iha oras pico), no tree sira aprende relasaun non-linear; (3) votu tree barak fó probability ne'ebé sai Anomaly Score; (4) robustu ba noise sensor no la presiza normaliza; (5) literatura hatudu família Random Forest mak klasifikadór di'ak liu (Fernández-Delgado et al., 2014). Iha komparasaun empíriku ho Decision Tree, Logistic Regression no Isolation Forest iha test set hanesan (dataset 20 GB), Random Forest mós mak accurate liu (F1 0.954); rezultadu kompletu iha seksaun 5.3.

2.1.15. Métrika Evaluasaun ba Dadus La Balansu

Iha dataset ne'e kuaze 96% readings Normal. Model ne'ebé dehan "Normal" ba buat hotu atinje Accuracy 96.2% maibé la deteta anomalia ida. Tan ne'e Accuracy la to'o; peskiza ne'e uza Precision, Recall, F1-score no **PR-AUC** ne'ebé informativu liu ROC iha dadus la balansu (Saito & Rehmsmeier, 2015). TP = anomalia ne'ebé deteta loos, FP = alarme falsu, FN = anomalia ne'ebé lakon, TN = normál loos.

Métrika	Definisaun / Objetivu
Precision	Husi alarme hotu, hira mak anomalia loos (hamenus alarme falsu).
Recall	Husi anomalia loos hotu, hira mak model hetan.
F1-Score	Média armóniku Precision no Recall.
PR-AUC (Average Precision)	Área iha kurva Precision–Recall ba threshold hotu; la depende ba threshold ida.
ROC-AUC	Probabilidade model fó score aas liu ba anomalia kompara ho normál.
False Positive Rate	Persentajen readings normál ne'ebé sai alarme falsu: FP/(FP+TN).
Macro F1 + Confusion Matrix	F1 média ba klase 7 (tipu anomalia) no tabela prediction vs atuál.

Tabela 2.3. Métrika evaluasaun deteksaun anomalia

$$Precision = \frac{TP}{TP+FP} \quad Recall = \frac{TP}{TP+FN} \quad F1 = \frac{2 \times Precision \times Recall}{Precision+Recall} \tag{2.1}$$

$$FPR = \frac{FP}{FP+TN} \quad Macro\ F1 = \frac{1}{K}\sum_{k=1}^{K} F1_k \tag{2.2}$$

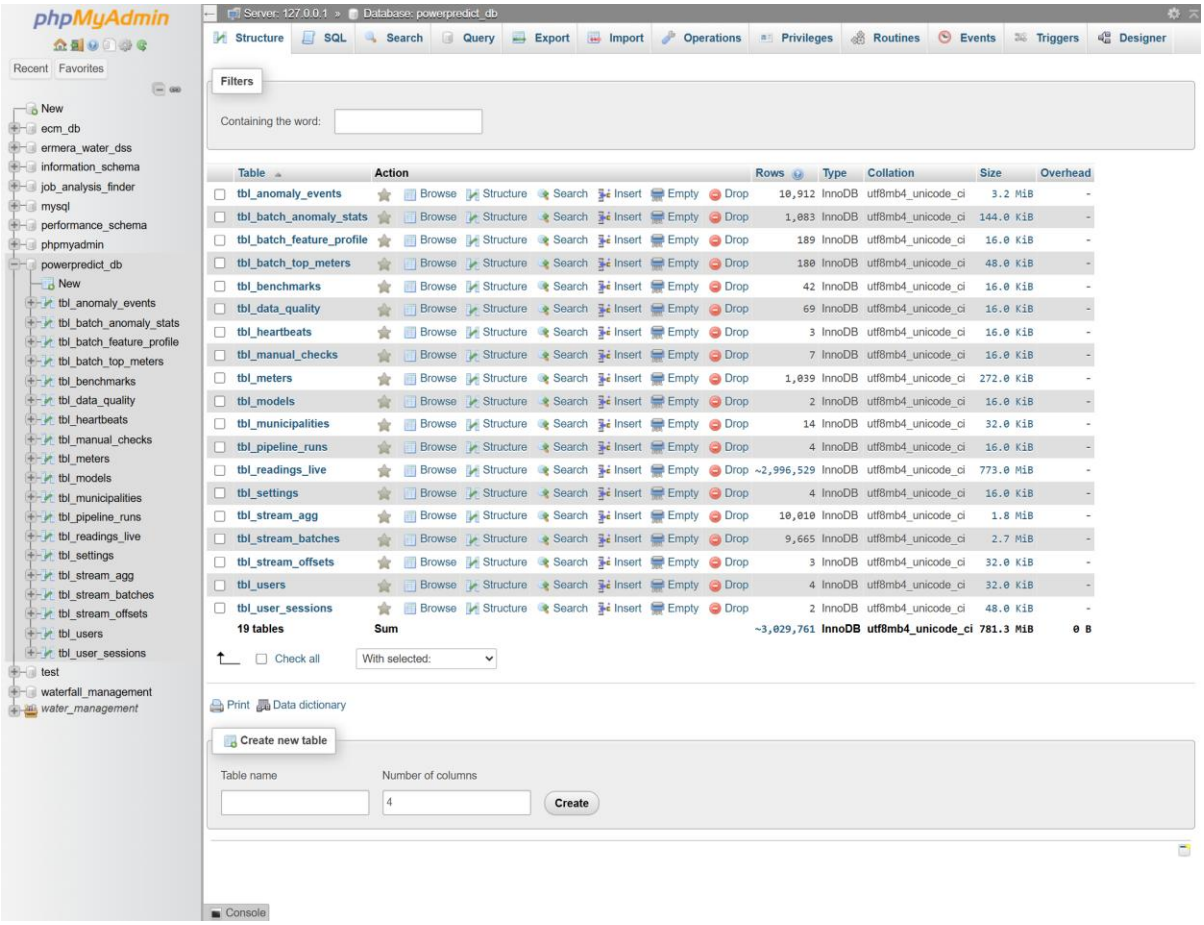
2.1.16. Tools no Linguajen Programasaun

Sistema PowerPredict TL dezenvolve iha **Visual Studio Code** ho teknolojia sira tuir mai:

- **Python 3.11** – jeradór dadus, bot, FastAPI, job Spark (PySpark), machine learning no website.
- **FastAPI 0.162 + Uvicorn** – REST API gateway ho validasaun Pydantic no dokumentasaun OpenAPI.
- **Apache Kafka 3.9.1** – data ingestion stream; topic `smart-meter-readings` ho partition 3.
- **Apache Spark 3.5.3 (PySpark)** – batch ETL, stream micro-batch no Spark MLlib.
- **scikit-learn 1.9** – Random Forest no métrika evaluasaun (Pedregosa et al., 2011).

- **Streamlit 1.63 + Altair** – website dashboard: login, pájina 9, Refresh, Night mode, ikon Material.
- **XAMPP (MariaDB/MySQL 10.4 + phpMyAdmin)** – database serving `powerpredict_db` ho table 19.
- **Apache Arrow / Parquet** – formatu data lake.
- **Cloudflare Tunnel (cloudflared)** – publika website ba internet ho HTTPS.
- **Git / GitHub / GitHub Pages** – kontrolu versaun, repozitóriu source code públiku no link permanente website.
- **Docker Compose + Caddy** – pakote deployment 24/7 (container no HTTPS automátiku).

MySQL mak *Relational Database Management System* (RDBMS) open-source. Imajen tuir mai hatudu database `powerpredict_db` iha phpMyAdmin ho table 19:



Imajen 2.1. Database MySQL powerpredict_db iha phpMyAdmin (XAMPP)

2.2. Peskiza Anterior

Peskiza husi (Wang et al., 2019), "Review of Smart Meter Data Analytics: Applications, Methodologies, and Challenges", revee analítika dadus smart meter ba análize karga, prediksaun karga, jestaun karga no **deteksaun anomalia**, no identifika dezafiu Big Data no privasidade dadus. Revizaun ne'e sai baze atu uza dadus smart meter ho teknolojia Big Data.

Peskiza daruak husi (Jokar et al., 2016), "Electricity Theft Detection in AMI Using Customers' Consumption Patterns", dezenvolve detetór na'ok eletrisidade bazeia ba padraun konsumu kliente iha AMI. Meter transformadór identifika área ho risku aas, depois anomalia konsumu identifika kliente suspeitu. Eksperimentu ho dadus reál kliente 5,000 hatudu dezempeñu aas. Ida ne'e hatudu katak padraun konsumu kompara ho normál (hanesan `baseline_ratio` iha peskiza ne'e) mak sinál xave ba na'ok.

Peskiza datoluk husi (Glauner et al., 2017), "The Challenge of Non-Technical Loss Detection Using Artificial Intelligence: A Survey", revee métodu deteksaun perda non-técnika. Autór sira hatudu katak NTL bele to'o 40% husi eletrisidade distribuída iha nasaun balun, no fahe métodu ba *expert system* no *machine learning*; dezafiu prinsipál mak klase la balansu no label ne'ebé limitadu.

Peskiza dahaat husi (Buzau et al., 2019), "Detection of Non-Technical Losses Using Smart Meter Data and Supervised Learning", propoin métodu supervised learning ba NTL ne'ebé uza dadus smart meter no informasaun auxiliar (lokalizasaun no karakterístika tékniku meter), testa ho dadus reál kliente industriál no komersiál Endesa. Estudu ne'e suporta abordajen supervised ho feature husi meter master hanesan iha PowerPredict TL.

Peskiza dalimak husi (Liu et al., 2008), "Isolation Forest", propoin métodu unsupervised ne'ebé izola anomalia ho tree aleatóriu, ho complexity lineár no memory ki'ik. Iha komparasaun peskiza ida ne'e (test set 20 GB), Isolation Forest atinje F1 0.144 de'it, tan ne'e la uza iha sistema.

Peskiza daneen husi (Fernández-Delgado et al., 2014), "Do we Need Hundreds of Classifiers to Solve Real World Classification Problems?", avalia klasifikadór 179 iha dataset 121; família Random Forest mak di'ak liu, atinje 94.1% husi accuracy máximu. Tanba kritika kona-ba bias, peskiza ida ne'e uza test set husi meters separadu.

Peskiza dahitu mak aprezentasaun individuál Reinaldo N. M. Alves Costa (Costa, 2026a), "Why and When Should We Move from Traditional to Big Data Processing?", ne'ebé kompara Pandas ho Apache Spark iha dadus 100 MB, 1 GB no 5 GB. Peskiza ida ne'e uza métodu eskala dadus no métodu sukat hanesan, maibé ba dadus eletrisidade no ho volume boot liu (5 GB, 10 GB no 20 GB).

No	Autór (Tinan)	Foku	Métodu	Rezultadu prinsipál
1	Wang et al. (2019)	Analítika smart meter	Revizaun	Aplikasaun inklui deteksaun anomalia; dezafiu Big Data
2	Jokar et al. (2016)	Na'ok eletrisidade	Padraun konsumu	Dezempeñu aas ho kliente 5,000
3	Glauner et al. (2017)	Perda non-técnika	Survey	NTL to'o 40%; klase la balansu
4	Buzau et al. (2019)	Deteksaun NTL	Supervised learning	Dadus smart meter + auxiliar (Endesa)
5	Liu et al. (2008)	Anomaly detection	Isolation Forest	Unsupervised, complexity lineár
6	Fernández-Delgado et al. (2014)	Klasifikadór 179	Komparasaun	Família RF di'ak liu
7	Costa (2026a)	Pandas vs Spark	Benchmark 100 MB–5 GB	Spark lalais liu iha dadus boot

Tabela 2.4. Rezumu peskiza anterior

KAPITULU III
METODOLOJIA PESKIZA

3.1. Fatin no Tempu Peskiza (Research Location and Time)

Peskiza ida ne'e hala'o iha Biblioteka Dili Institute of Technology (DIT) no iha uma membru grupu nian iha Dili, durante semestre unidade kurikuler Big Data (semana 16) tuir planu Capstone Project DIT. Dezenvolvimentu, testing no evaluasaun sistema remata iha fulan Novembru 2026.

3.2. Etapa no Oráriu Peskiza

Peskiza tuir métodu *Project-Based Learning* no etapa dezenvolvimentu projetu iha manual Capstone Project (seksaun 8): husi identifikasaun problema to'o demonstrasaun live. Tabela tuir mai hatudu planu semana 16 (manual seksaun 9) no output ne'ebé PowerPredict TL produz iha kada semana:

Semana	Atividade (manual)	Output PowerPredict TL
1	Introdusaun no formasaun grupu	Grupu no tema: deteksaun anomalia rede eletrisidade
2	Identifikasaun problema	Problem statement (seksaun 1.2)
3	Literature Review	Referénsia no peskiza anterior (Kapitulu II)
4	Preparasaun Project Proposal	Proposta projetu
5	Data Source no API	UCI, Open-Meteo API, bot smart meter
6	Data Collection	Raw dataset 5 GB / 10 GB / 20 GB
7	Kafka / Data Ingestion	FastAPI → Kafka (streaming pipeline)
8	Data Storage	Data Lake (D:) + MySQL (table 19)
9	Spark Processing	Spark batch ETL + Spark Streaming
10	Data Analytics	Analítika deskritiva, tendénsia no estatístika (Spark)
11	Machine Learning	Random Forest (scikit-learn + Spark MLlib)
12	Dashboard Development	Website Streamlit: login, pájina 9, Night mode
13	System Integration	Protótipu: start_all.bat (sistema tomak) + publikasaun website (internet)
14	Testing no Evaluation	Functional, performance, data quality, scalability (Kapitulu V)
15	Final Report no Presentation	Relatóriu finál no aprezentasaun
16	Final Demo & Defense	Demonstrasaun live no vídeo YouTube + website publiku live

Tabela 3.1. Etapa no oráriu peskiza (semana 16)

3.3. Komponente Hardware no Software

3.3.1. Komponente Hardware

No	Hardware	Deskrisaun
1.	Laptop	Lenovo Legion (model 83NN)
2.	Processor (CPU)	Intel Core i9-14900HX (thread 32)
3.	Memory (RAM)	32 GB
4.	Storage	SSD 1 TB (C:, sistema no kódigu) + SSD 512 GB (D:, data lake)
5.	Konektividade	Internet ba Open-Meteo API

Tabela 3.2. Komponente hardware

3.3.2. Komponente Software

No	Software	Versaun	Deskrisaun
1.	Windows 11 Home	26H2	Operating system
2.	Python	3.11	Linguajen programasaun prinsipál
3.	Java (Eclipse Temurin)	17	Runtime ba Spark no Kafka
4.	Apache Kafka	3.9.1 (KRaft)	Data ingestion stream
5.	Apache Spark (PySpark)	3.5.3	Batch ETL, streaming, MLlib
6.	FastAPI + Uvicorn	0.162 / 0.52	REST API gateway
7.	XAMPP (MariaDB + phpMyAdmin)	10.4.32	Database serving
8.	scikit-learn / pandas / PyArrow	1.9 / 3.0 / 25	Random Forest no dadus
9.	Streamlit + Altair	1.63 / 6.2	Website dashboard
10.	Playwright + Microsoft Edge	2024	Teste website automátiku no screenshot
11.	Visual Studio Code	2025/1.145.2	Code editor
12.	Microsoft Office	2024	Hakerek relatóriu
13.	Cloudflare cloudflared	2026.10.7	Tunnel HTTPS ba website publiku
14.	Git + GitHub / GitHub Pages	2.55	Repozitóriu source code no link permanente
15.	Docker Compose + Caddy	Caddy 2, MariaDB 11.4	Pakote deployment 24/7 (Oracle Cloud)

Tabela 3.3. Komponente software

3.4. Data Source

Tuir rekezitu manual Capstone Project (seksaun 5.1), sistema uza data source online no real-time. Dadus iha peskiza ida ne'e mai husi fonte haat:

1. **UCI ElectricityLoadDiagrams20112014** (Trindade, 2015) – dataset reál husi UCI Machine Learning Repository (Open Data) ho konsumu eletrisidade kliente 264. Uza nu'udar seed ba padraun fulan-fulan no nivel konsumu meters sira.

2. **Open-Meteo Historical Weather API** (Zippenfenig, 2023) – REST API ne'ebé fornese temperature oras-oras reál tinan 2025 ba kapitál municipality 14 (valór 122,640).
3. **Open-Meteo Forecast API (live)** – temperature real-time ba municipality 14, ne'ebé bot uza kada minutu 15.
4. **Bot smart meter (Simulated Real-Time Data)** – meters live 281 ne'ebé haruka readings JSON ba FastAPI kada segundu 2, ho events anomalia no ground truth.

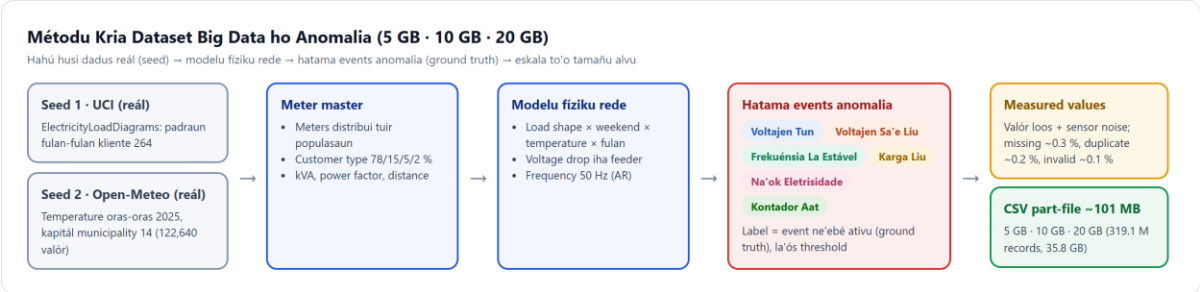
Fonte	Tipu	Formatu	Frekuénsia	Uza ba
UCI ElectricityLoadDiagrams	Open data (reál)	CSV	Fulan-fulan	Seed konsumu
Open-Meteo Archive	REST API (reál)	JSON	Oras-oras 2025	Seed temperature
Open-Meteo Forecast	REST API (live)	JSON	Minutu 15	Temperature stream
Bot smart meter	Simulated real-time	JSON → FastAPI → Kafka	Segundu 2	Speed layer
Jeradór dataset	Big data sintétiku	CSV (part-file)	Oras-oras 2025	Batch layer

Tabela 3.4. Data source PowerPredict TL

3.5. Data Collection

3.5.1. Métodu Kria Dataset Big Data ho Anomalia (5 GB, 10 GB no 20 GB)

Tanba dadus kliente EDTL reál privadu, peskiza ne'e uza métodu aprezentasaun individuál: **hahú husi dadus reál, depois eskala to'o volume alvu**. Jeradór (`data_generator/generate_dataset.py`) kria meters ne'ebé distribui tuir populasaun municipality, ho customer type (78% residensiál, 15% komersiál, 5% governu, 2% indústria), contracted kVA no power factor. Kada meter kopia padraun fulan-fulan husi kliente UCI ida no uza temperature Open-Meteo husi nia municipality. Modelu fíziku rede kalkula power (load shape × weekend × temperature × fulan), voltage drop iha feeder no frequency 50 Hz. Depois jeradór **hatama events anomalia** no rai label ground truth. Meters aumenta to'o file CSV to'o tamañu alvu (meters 5,194 ba 5 GB to'o 20,776 ba 20 GB, kada meter ho oras 8,760 iha tinan 2025); file fahe ba part-file ~101 MB ne'ebé jera paralelu ho process 12–16. Rezultadu data collection (tamañu, records, kualidade) iha seksaun 5.1.



Imajen 3.1. Métodu kria dataset big data ho anomalia husi seed reál

Tipu anomalia	Oinsá jeradór hatama	% records (20 GB)
Voltajen Tun	12-48 feeder events / municipality / year, -14..-35 V, 1-6 h (70 % at the evening peak)	1.160%
Voltajen Sa'e Liu	~10 night-time events / municipality / year, +22..+34 V, 1-3 h	0.198%
Frekuénsia La Estável	38-68 generator-trip events / grid / year, 0.6-1.6 Hz away from 50 Hz, 1-3 h	0.775%
Karga Liu	~6 events / meter / year at 1.05-1.6 x contract, plus natural peaks above 100 %	1.065%
Na'ok Eletrisidade	6 % of meters, 1-3 episodes of 3-30 days, the meter registers only 5-55 %	0.513%
Kontador Aat	8 % of meters, one 1-7 day current-channel fault (x0.15-0.5 or x1.8-3.0)	0.087%

Tabela 3.5. Events anomalia ne'ebé jeradór hatama (ground truth)

3.5.2. Deskrisaun Dataset

Dataset 20 GB (records 182,361,755) mak dataset prinsipál ba training Random Forest no análize batch, enkuantu 5 GB no 10 GB uza ba scalability testing no Spark ETL komparativu. Target label mak koluna `anomaly_type` ho klase 7: Normal no tipu anomalia neen. Dataset iha tipu dadus barak (numeric, categorical, datetime, text) — liu rekezitu mínimu tipu 2 iha manual.

No	Tempu	Municipality	Tipu	kVA	V	A	kW	PF	Hz	Anomalia
1	2025-06-08 16:00	Aileu	comme	7.7	220.42	5.56	2.445	0.678	49.93	Normal
2	2025-03-25 23:00	Aileu	resid	1.3	200.65	3.56	0.663	0.939	49.9	Voltajen Tun
3	2025-05-14 02:00	Aileu	resid	3.5	249.5	2.53	0.583	0.949	50.01	Voltajen Sa'e Liu
4	2025-03-15 09:00	Aileu	resid	2.2	218.16	3.57	0.683	0.892	49.26	Frekuénsia La Estável
5	2025-03-08 06:00	Aileu	comme	16.5	214.32	35.78	20.629	0.891	49.94	Karga Liu
6	2025-02-25 23:00	Aileu	comme	7.7	218.35	0.15	0.084	0.843	50.07	Na'ok Eletrisidade
7	2025-03-21 02:00	Aileu	comme	11	221.47	0.31	0.917	0.902	49.95	Kontador Aat

Tabela 3.6. Raw data readings smart meter (sample husi dataset 20 GB, tipu ida-idak)

No	Atributu	Deskrisaun	Data type
1	reading_id	Identifikadór úniku reading	Numeric

No	Atributu	Deskrisaun	Data type
2	timestamp	Data no oras reading (Asia/Dili)	Datetime
3	meter_id	Kódigu meter (ezemplu TL-DIL-000412)	Categorical
4	municipality	Municipality (14)	Categorical
5	customer_type	Residential, commercial, government, industrial	Categorical
6	contracted_kva	Contracted power (kVA)	Numeric
7	phase	Single ka three phase	Categorical
8	voltage_v	Voltage (V), nominál 220 V	Numeric
9	current_a	Current (A)	Numeric
10	power_kw	Active power (kW)	Numeric
11	energy_kwh	Energy (kWh)	Numeric
12	power_factor	Power factor (0–1)	Numeric
13	frequency_hz	Frequency (Hz), nominál 50 Hz	Numeric
14	temperature_c	Temperature (°C) husi Open-Meteo	Numeric
15	anomaly_type	Normal ka tipu anomalia 6 (target, ground truth)	Categorical

Tabela 3.7. Deskrisaun atributu dataset

3.6. Data Ingestion

Tuir manual seksaun 5.2 (ezemplu API → Kafka), dadus real-time tama liuhusi dalan: **Bot → FastAPI → Apache Kafka**. Bot (`bot/bot_live.py`) hamosu readings husi meters live 281 kada segundu 2 no haruka batch ida ba FastAPI (`POST /api/readings/batch`). FastAPI valida structure (Pydantic) no publish kada reading ba topic Kafka `smart-meter-readings` (partition 3) ho key = meter_id, entaun readings husi meter ida sempre iha partition hanesan no tuir orden. Ezemplu mensajen:

```
{"reading_id": "9f2c…", "meter_id": "TL-OEC-000917", "timestamp": "2026-11-07 09:43:17", "municipality": "Oecusse", "customer_type": "residential", "contracted_kva": 2.2, "phase": "single", "voltage_v": 191.4, "current_a": 4.21, "power_kw": 0.76, "power_factor": 0.94, "frequency_hz": 50.02, "temperature_c": 31.8, "true_type": "Voltage Sag", "producer_ts": 1794012197000}
```

Ba batch layer, dataset CSV (5 GB / 10 GB / 20 GB) tama direta ba data lake raw zone; bot replay (`bot/bot_replay.py`) bele haruka dataset ne'e ba Kafka hanesan stream.

3.7. Data Storage

Tuir manual seksaun 5.3, sistema uza teknolojia storage rua: **Data Lake** no **MySQL**. Data lake rai iha disk D: (`PP_DATA_DIR`) ho zona tolu — raw zone (CSV dataset), curated zone (Parquet husi Spark ETL, partition tuir fulan, no sample training/test) no stream zone (Parquet

husi readings real-time). MySQL `powerpredict_db` (XAMPP) mak serving layer ho table 19: readings live, anomaly events, model, rezultadu Spark, benchmark no konta website. Dezeñu database detalla iha seksaun 4.4.

3.8. Data Processing (Apache Spark)

Tuir manual seksaun 5.4, Apache Spark halo Data Cleaning, Transformation, Aggregation, Filtering, Join no Feature Engineering. Batch ETL (`pipeline/spark_batch_etl.py`) no Spark Streaming (`pipeline/spark_streaming.py`) uza funsaun feature hanesan.

3.8.1. Data Cleaning

- **Missing value:** reading ho valór mamuk iha koluna krítiku (voltage, current, power factor, temperature) hamoos (`dropna`).
- **Duplicate:** reading ho meter_id no timestamp hanesan hamoos (`dropDuplicates`).
- **Invalid value:** valór imposível (voltage 0 ka 999.9 V, power factor > 1, power negativu) hamoos tuir limit validasaun (Filtering).
- **Anomalia la hamoos:** voltage ne'ebé tun ka sa'e maka'as mak anomalia reál ne'ebé model tenke aprende — outliers la'ós erru (analiza iha seksaun 5.1.3).

3.8.2. Feature Engineering no Join

Spark kria feature foun husi koñesimentu domíniu, no halo broadcast join ho meter master (peak utilisation, power factor base) no table load shape ki'ik (customer type × oras). Feature rua ikus mak xave ba anomalia kliente no meter:

Feature	Formula	Deteta
load_ratio	(power_kw / power_factor) / contracted_kva	Karga Liu
voltage_dev_pct	(voltage_v − 220) / 220 × 100	Voltajen Tun, Voltajen Sa'e Liu
freq_dev_hz	frequency_hz − 50	Frekuénsia La Estável
baseline_ratio	power_kw / (kVA × peak util × load shape(oras) × weekend × temperature × PF base)	Na'ok Eletrisidade, Karga Liu
consistency_ratio	power_kw / (V × I × PF × faze) ×1000	Kontador Aat
hour, is_weekend	hour(timestamp), dayofweek	Kontestu tempu

Tabela 3.8. Feature engineering iha Spark

Funsaun feature hanesan iha Spark (batch no stream) no pandas (Deteksaun Manual); teste ida konfirma katak diferensa entre rua mak 0 ba feature hotu. Model uza feature 16 (numeric 14 + categorical 2).

3.8.3. Data Transform

Feature kategóriku (customer_type, municipality) transforma ho **one-hot encoding**. Random Forest la presiza normalizasaun (scaling) tanba split tree depende ba orden valór, la'ós ba eskala. Valór numériku rai nu'udar float; output curated rai nu'udar Parquet (columnar).

3.8.4. Data Labeling (Ground Truth)

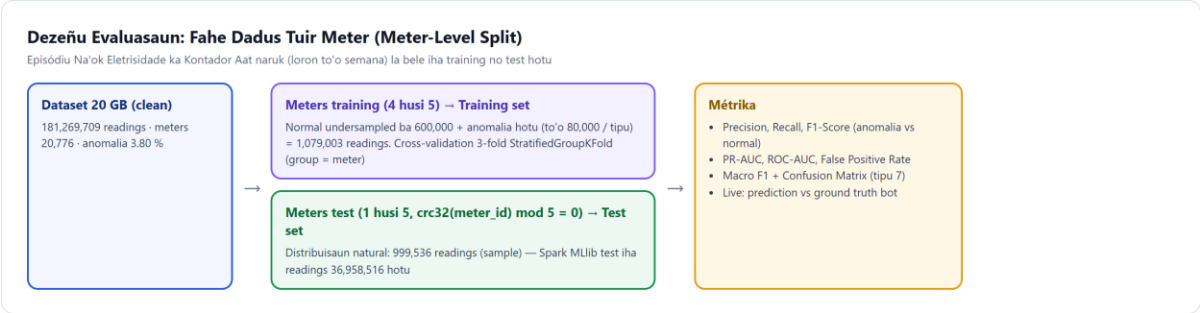
Label la kalkula husi threshold, maibé husi **events ne'ebé jeradór hatama**. Se events barak akontese iha oras hanesan, label tuir prioridade: Kontador Aat > Na'ok Eletrisidade > Frekuénsia La Estável > Voltajen Tun > Voltajen Sa'e Liu > Karga Liu. Dataset rai valór **medidu** (valór loos + noise sensor), tan ne'e model tenke aprende padraun ho noise:

Label	Deskrisaun	Severity
Normal	La iha event	–
Voltajen Tun	Event feeder: voltage tun 14–35 V, oras 1–6	High
Voltajen Sa'e Liu	Event kalan: voltage sa'e 22–34 V, oras 1–3	Medium
Frekuénsia La Estável	Generator trip: frequency 0.6–1.6 Hz husi 50 Hz	High
Karga Liu	Load ratio loos > 1.0 (event ka pico naturál)	High
Na'ok Eletrisidade	Meter rejista 5–55% husi konsumu loos durante loron 3–30	High
Kontador Aat	Current channel ×0.15–0.5 ka ×1.8–3.0 durante loron 1–7	Medium

Tabela 3.9. Data labeling (target classes)

3.8.5. Divizaun Training no Test (Meter-Level Split)

Episódiu Na'ok Eletrisidade ka Kontador Aat naruk (loron to'o semana). Se fahe tuir reading aleatóriu, readings husi episódiu hanesan bele mosu iha training no test, no rezultadu sai di'ak liu realidade. Tan ne'e Spark fahe **tuir meter**: meter ne'ebé crc32(meter_id) mod 5 = 0 sai meter test. Training set: Normal undersampled ba 600,000 + anomalia hotu (to'o 80,000 kada tipu) = 1,079,003 readings. Test set: sample 999,536 readings husi meters test ho **distribuisaun natural**; meters test iha readings 36,958,516 hotu (uza ba test Spark MLlib).



Imajen 3.2. Dezeñu evaluasaun: meter-level split no test set ho distribuisaun natural

Klase	Training set (maizumenus)	Test set (natural)
Normal	600,000	960,307
Voltajen Tun	80,000	11,535
Voltajen Sa'e Liu	80,000	1,966
Frekuénsia La Estável	80,000	7,807
Karga Liu	80,000	11,101
Na'ok Eletrisidade	80,000	5,868
Kontador Aat	80,000	952

Tabela 3.10. Distribuisaun klase iha training set no test set (dataset 20 GB)

Nota: númeru training set ba kada klase mak maizumenus, tanba Spark `sampleBy` foti sample aleatóriu tuir fraksaun.

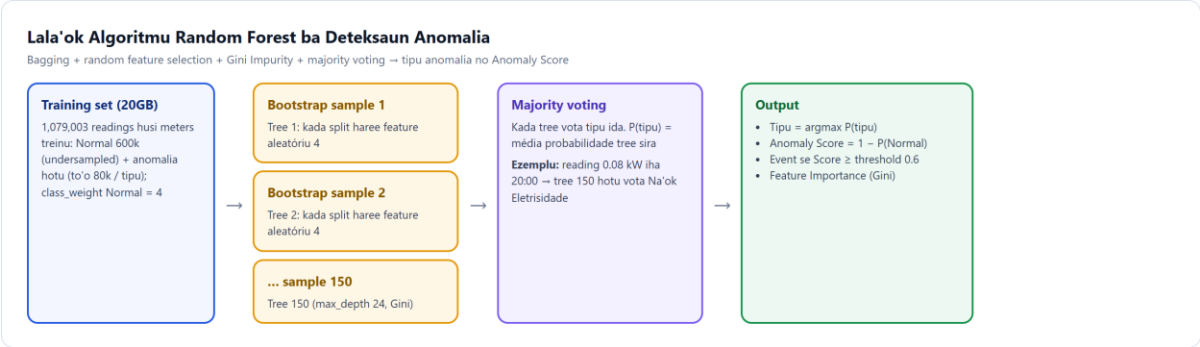
3.9. Data Analytics no Métodu Random Forest

3.9.1. Analítika Deskritiva, Tendénsia no Estatístika

Tuir manual seksaun 5.5, Spark halo analítika tolu iha dataset 20 GB (`4_anomaly_analytics` iha batch ETL) no iha stream (window 1 minutu): (1) **descriptive** — readings no anomalia kada municipality, tipu no customer type; (2) **trend** — anomalia kada oras, kada fulan no kada minutu; (3) **statistical** — média no percentile (p05, p50, p95) feature kada tipu anomalia (`percentile_approx`), no top meters ho oras anómala barak liu. Rezultadu rai iha `tbl_batch_anomaly_stats`, `tbl_batch_feature_profile` no `tbl_batch_top_meters`; rezultadu iha seksaun 5.2. Analítika avansadu mak klasifikasaun anomalia ho Random Forest.

3.9.2. Random Forest

Random Forest mak métodu *ensemble learning* ne'ebé hari'i decision tree barak no kombina rezultadu liuhusi votasaun. Kalkulasaun iha sistema bazeia ba etapa tuir mai:



Imajen 3.3. Lala'ok algoritmu Random Forest ba deteksaun anomalia

a) Bootstrap Aggregating (Bagging).

Ba kada tree (150), sistema foti random sample ho replacement husi training set (1,079,003 readings). Readings balun mosu dala barak no balun la mosu (kuaze 36.8%), entaun tree ida-idak iha variasaun.

b) Random Feature Selection.

Iha kada node, tree la haree feature hotu, maibé sub-konjuntu aleatóriu $\sqrt{16} \approx 4$ (parameter `max_features = sqrt`). Ida ne'e hamenus korelasaun entre tree no variance model. Tree ida-idak bele to'o kle'an 24 (`max_depth`) ho folla minimu readings 2.

c) Gini Impurity no Class Weight.

$$Gini(t) = 1 - \sum_{k=1}^{K} p_k{}^2 \qquad (3.1)$$

$$\Delta Gini = Gini(parent) - \frac{n_L}{n} Gini(L) - \frac{n_R}{n} Gini(R) \qquad (3.2)$$

- p_k: proporsaun klase k iha node t; K = 7 (Normal + tipu anomalia 6). Split ho ΔGini boot liu mak hili.
- **Class weight:** kada reading Normal konta dala 4 iha kalkulasaun p_k (`class_weight = {Normal: 4}`). Tanba training set undersample Normal, peza ne'e fó fali importánsia ba Normal hodi model la fó alarme falsu barak iha distribuisaun natural.

d) Majority Voting no Anomaly Score.

Kada tree vota klase ida. Probability klase mak média probability tree hotu; klase finál mak probability aas liu. **Anomaly Score** mak probability katak reading la'ós Normal:

$$P(k|x) = \frac{1}{T}\sum_{t=1}^{T} p_{t,k}(x) \quad Anomaly\ Score = 1 - P(Normal|x) \qquad (3.3)$$

Ezemplu reál husi sistema: meter residensiál iha Dili ho contracted 3.5 kVA, iha oras 20:00, rejista power 0.08 kW (baseline ratio 0.037), voltage 221 V no frequency 50 Hz. Husi tree 150, tree 150 vota **Na'ok Eletrisidade**; Anomaly Score 1.00. Konsumu ki'ik tebes iha oras pico kalan mak padraun na'ok eletrisidade.

e) Feature Importance.

$$Imp(f) = \frac{1}{T}\sum_{tree}\sum_{t:\, split\ ho\ f}\frac{n_t}{n}\Delta Gini(t) \qquad (3.4)$$

Feature Importance mak redusaun Gini totál husi split ne'ebé uza feature f, média ba tree hotu; rezultadu iha seksaun 5.3.

3.9.3. Data Training: Kalkulasaun Manuál Root Node

Hodi hatudu oinsá decision tree hili root node, tuir mai amostra readings reál **12** husi training set 20 GB ho klase tolu (Normal 5, Voltajen Tun 4, Na'ok Eletrisidade 3) no atributu diskretizadu tolu: Voltage (Low se voltage deviation < −6%), Consumption (Low se baseline ratio < 0.5) no Period (Peak 18:00–22:00):

No	Meter	Municipality	V	Dev %	Baseline	Oras	Voltage	Consumption	Period	Klase
1	TL-ERM-012858	Ermera	212.09	-3.595	0.823	20	Normal	Normal	Peak	Normal
2	TL-DIL-007097	Dili	219.63	-0.168	0.961	0	Normal	Normal	Off-peak	Normal
3	TL-ERM-012372	Ermera	190.91	-13.223	1.001	18	Low	Normal	Peak	Voltajen Tun
4	TL-MNT-016559	Manatuto	220.56	0.255	0.095	15	Normal	Low	Off-peak	Na'ok Eletrisidade
5	TL-COV-006699	Cova Lima	187.72	-14.673	0.86	0	Low	Normal	Off-peak	Voltajen Tun
6	TL-AIL-000420	Aileu	221.77	0.805	0.873	14	Normal	Normal	Off-peak	Normal
7	TL-MNT-017292	Manatuto	219.86	-0.064	0.245	17	Normal	Low	Off-peak	Na'ok Eletrisidade
8	TL-BAU-003763	Baucau	211.54	-3.845	0.792	20	Normal	Normal	Peak	Normal
9	TL-ERM-012829	Ermera	219.35	-0.295	0.782	16	Normal	Normal	Off-peak	Normal
10	TL-VIQ-020418	Viqueque	201.03	-8.623	1.553	0	Low	Normal	Off-peak	Voltajen Tun
11	TL-LAU-015110	Lautém	205.5	-6.591	0.812	0	Low	Normal	Off-peak	Voltajen Tun
12	TL-ERM-012180	Ermera	220.73	0.332	0.306	16	Normal	Low	Off-peak	Na'ok Eletrisidade

Tabela 3.11. Amostra training data 12 (reál)

Kalkulasaun Entropy

$$Entropy(S) = -\sum_{i=1}^{3} p_i log_2 p_i \qquad (3.5)$$

$$Entropy(S) = -(\frac{5}{12} \times log_2 \frac{5}{12}) - (\frac{4}{12} \times log_2 \frac{4}{12}) - (\frac{3}{12} \times log_2 \frac{3}{12})$$

$$Entropy(S) = 0.5263 + 0.5283 + 0.5000 = 1.5546$$

A. Atributu: Voltage

• **Voltage = Low** (4 records: Voltajen Tun 4) → Entropy = −(4/4)·log₂(4/4) = **0.0000**

• **Voltage = Normal** (8 records: Normal 5, Na'ok Eletrisidade 3) → Entropy = −(5/8)·log₂(5/8) −(3/8)·log₂(3/8) = **0.9544**

B. Atributu: Consumption

• **Consumption = Low** (3 records: Na'ok Eletrisidade 3) → Entropy = −(3/3)·log₂(3/3) = **0.0000**

• **Consumption = Normal** (9 records: Normal 5, Voltajen Tun 4) → Entropy = −(5/9)·log₂(5/9) −(4/9)·log₂(4/9) = **0.9911**

C. Atributu: Period

• **Period = Off-peak** (9 records: Normal 3, Voltajen Tun 3, Na'ok Eletrisidade 3) → Entropy = −(3/9)·log₂(3/9) −(3/9)·log₂(3/9) −(3/9)·log₂(3/9) = **1.5850**

• **Period = Peak** (3 records: Normal 2, Voltajen Tun 1) → Entropy = −(2/3)·log₂(2/3) −(1/3)·log₂(1/3) = **0.9183**

Kalkula Gain

$$Gain(S, A) = Entropy(S) - \sum_{v \in A} \frac{|Sv|}{|S|} Entropy(S_v) \qquad (3.6)$$

$$Gain(Voltage) = 1.5546 - [(\frac{4}{12} \times 0.0000) + (\frac{8}{12} \times 0.9544)] = 0.9183$$

Atributu	Totál	Normal	Voltajen Tun	Na'ok	Entropy	Gain	Gini / ΔGini
Voltage	12	5	4	3		0.9183	0.3403
Low	4	0	4	0	0.0000		0.0000
Normal	8	5	0	3	0.9544		0.4688
Consumption	12	5	4	3		0.8113	0.2824
Low	3	0	0	3	0.0000		0.0000
Normal	9	5	4	0	0.9911		0.4938
Period	12	5	4	3		0.1363	0.0417
Off-peak	9	3	3	3	1.5850		0.6667
Peak	3	2	1	0	0.9183		0.4444

Tabela 3.12. Kalkula Gain root node (summary)

Atributu **Voltage** sai root node tanba nia Information Gain boot liu (0.9183) no redusaun Gini boot liu (0.3403). Depois Consumption (Gain 0.8113) fahe Na'ok Eletrisidade husi Normal. Period fó informasaun uitoan de'it (Gain 0.1363). Ida ne'e konsistente ho Feature Importance model tomak (seksaun 5.3), iha ne'ebé voltage no baseline ratio mak feature importante.

3.10. Visualization

Tuir manual seksaun 5.6, dashboard tenke hatudu KPI, gráfiku, tendénsia, monitorizasaun real-time no alerta. Website Streamlit PowerPredict TL implementa elementu hotu ne'e:

Rekezitu manual	Implementasaun iha website
KPI	Readings, meters, municipalities, Anomaly Rate, anomaly events, Precision / Recall / F1 live
Charts / Grafiku	Bar empilhadu tuir tipu, histograma Anomaly Score, Confusion Matrix, Feature Importance, mapa meters
Trend	Anomalia kada minutu (window Spark 1 min / 5 min), tuir oras no fulan
Real-Time Monitoring	Monitor Real-Time (refresh 5 s), Dashboard (refresh 10 s), feed anomalia, Voltage no Frequency
Alert / Notification	Alerta meters ho anomalia agora no suspeita Na'ok Eletrisidade; anomaly events ho severity

Tabela 3.13. Rekezitu visualization no implementasaun

Dezeñu dashboard no screenshot interface hotu iha seksaun 4.6 no 4.7.

3.11. Métodu Testing no Evaluasaun

Tuir manual seksaun 14, testing hala'o iha aspeitu haat, plus evaluasaun model machine learning:

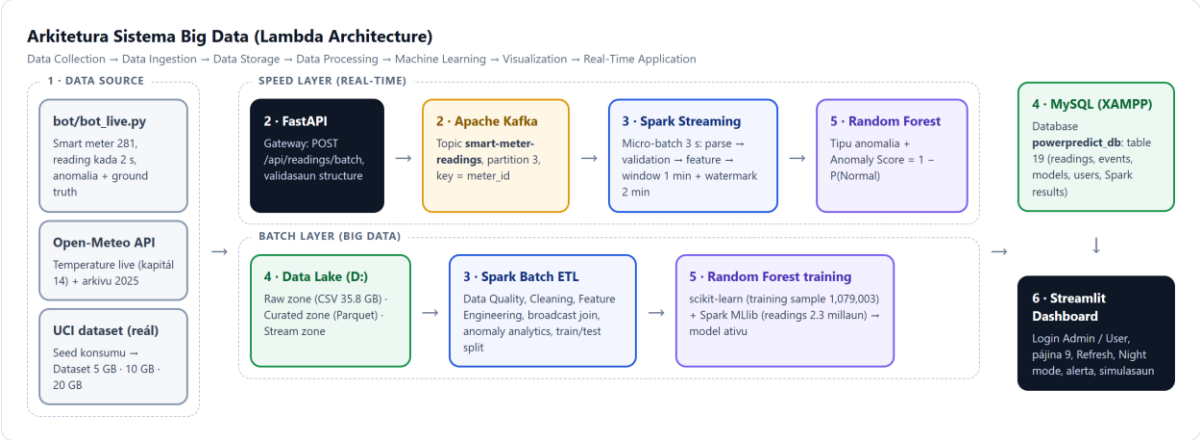
Aspeitu	Saida mak sukat	Oinsá
Functional Testing	Dadus tama? Prosesa loos? Rai loos? Dashboard funsiona?	Kontajen Kafka / MySQL, simulasaun anomalia, pytest, teste website automátiku (Playwright)
Performance Testing	Processing time, throughput, latency, CPU / memory usage	Timer kada etapa Spark ETL, latency kada micro-batch, psutil (CPU % no RAM)
Data Quality Testing	Missing values, duplicate, invalid records, outliers	Spark profiling iha kada ETL; validasaun stream; regra IQR ba outliers
Scalability Testing	Processing time tuir volume 10,000 – 1,000,000 + 5 / 10 / 20 GB	Pandas vs Spark, servisu hanesan, run 3 kada engine, memory guard 85% RAM
Evaluasaun model	Precision, Recall, F1, PR-AUC, ROC-AUC, FPR, Macro F1	Test set husi meters separadu (seksaun 3.8.5), cross-validation 3-fold, stream live
Public Access Testing	Website publiku loke husi internet? Konvidadu read-only? Pájina Admin taka? Offline hatene?	Playwright liuhusi link permanente → tunnel (la ho konta); HTTP status, navigasaun, URL Admin direta

Tabela 3.14. Métodu testing no evaluasaun

KAPITULU IV
DEZEÑU SISTEMA NO IMPLEMENTASAUN

4.1. Arkitetura Sistema (System Architecture)

Tuir manual seksaun 7, imajen tuir mai hatudu fluxu husi fonte dadus to'o dashboard real-time. Sistema uza arkitetura Lambda: *speed layer* ba readings real-time no *batch layer* ba dataset boot; MySQL mak *serving layer*. Estrutura folder tuir etapa pipeline (bot → api → pipeline → dashboard):

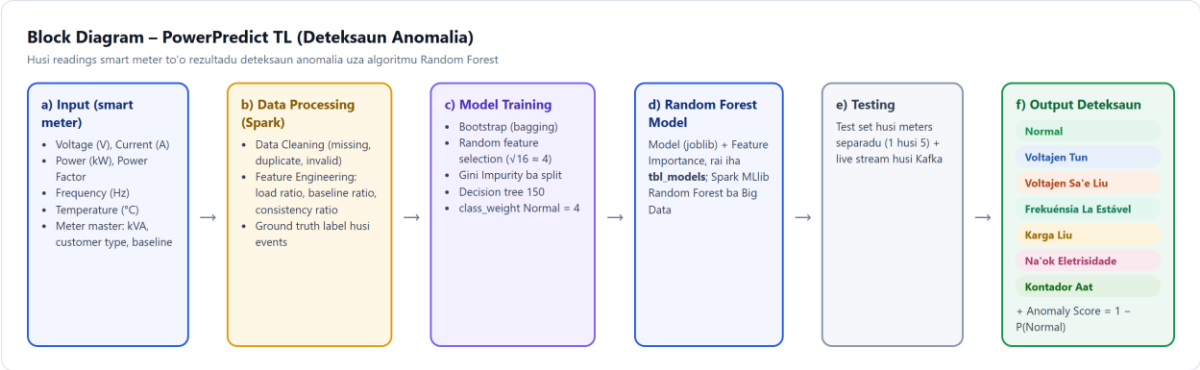


Imajen 4.1. Arkitetura sistema Big Data (Lambda Architecture)

Etapa	Implementasaun
Data Collection	Bot smart meter (meters 281, Open-Meteo live) + jeradór dataset (seed UCI + Open-Meteo)
Data Ingestion	FastAPI gateway (`POST /api/readings/batch`) → Apache Kafka topic `smart-meter-readings`, partition 3, key = meter_id
Data Storage	Data Lake iha D: (raw CSV, curated Parquet, stream Parquet) + MySQL `powerpredict_db` (table 19)
Data Processing	Spark batch ETL (cleaning, feature engineering, broadcast join, aggregation) + Spark Streaming (micro-batch 3 s, window 1 min, watermark 2 min)
Data Analytics / ML	Descriptive, trend, statistical analysis + Random Forest (scikit-learn no Spark MLlib)
Visualization	Website Streamlit: login Admin / User, pájina 9, Refresh, Night mode (KPI, gráfiku, mapa, table)
Real-Time Application	Anomaly Score kada reading, anomaly events, lista suspeita Na'ok Eletrisidade, simulasaun

Tabela 4.1. Etapa pipeline Big Data

Block diagram tuir mai hatudu prosesu husi input readings to'o output deteksaun anomalia uza Random Forest:

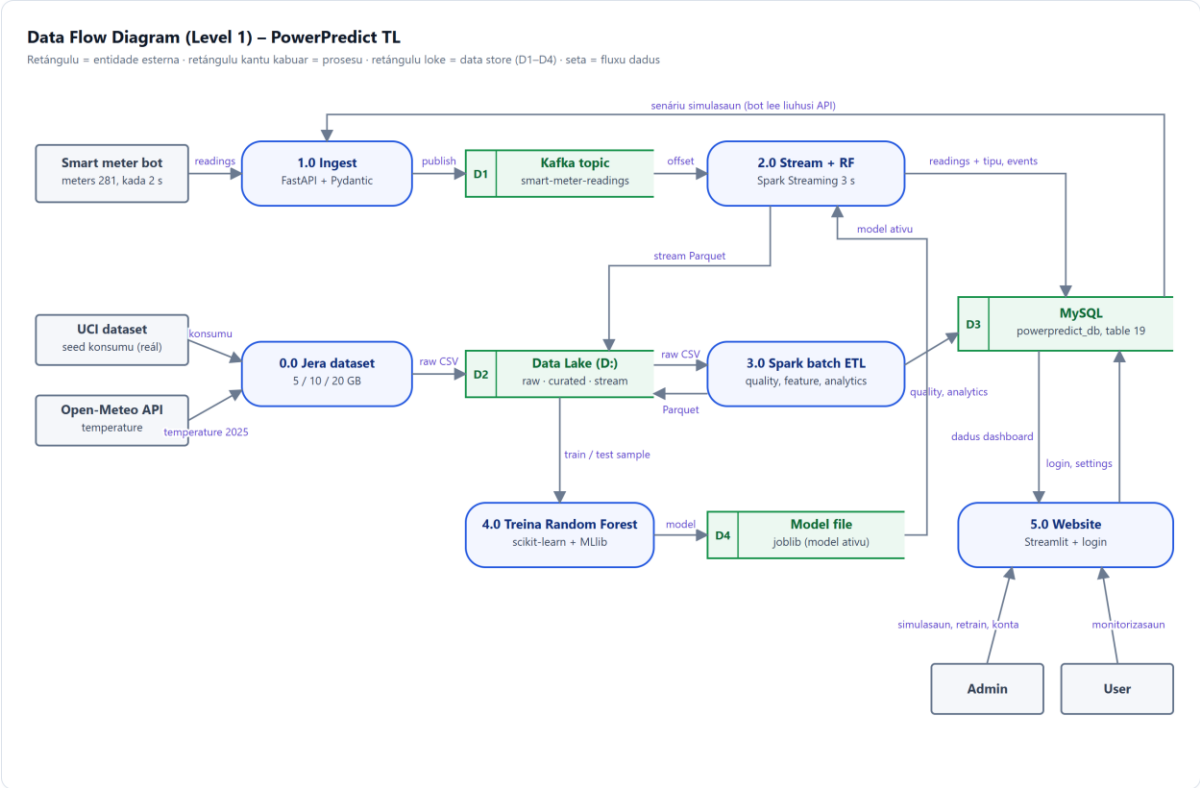


Imajen 4.2. Block diagram Sistema Deteksaun Anomalia Rede Eletrisidade

- **a) Input:** reading smart meter (voltage, current, power, power factor, frequency), temperature no meter master (kVA, customer type, baseline).
- **b) Data Processing:** Spark hamoos dadus, kria feature foun (load ratio, baseline ratio, consistency ratio) no uza label ground truth husi events.
- **c) Model Training:** bootstrap (bagging), random feature selection, Gini Impurity, decision tree 150, class weight Normal = 4.
- **d) Random Forest Model:** model rai iha file joblib no rejista iha `tbl_models` ho métrika no Feature Importance.
- **e) Testing:** test set husi meters separadu no readings real-time husi Kafka.
- **f) Output:** Normal ka tipu anomalia neen (Voltajen Tun, Voltajen Sa'e Liu, Frekuénsia La Estável, Karga Liu, Na'ok Eletrisidade, Kontador Aat) + Anomaly Score.

4.2. Data Flow Diagram

Data Flow Diagram (level 1) hatudu entidade esterna (bot, UCI, Open-Meteo, Admin, User), prosesu (0.0–5.0), data store (D1 Kafka, D2 Data Lake, D3 MySQL, D4 file model) no dadus ne'ebé sulin entre sira:



Imajen 4.3. Data Flow Diagram (level 1) PowerPredict TL

Prosesu	Input	Output
0.0 Jera dataset	Seed UCI, temperature Open-Meteo	CSV raw 5 / 10 / 20 GB (D2)
1.0 Ingest	Readings JSON husi bot; senáriu husi D3	Mensajen Kafka (D1)
2.0 Stream + Random Forest	Offset foun husi D1; model ativu (D4)	Readings + tipu anomalia, events, batch stats (D3); Parquet stream (D2)
3.0 Spark batch ETL	CSV raw (D2)	Parquet curated, sample train / test (D2); Data Quality no analítika (D3)
4.0 Treina Random Forest	Sample train / test (D2)	File model (D4); métrika model (D3)
5.0 Website	Dadus dashboard (D3); asaun Admin / User	Login, settings, konfirma events (D3)

Tabela 4.2. Prosesu iha Data Flow Diagram

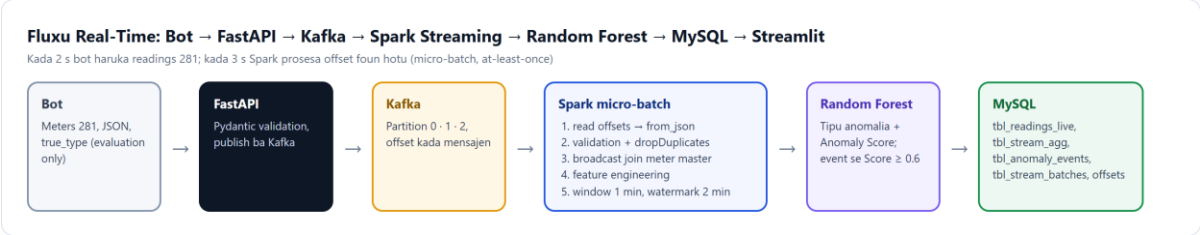
4.3. Processing Pipeline

4.3.1. Batch Pipeline (Spark ETL)

Job `pipeline/spark_batch_etl.py` hala'o etapa neen: (1) read CSV (part-file hotu, schema explícitu); (2) Data Quality profile (missing, duplicate, invalid kada koluna); (3) cleaning + feature engineering + broadcast join ho meter master; (4) anomaly analytics (agregasaun tuir municipality, oras, fulan, feature profile, top meters); (5) sample training / test

tuir meter; (6) write curated Parquet (partition tuir fulan). Ba dataset 20 GB, Spark uza `local[16]` no shuffle partitions = 2 × partition input (416) hodi Java heap la to'o limite; spill no ficheiru temporáriu rai iha D: (`PP_SPARK_LOCAL_DIR`).

4.3.2. Streaming Pipeline (Spark Streaming)



Imajen 4.4. Fluxu real-time Bot → FastAPI → Kafka → Spark → Random Forest → MySQL

Spark Streaming (`pipeline/spark_streaming.py`) hala'o kada segundu 3: (1) lee offset foun husi partition hotu; (2) parse JSON ho schema; (3) validation no dropDuplicates; (4) broadcast join ho meter master no feature engineering; (5) window 1 minutu kada municipality ho **watermark 2 minutu** (reading tarde liu la tama iha window count); (6) Random Forest klasifika kada reading; (7) rai rezultadu iha MySQL, kria anomaly event (Anomaly Score ≥ 0.6, máximu 1 kada meter/tipu kada minutu 5), rai iha data lake stream zone; (8) commit offset ba `tbl_stream_offsets` (at-least-once). `true_type` uza de'it ba evaluasaun, nunka sai input model. Topic Kafka iha mensajen 2,024,886 iha partition 3.

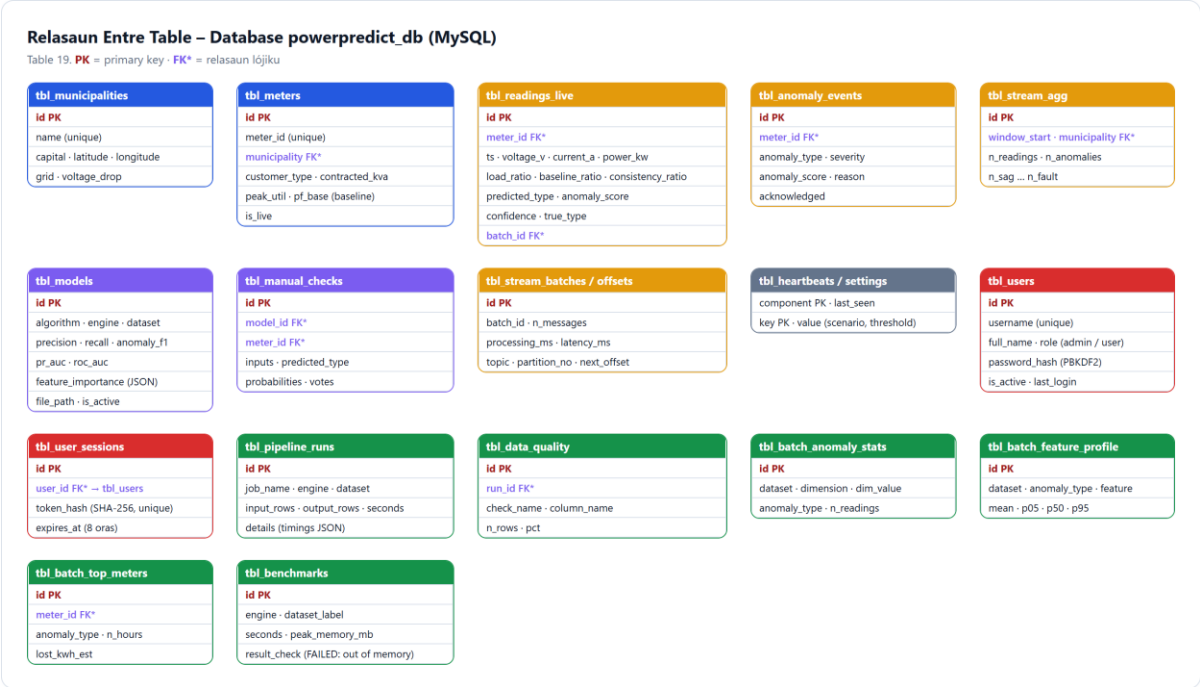
4.3.3. Integrasaun Sistema (Launcher)

Launcher `start_all.bat` hahú Kafka (KRaft), FastAPI, Spark Streaming, bot no website iha janela rasik. Script ne'e verifika uluk se komponente ida ativu ona (`setup/check_running.ps1`) no la hahú dala rua, tanba Kafka broker rua iha log dir hanesan bele estraga dadus Kafka. Se dadus Kafka estraga (`CorruptRecordException`), `setup\start_kafka.bat reset` hamoos dadus Kafka de'it — readings iha MySQL la lakon. Pasu 6 (foun) hahú website publiku: setup/public_site.ps1 hahú kópia website konvidadu iha port 8510, Cloudflare Tunnel, no atualiza link permanente iha GitHub Pages (seksaun 4.9).

4.4. Dezeñu Database (Database Design)

Imajen tuir mai hatudu oinsá table sira liga ba malu iha database `powerpredict_db`: meter liga ba municipality; readings live no anomaly events liga ba meter; model rejista iha

tbl_models; rezultadu Spark (Data Quality, statistics anomalia, top meters) rai tuir dataset; konta website (tbl_users) iha sesaun login (tbl_user_sessions).



Imajen 4.5. Relasaun entre table database powerpredict_db

Database iha table 19. Table importante liu mak:

No	Field	Type	Deskrisaun
1	meter_id	Varchar(20)	Kódigu meter (unique)
2	customer_type	Varchar(20)	Customer type
3	municipality	Varchar(40)	Municipality
4	contracted_kva	Float	Contracted kVA
5	peak_util, pf_base	Float	Ba baseline meter
6	is_live	Boolean	Haruka stream real-time

Tabela 4.3. tbl_meters

No	Field	Type	Deskrisaun
1	id	BigInt	Primary key
2	meter_id, ts	Varchar / Datetime	Meter no tempu
3	voltage_v … temperature_c	Float	Valór medidu
4	load_ratio, baseline_ratio, consistency_ratio	Float	Feature engineering
5	predicted_type	Varchar(24)	Output Random Forest
6	anomaly_score, confidence	Float	1 − P(Normal), P(tipu)
7	true_type	Varchar(24)	Ground truth (evaluation only)
8	latency_ms	Float	Latency bot → MySQL

Tabela 4.4. tbl_readings_live

No	Field	Type	Deskrisaun
1	id	Integer	Primary key
2	meter_id, municipality	Varchar	Fatin
3	anomaly_type, severity	Varchar	Tipu no severity
4	anomaly_score	Float	Score Random Forest
5	reason	Text	Evidénsia (ezemplu voltage −14%)
6	acknowledged	Boolean	Admin konfirma ona

Tabela 4.5. tbl_anomaly_events

No	Field	Type	Deskrisaun
1	id	Integer	Primary key
2	algorithm, engine, dataset	Varchar	random_forest, scikit-learn / spark-mllib
3	anomaly_precision, anomaly_recall, anomaly_f1	Float	Métrika binary
4	pr_auc, roc_auc, f1_macro	Float	Métrika seluk
5	feature_importance, class_report	LongText	JSON
6	file_path, is_active	Varchar / Boolean	File joblib, model ativu

Tabela 4.6. tbl_models

No	Field	Type	Deskrisaun
1	id	Integer	Primary key
2	username	Varchar(40)	Naran login (unique)
3	full_name	Varchar(80)	Naran kompletu
4	role	Varchar(10)	admin ka user
5	password_hash	Varchar(200)	pbkdf2_sha256$260000$salt$hash (la rai password)
6	is_active, last_login	Boolean / Datetime	Konta ativu, login ikus

Tabela 4.7. tbl_users

No	Field	Type	Deskrisaun
1	id	Integer	Primary key
2	token_hash	Varchar(64)	SHA-256 husi token cookie (unique)
3	user_id	Integer	Konta (tbl_users)
4	expires_at	Datetime	Sesaun remata (oras 8)

Tabela 4.8. tbl_user_sessions

4.5. Implementasaun Random Forest

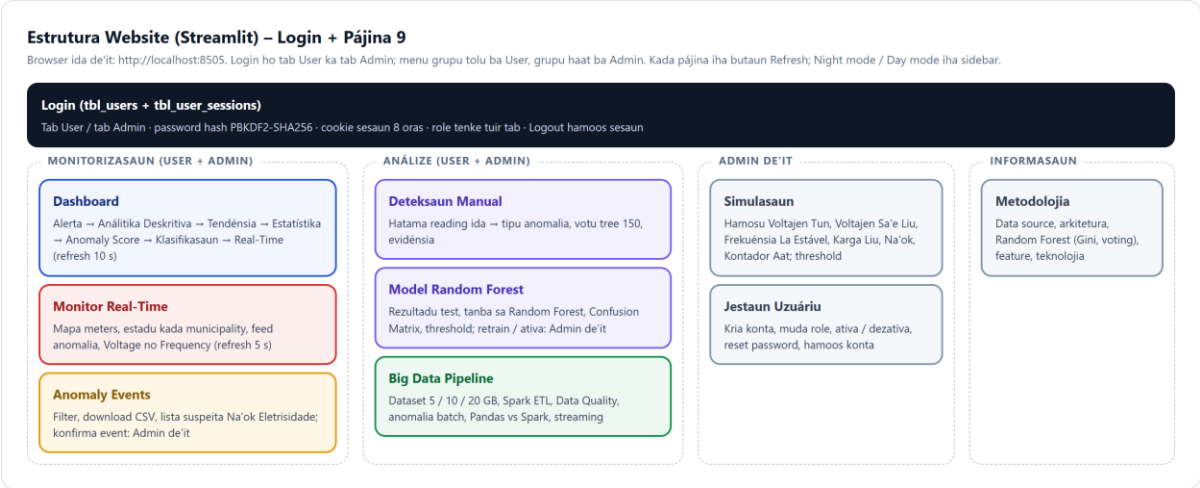
Random Forest implementa dala rua: **scikit-learn** (`ml/train.py`) ba model ne'ebé sistema uza iha stream no Deteksaun Manual, no **Spark MLlib** (`pipeline/spark_ml_train.py`) ba training distribuídu iha readings millaun. Model ida-idak rejista iha `tbl_models`; Admin bele ativa model ida husi pájina Model Random Forest.

Parameter	scikit-learn (#13, ativu)	Spark MLlib (#12)
Tree	150	60
Kle'an máximu	24	16
Feature kada split	sqrt	sqrt
Folla minimu / bins	min_samples_leaf 2	maxBins 64
Class weight	Normal = 4	la uza
Training data	1,079,003 readings (sample)	2,298,650 readings (Normal 1,700,000 + to'o 100,000 / tipu)
Test data	999,536 readings (sample natural)	36,958,516 readings (meters test hotu)

Tabela 4.9. Konfigurasaun Random Forest

4.6. Dezeñu Dashboard (Dashboard Design)

Website dezenvolve ho **Streamlit** hodi implementasaun fasil: operadór EDTL, ekipa inspesaun no estudante loke browser de'it (http://localhost:8505). Website iha pájina login no pájina 9 ne'ebé grupu iha menu haat: Monitorizasaun, Análize, Admin no Informasaun. Pájina prinsipál tuir estrutura projetu capstone referénsia (alerta → deskritiva → tendénsia → estatístika → machine learning → real-time).



Imajen 4.6. Estrutura website Streamlit (login + pájina 9)

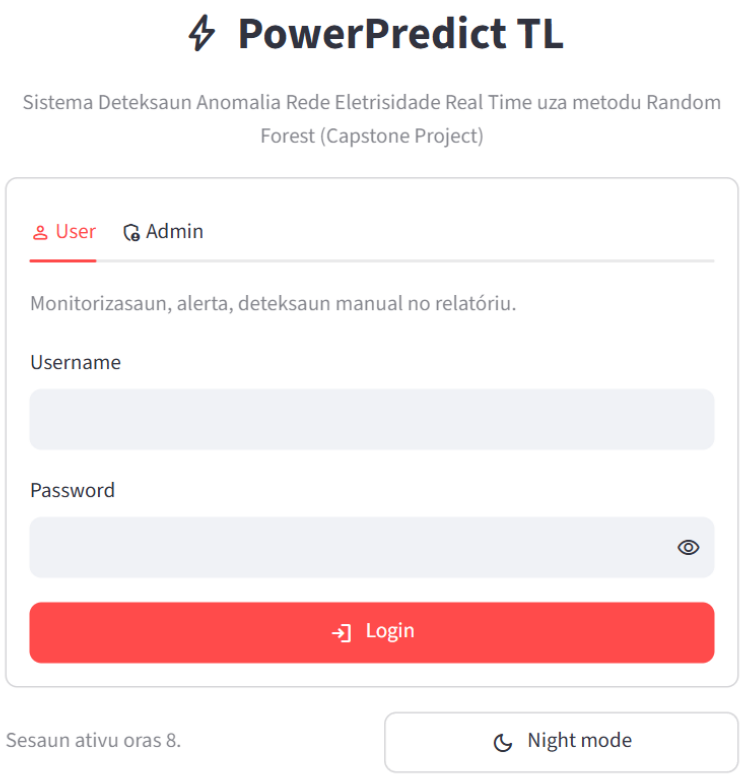
Funsaun komún iha website:

- **Login no role:** konta iha `tbl_users` ho role **Admin** ka **User**. Pájina login iha tab rua (User no Admin); konta tenke uza tab tuir nia role. User haree no analiza (Dashboard, Monitor, Events, Deteksaun Manual, Model, Big Data Pipeline, Metodolojia); Admin bele mós halo simulasaun, retrain no ativa model, konfirma events no jere konta.
- **Seguransa:** password rai nu'udar hash PBKDF2-SHA256 (iterasaun 260,000, salt aleatóriu) — sistema nunka rai password. Login kria token aleatóriu; browser rai token iha cookie `pp_session` (oras 8) no database rai de'it nia hash SHA-256. Logout ka muda password hamoos sesaun. Konta Admin no User dahuluk kria husi `.env` (la hakerek iha kódigu).
- **Refresh:** kada pájina iha butaun Refresh ho oras atualizasaun ikus; butaun ne'e hamoos cache no foti dadus foun husi MySQL.
- **Night mode / Day mode:** butaun iha sidebar muda tema website tomak; login la lakon bainhira pájina reload.
- **Ikon profisionál:** ikon Material Design iha menu, títulu no KPI (la uza emoji); sidebar hatudu konta ne'ebé login no estadu sistema (MySQL, FastAPI, bot, Spark Streaming).

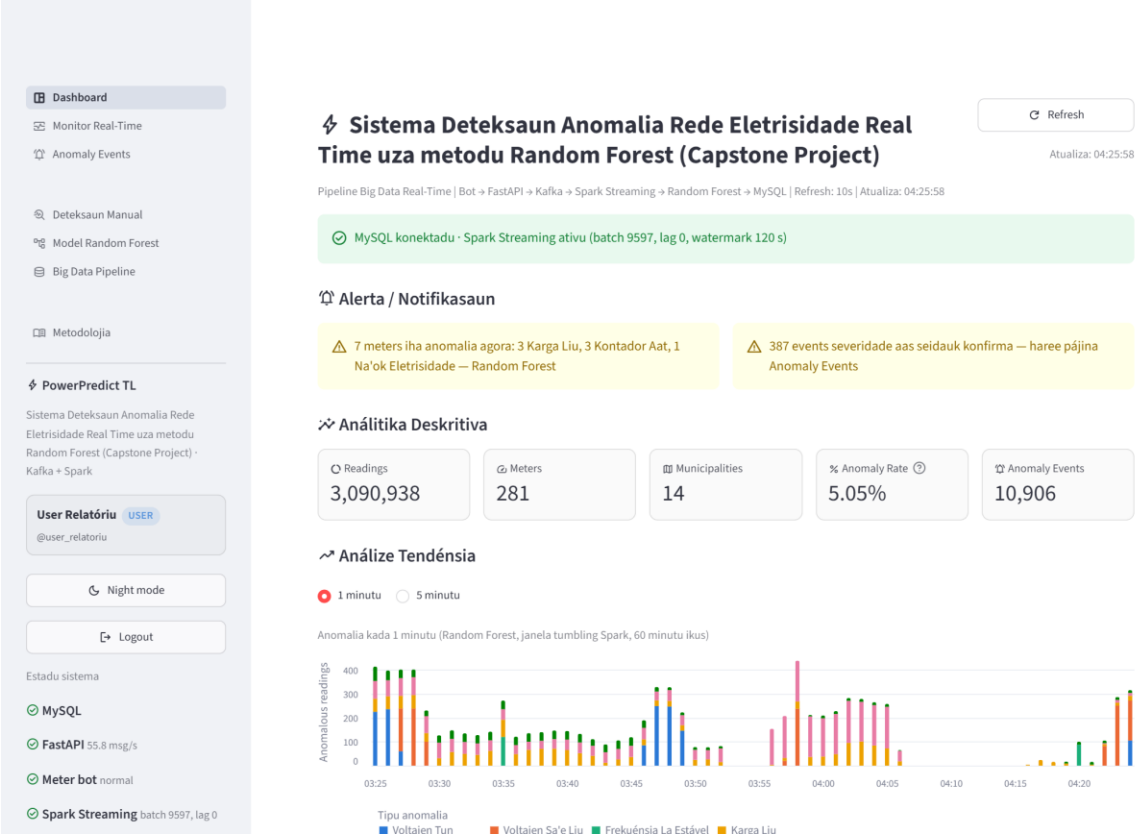
4.7. Implementasaun Interface

4.7.1. Login no Asesu tuir Role

Pájina login hatudu tab User no tab Admin, username no password, no butaun Night mode. Depois login, sidebar hatudu naran konta no nia role. Ba User, menu Admin (Simulasaun, Jestaun Uzuáriu) la mosu no URL pájina Admin la loke.



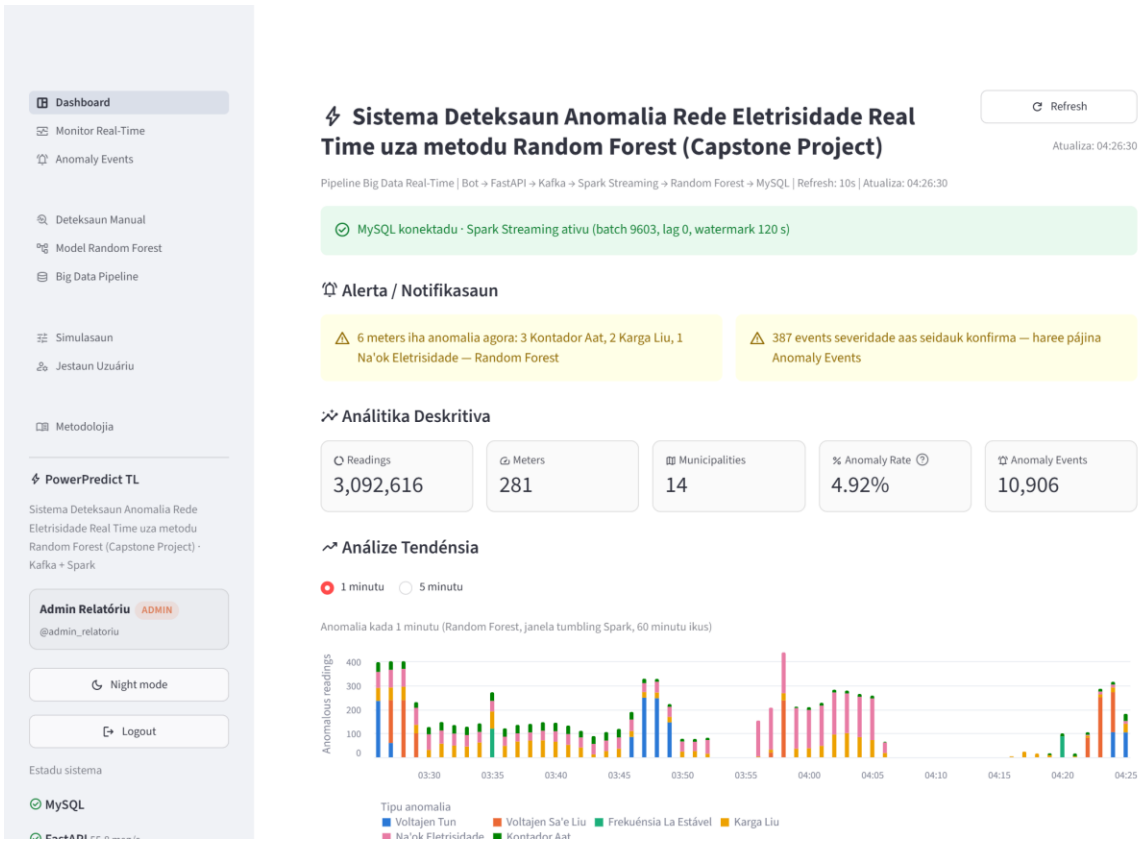
Imajen 4.7. Pájina login (tab User / Admin)



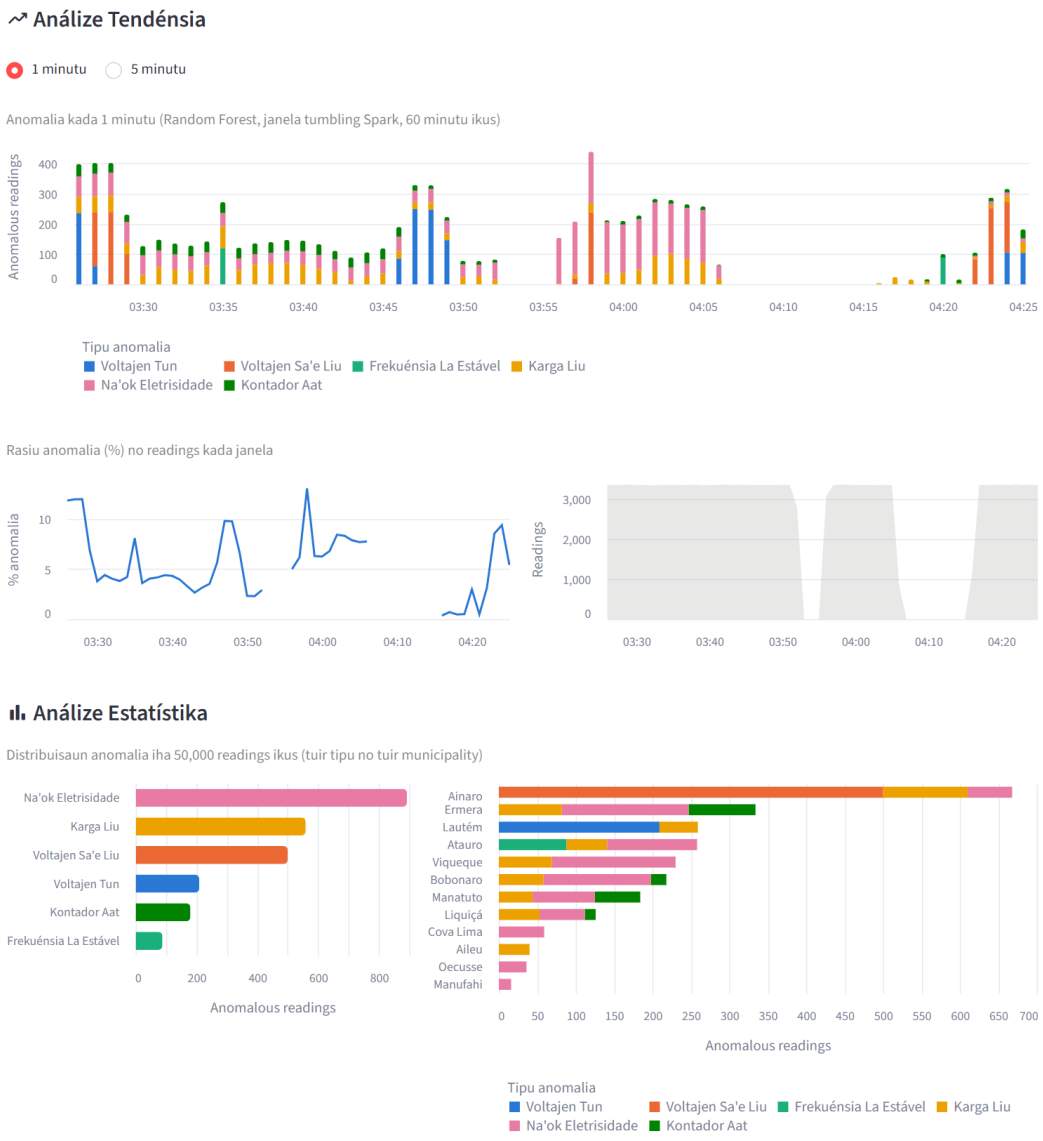
Imajen 4.8. Website ba konta User: menu Admin la mosu

4.7.2. Dashboard (Pájina Prinsipál)

Dashboard hatudu header ho butaun Refresh no status MySQL no Spark Streaming, alerta (meters ho anomalia agora no suspeita Na'ok Eletrisidade), análitika deskritiva (readings, meters, municipalities, Anomaly Rate, anomaly events), tendénsia anomalia kada minutu, no estatística tuir tipu no municipality. Refresh automátiku kada segundu 10.



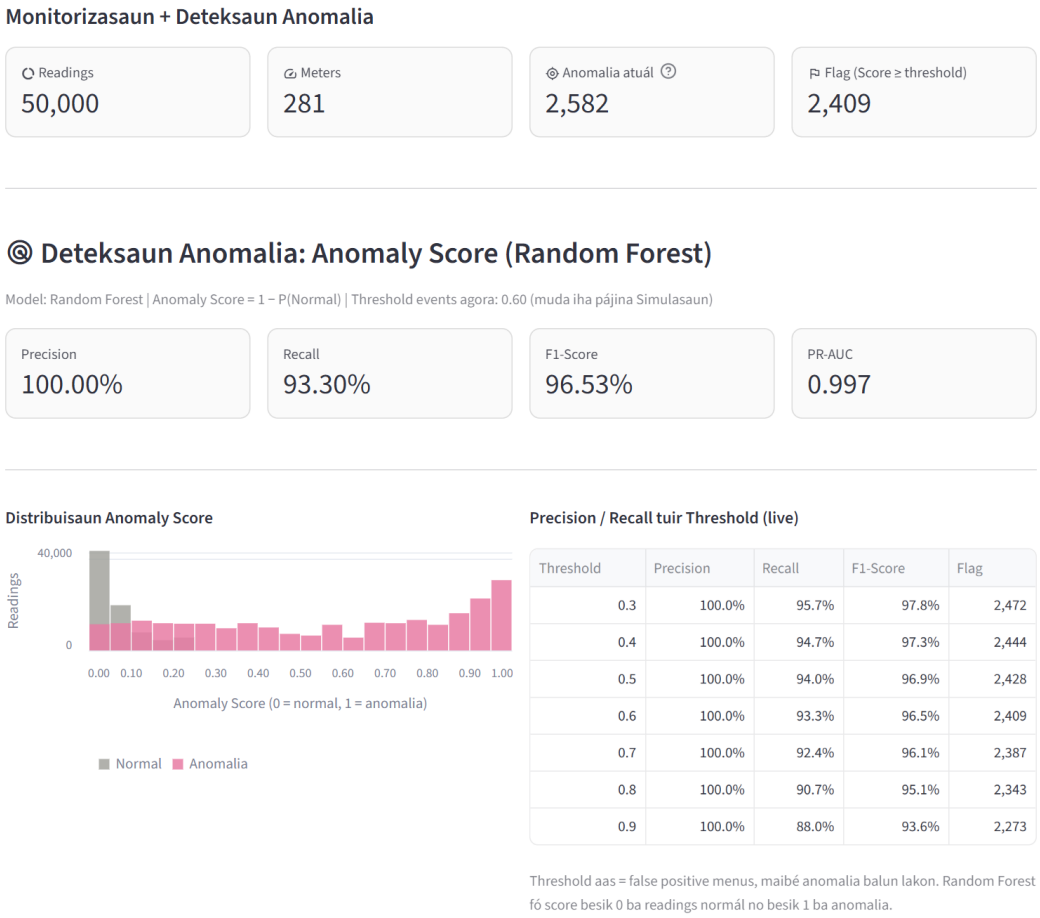
Imajen 4.9. Dashboard ho sidebar (menu, konta, Night mode, estadu sistema)



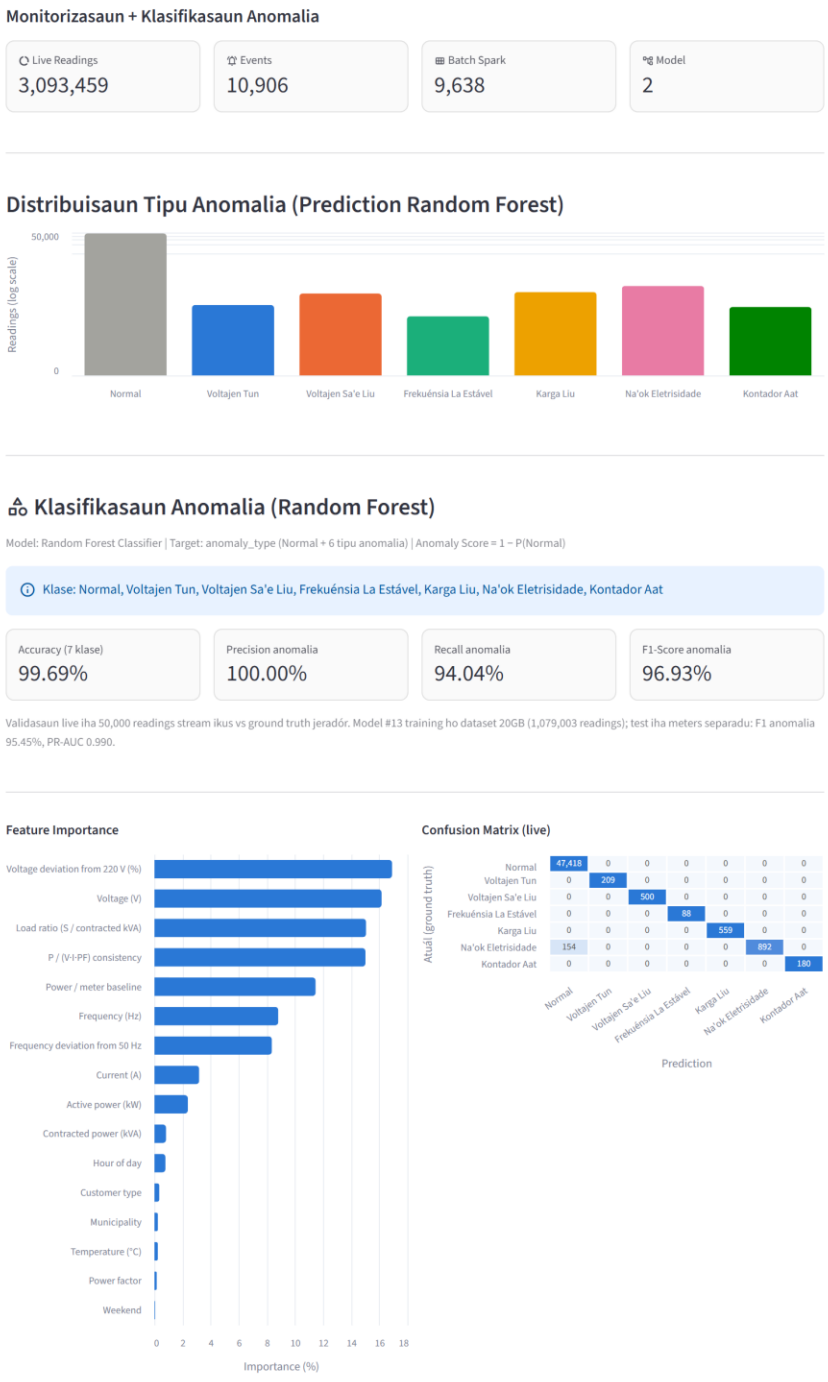
Imajen 4.10. Dashboard: análize tendénsia no estatístika

4.7.3. Dashboard: Machine Learning no Real-Time

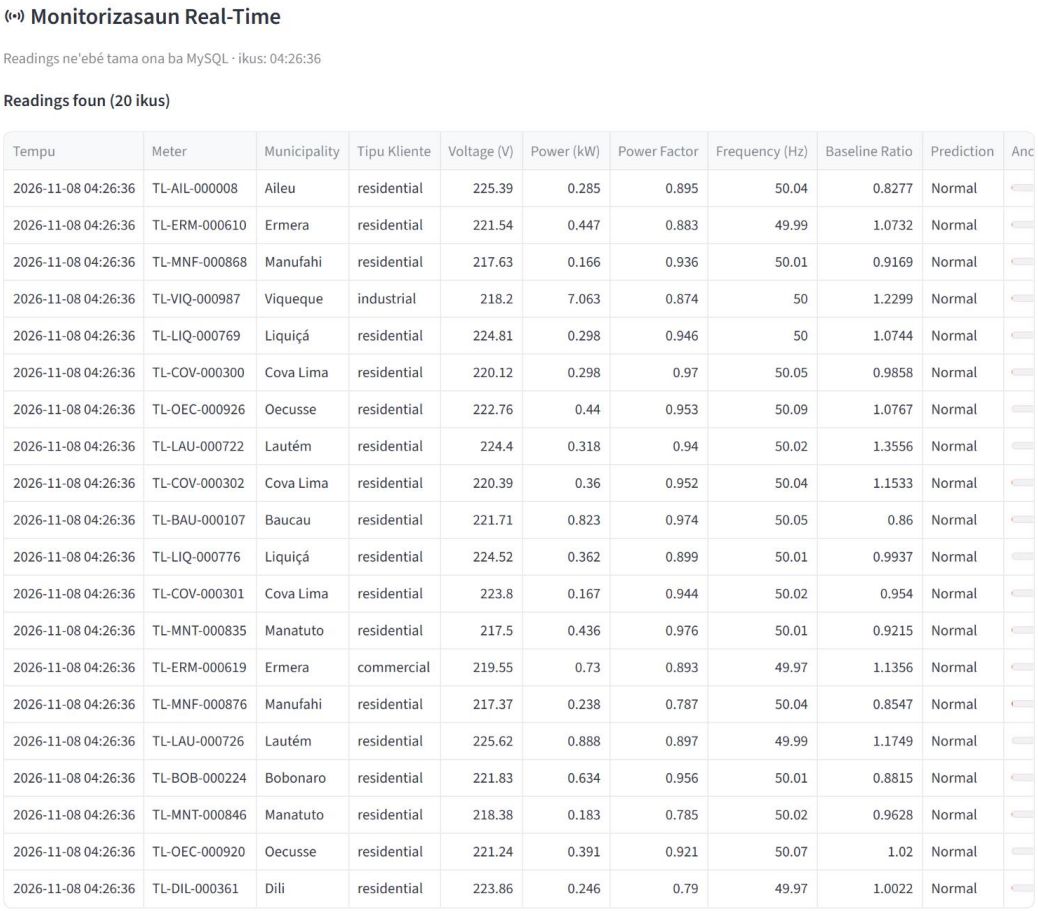
Seksaun Deteksaun Anomalia hatudu distribuisaun Anomaly Score no Precision/Recall tuir threshold; seksaun Klasifikasaun hatudu Accuracy, Precision, Recall, F1-Score live, Feature Importance, Confusion Matrix no table readings foun.



Imajen 4.11. Dashboard: Anomaly Score Random Forest



Imajen 4.12. Dashboard: klasifikasaun tipu anomalia (Random Forest)

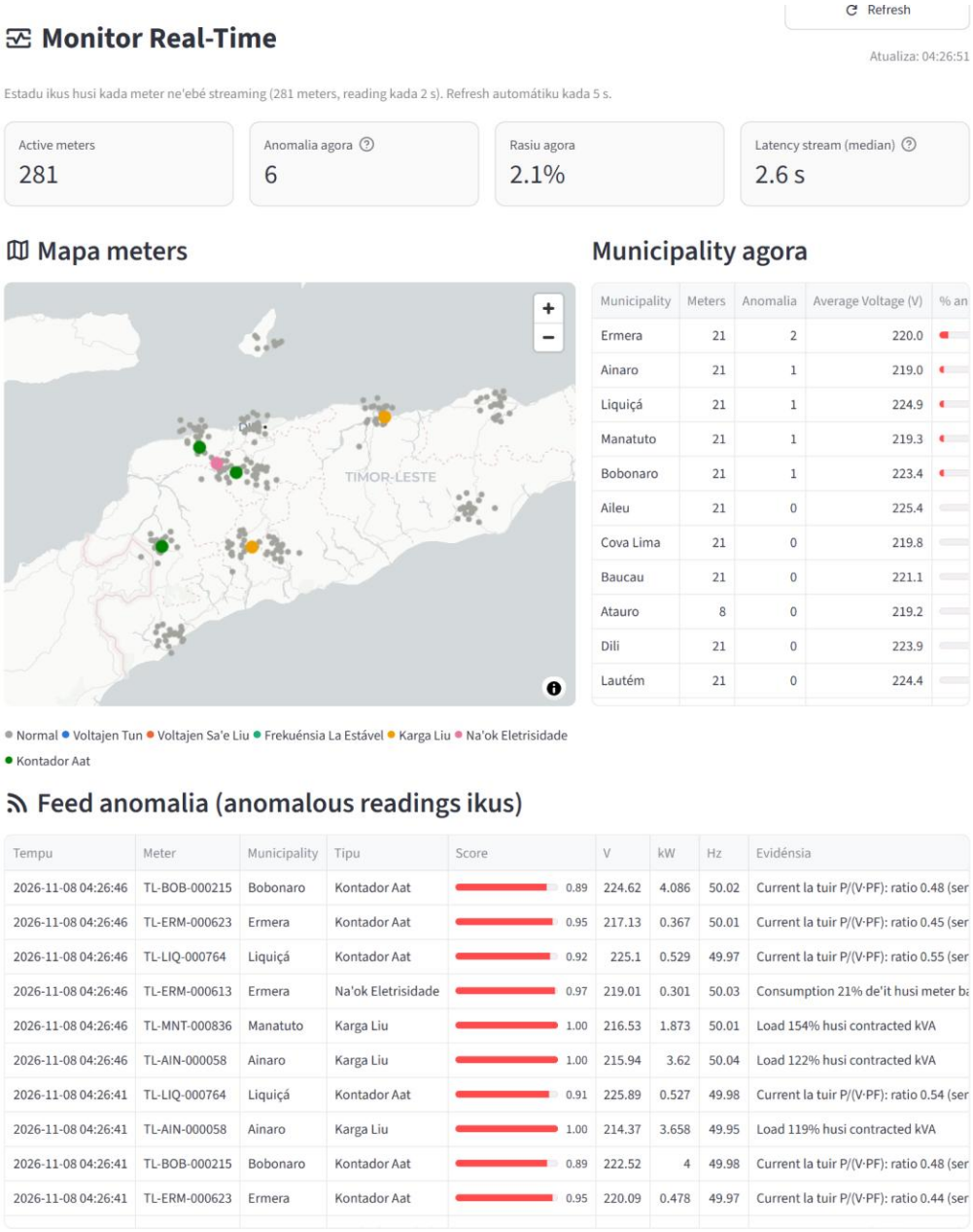


Tempu	Meter	Municipality	Tipu Kliente	Voltage (V)	Power (kW)	Power Factor	Frequency (Hz)	Baseline Ratio	Prediction	Anc
2026-11-08 04:26:36	TL-AIL-000008	Aileu	residential	225.39	0.285	0.895	50.04	0.8277	Normal	
2026-11-08 04:26:36	TL-ERM-000610	Ermera	residential	221.54	0.447	0.883	49.99	1.0732	Normal	
2026-11-08 04:26:36	TL-MNF-000868	Manufahi	residential	217.63	0.166	0.936	50.01	0.9169	Normal	
2026-11-08 04:26:36	TL-VIQ-000987	Viqueque	industrial	218.2	7.063	0.874	50	1.2299	Normal	
2026-11-08 04:26:36	TL-LIQ-000769	Liquiçá	residential	224.81	0.298	0.946	50	1.0744	Normal	
2026-11-08 04:26:36	TL-COV-000300	Cova Lima	residential	220.12	0.298	0.97	50.05	0.9858	Normal	
2026-11-08 04:26:36	TL-OEC-000926	Oecusse	residential	222.76	0.44	0.953	50.09	1.0767	Normal	
2026-11-08 04:26:36	TL-LAU-000722	Lautém	residential	224.4	0.318	0.94	50.02	1.3556	Normal	
2026-11-08 04:26:36	TL-COV-000302	Cova Lima	residential	220.39	0.36	0.952	50.04	1.1533	Normal	
2026-11-08 04:26:36	TL-BAU-000107	Baucau	residential	221.71	0.823	0.974	50.05	0.86	Normal	
2026-11-08 04:26:36	TL-LIQ-000776	Liquiçá	residential	224.52	0.362	0.899	50.01	0.9937	Normal	
2026-11-08 04:26:36	TL-COV-000301	Cova Lima	residential	223.8	0.167	0.944	50.02	0.954	Normal	
2026-11-08 04:26:36	TL-MNT-000835	Manatuto	residential	217.5	0.436	0.976	50.01	0.9215	Normal	
2026-11-08 04:26:36	TL-ERM-000619	Ermera	commercial	219.55	0.73	0.893	49.97	1.1356	Normal	
2026-11-08 04:26:36	TL-MNF-000876	Manufahi	residential	217.37	0.238	0.787	50.04	0.8547	Normal	
2026-11-08 04:26:36	TL-LAU-000726	Lautém	residential	225.62	0.888	0.897	49.99	1.1749	Normal	
2026-11-08 04:26:36	TL-BOB-000224	Bobonaro	residential	221.83	0.634	0.956	50.01	0.8815	Normal	
2026-11-08 04:26:36	TL-MNT-000846	Manatuto	residential	218.38	0.183	0.785	50.02	0.9628	Normal	
2026-11-08 04:26:36	TL-OEC-000920	Oecusse	residential	221.24	0.391	0.921	50.07	1.02	Normal	
2026-11-08 04:26:36	TL-DIL-000361	Dili	residential	223.86	0.246	0.79	49.97	1.0022	Normal	

Imajen 4.13. Dashboard: monitorizasaun real-time

4.7.4. Monitor Real-Time

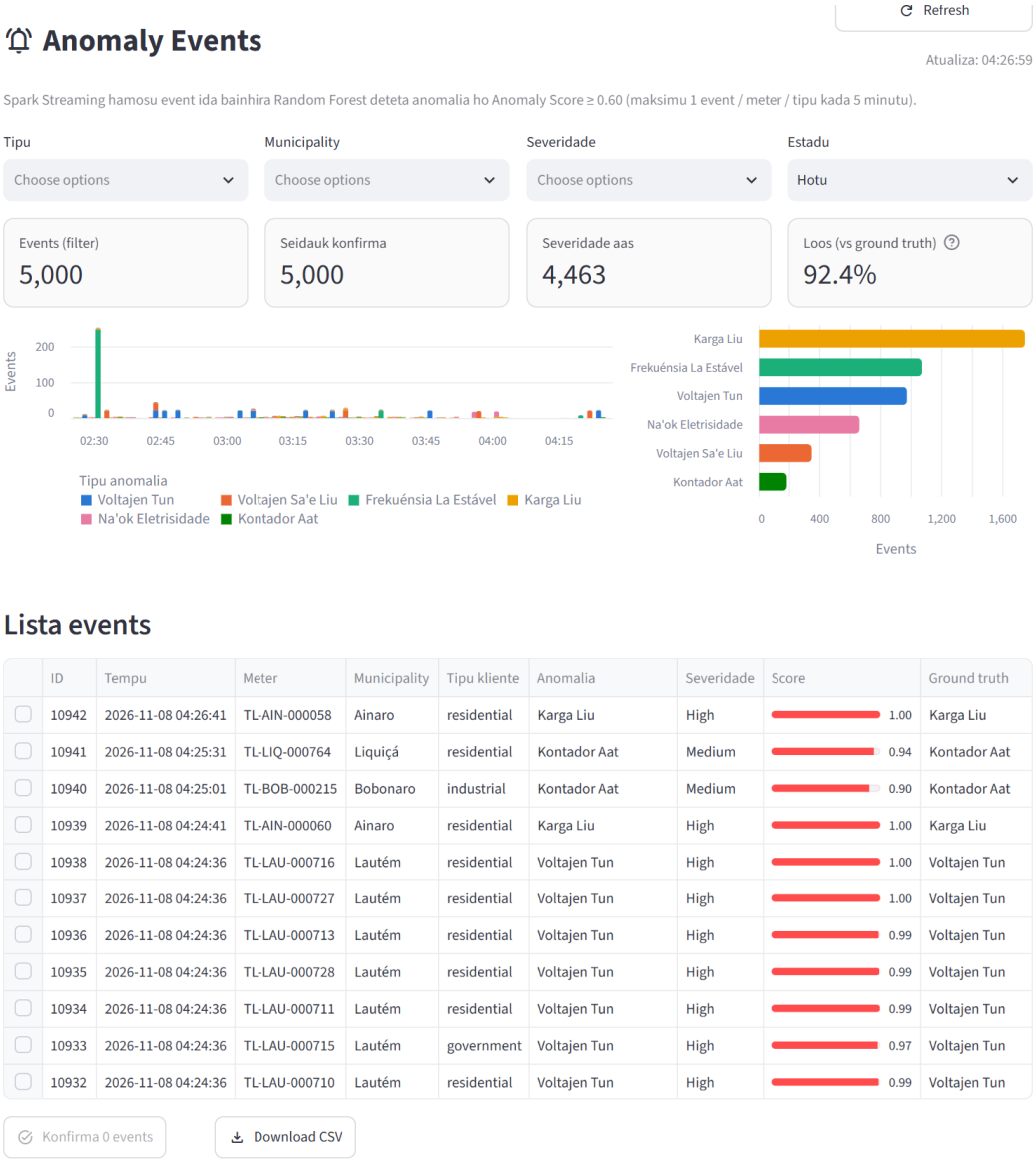
Mapa meters 281 ho kór tuir tipu anomalia, estadu kada municipality, feed anomalia ho evidénsia, no gráfiku voltage no frequency. Refresh kada segundu 5.



Imajen 4.14. Monitor Real-Time

4.7.5. Anomaly Events

Lista events ho filter (tipu, municipality, severity, estadu), download CSV no lista suspeita Na'ok Eletrisidade ba ekipa inspesaun. Konfirma event: Admin de'it.



Lista events

	ID	Tempu	Meter	Municipality	Tipu kliente	Anomalia	Severidade	Score	Ground truth
☐	10942	2026-11-08 04:26:41	TL-AIN-000058	Ainaro	residential	Karga Liu	High	1.00	Karga Liu
☐	10941	2026-11-08 04:25:31	TL-LIQ-000764	Liquiçá	residential	Kontador Aat	Medium	0.94	Kontador Aat
☐	10940	2026-11-08 04:25:01	TL-BOB-000215	Bobonaro	industrial	Kontador Aat	Medium	0.90	Kontador Aat
☐	10939	2026-11-08 04:24:41	TL-AIN-000060	Ainaro	residential	Karga Liu	High	1.00	Karga Liu
☐	10938	2026-11-08 04:24:36	TL-LAU-000716	Lautém	residential	Voltajen Tun	High	1.00	Voltajen Tun
☐	10937	2026-11-08 04:24:36	TL-LAU-000727	Lautém	residential	Voltajen Tun	High	1.00	Voltajen Tun
☐	10936	2026-11-08 04:24:36	TL-LAU-000713	Lautém	residential	Voltajen Tun	High	0.99	Voltajen Tun
☐	10935	2026-11-08 04:24:36	TL-LAU-000728	Lautém	residential	Voltajen Tun	High	0.99	Voltajen Tun
☐	10934	2026-11-08 04:24:36	TL-LAU-000711	Lautém	residential	Voltajen Tun	High	0.99	Voltajen Tun
☐	10933	2026-11-08 04:24:36	TL-LAU-000715	Lautém	government	Voltajen Tun	High	0.97	Voltajen Tun
☐	10932	2026-11-08 04:24:36	TL-LAU-000710	Lautém	residential	Voltajen Tun	High	0.99	Voltajen Tun

Konfirma 0 events Download CSV

Imajen 4.15. Anomaly Events

Suspeita na'ok eletrisidade (meters ho konsumu ki'ik la normál)

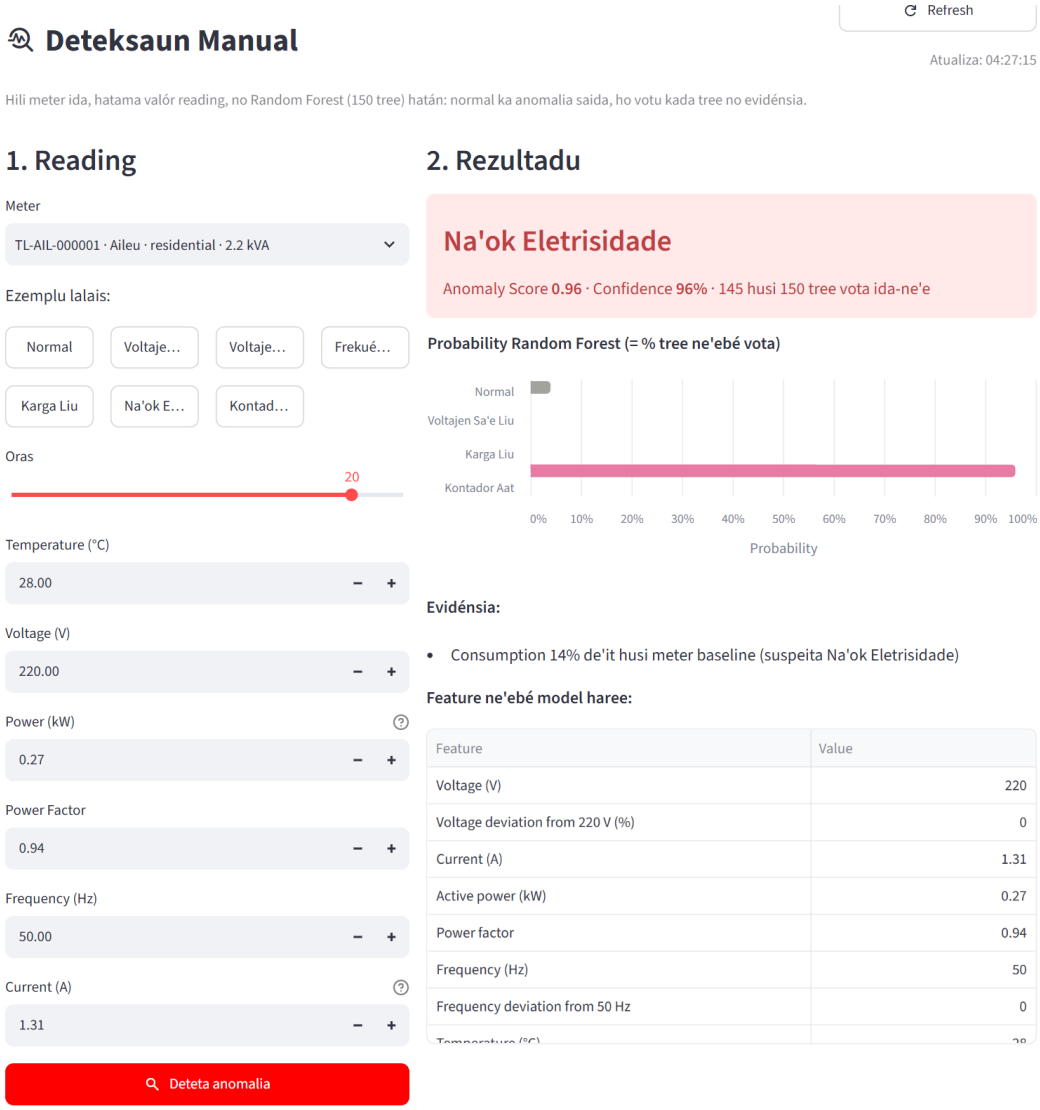
Na'ok la'ós reading ida de'it: meter ida sai suspeita bainhira Random Forest klasifika ≥ 50 % husi nia readings ikus hanesan Na'ok Eletrisidade. Ida-ne'e mak lista ba ekipa inspesaun.

La iha suspeita agora

Imajen 4.16. Lista suspeita Na'ok Eletrisidade

4.7.6. Deteksaun Manual

Utilizadór hili meter, hatama valór reading (ka hili ezemplu) no Random Forest hatán tipu anomalia, Anomaly Score, votu tree 150 no evidénsia.



Imajen 4.17. Deteksaun Manual (ezemplu Na'ok Eletrisidade)

4.7.7. Model Random Forest

Pájina model hatudu rezultadu test model Random Forest (scikit-learn no Spark MLlib), tanba sa Random Forest, Precision/Recall tuir tipu, Confusion Matrix, threshold no Feature Importance. Admin bele retrain ho parameter foun no ativa model; User haree de'it.

Model Random Forest (test iha meters separadu)

Model	Training	Test	Precision	Recall	F1 anomalia	PR-AUC	ROC-AUC	False Positive Rate	Macro F1 (tipu)	T
#12 Random Forest (Spark MLlib, 20GB)	2,298,650	36,958,516	78.5%	96.8%	86.7%	None	None	1.1%	81.5%	
#13 Random Forest (scikit-learn, 20GB)	1,079,003	999,536	94.2%	96.8%	95.4%	0.990	0.999	0.2%	96.3%	

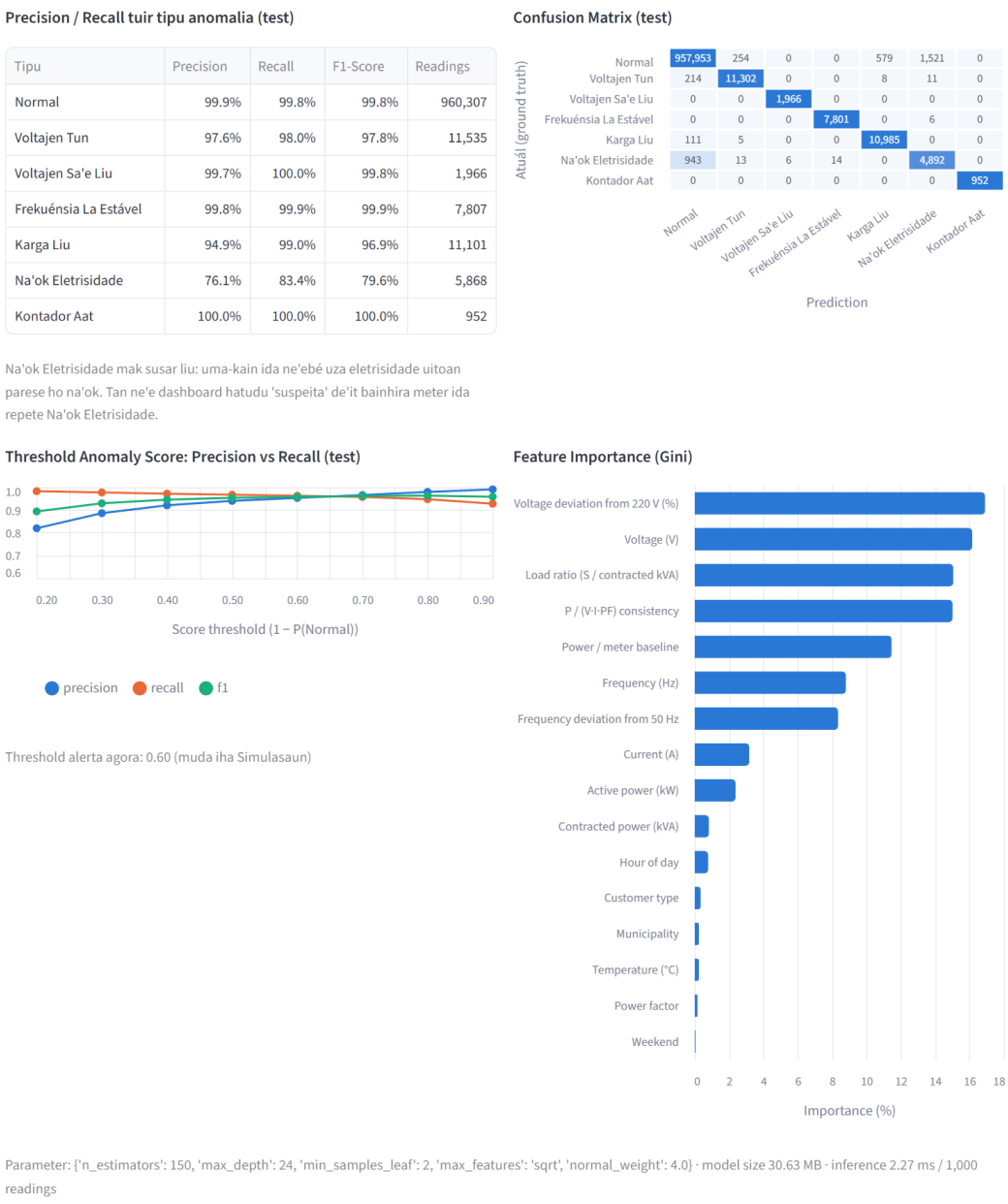
Spark MLlib Random Forest = método hanesan, treina distribuídu iha 2,298,650 readings no test iha readings hotu husi meters test (36,958,516 readings, dataset 20GB).

Tanba sa Random Forest?

1. **Accurate liu.** Iha test hanesan (dataset 20 GB, readings 1 millaun husi meters separadu), Random Forest iha F1 anomalia **0.954** (PR-AUC 0.990), aas liu Decision Tree (0.925), Logistic Regression (0.896) no Isolation Forest (0.144). Tan ne'e sistema uza **Random Forest de'it**.
2. **Dadus iha label (ground truth).** Jeradór rejista anomaly events ba kada reading, entaun ita bele uza supervised learning ne'ebé fó mós **tipu** anomalia (la'ós de'it "anomalia ka lae").
3. **Anomalia depende ba feature barak hamutuk.** Na'ok Eletrisidade = power ki'ik *kompara ho baseline meter iha oras ne'e*; Kontador Aat = current la tuir P/(V·PF). Tree sira aprende decision boundary non-linear no interasaun.
4. **Votu tree 150 estável.** Probability husi votu sai **Anomaly Score** (1 − P(Normal)) ne'ebé bele hili threshold.
5. **Feature numériku + kategóriku, la presiza normaliza, robustu ba noise sensor.**
6. **Model estável:** cross-validation 3-fold grupu tuir meter: F1 anomalia **0.981 ± 0.001**.
7. **Eskala ba Big Data:** Spark MLlib Random Forest treina distribuídu iha readings millaun.

Random Forest #13 (ativu) - detalla

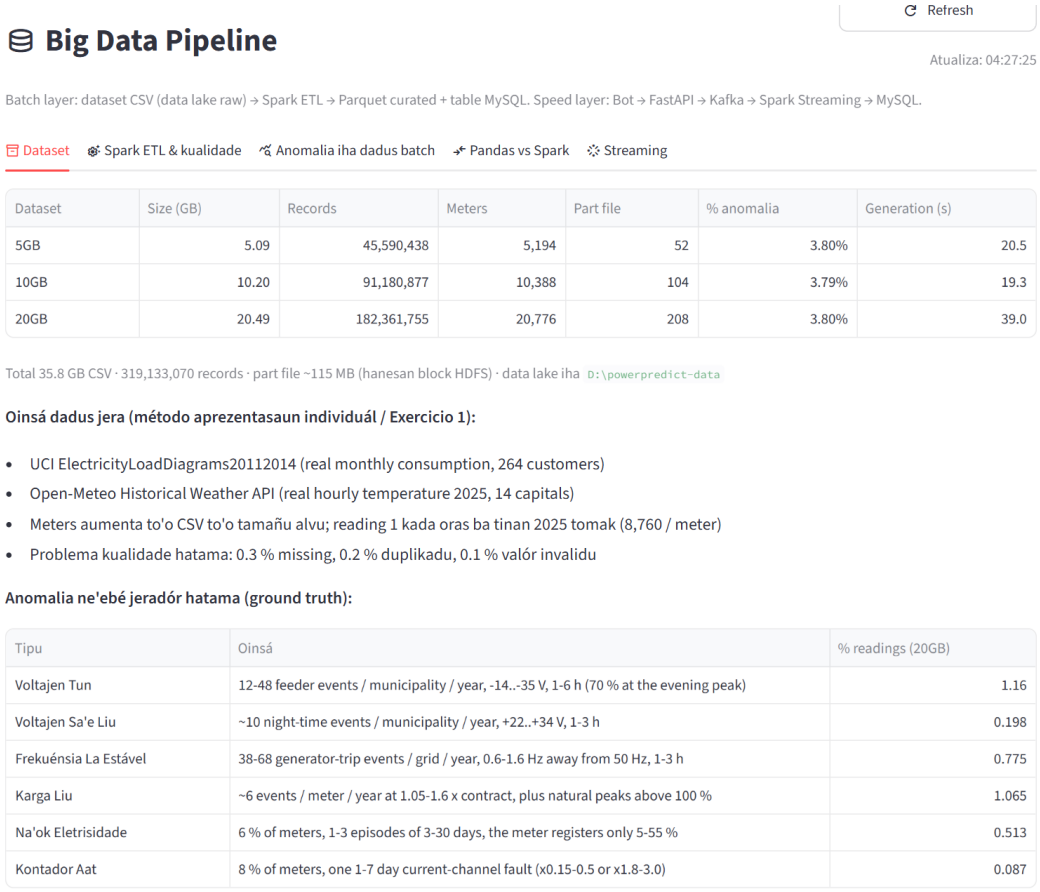
Imajen 4.18. Model Random Forest: rezultadu test no tanba sa Random Forest



Imajen 4.19. Model Random Forest: evaluasaun detalla

4.7.8. Big Data Pipeline

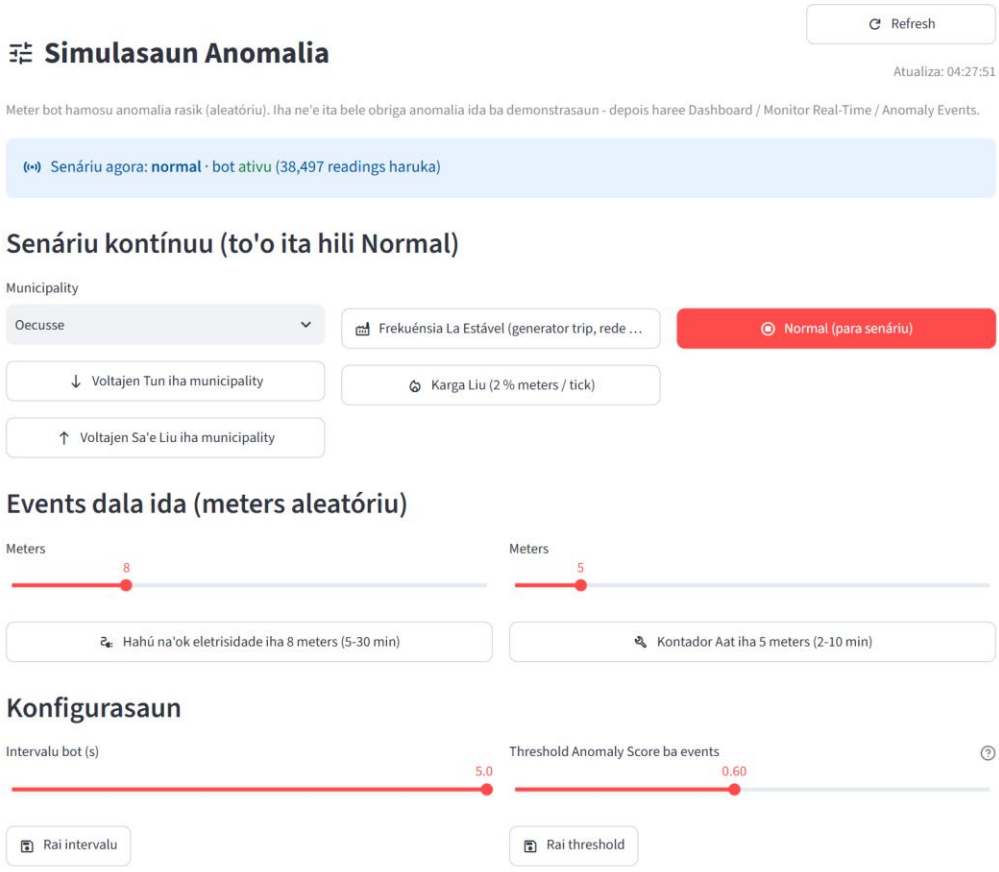
Pájina ne'e hatudu dataset 5 GB / 10 GB / 20 GB, Spark ETL no Data Quality, anomalia iha dadus batch (oras, fulan, municipality, feature profile, top meters), benchmark Pandas vs Spark no estatístika streaming.



Imajen 4.20. Big Data Pipeline: dataset no anomalia ne'ebé hatama

4.7.9. Simulasaun (Admin)

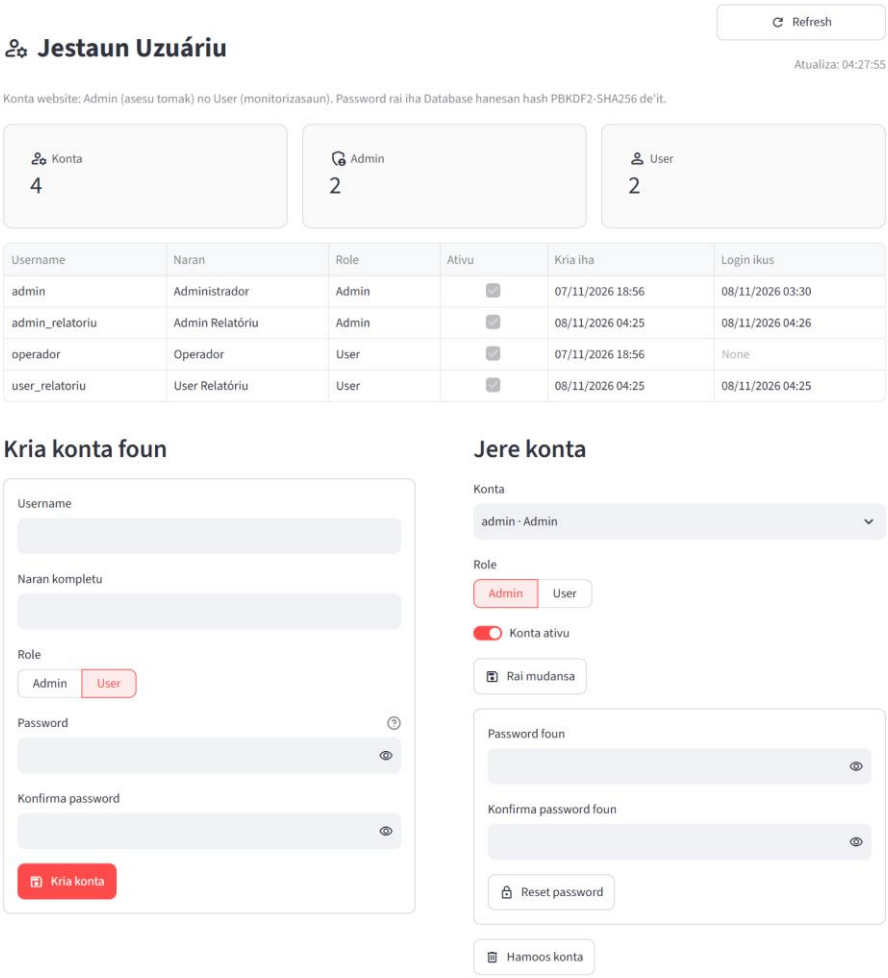
Simulasaun permite Admin hamosu Voltajen Tun, Voltajen Sa'e Liu, Frekuénsia La Estável, Karga Liu, Na'ok Eletrisidade no Kontador Aat ba demonstrasaun, no muda intervalu bot no threshold events.



Imajen 4.21. Simulasaun anomalia

4.7.10. Jestaun Uzuáriu (Admin)

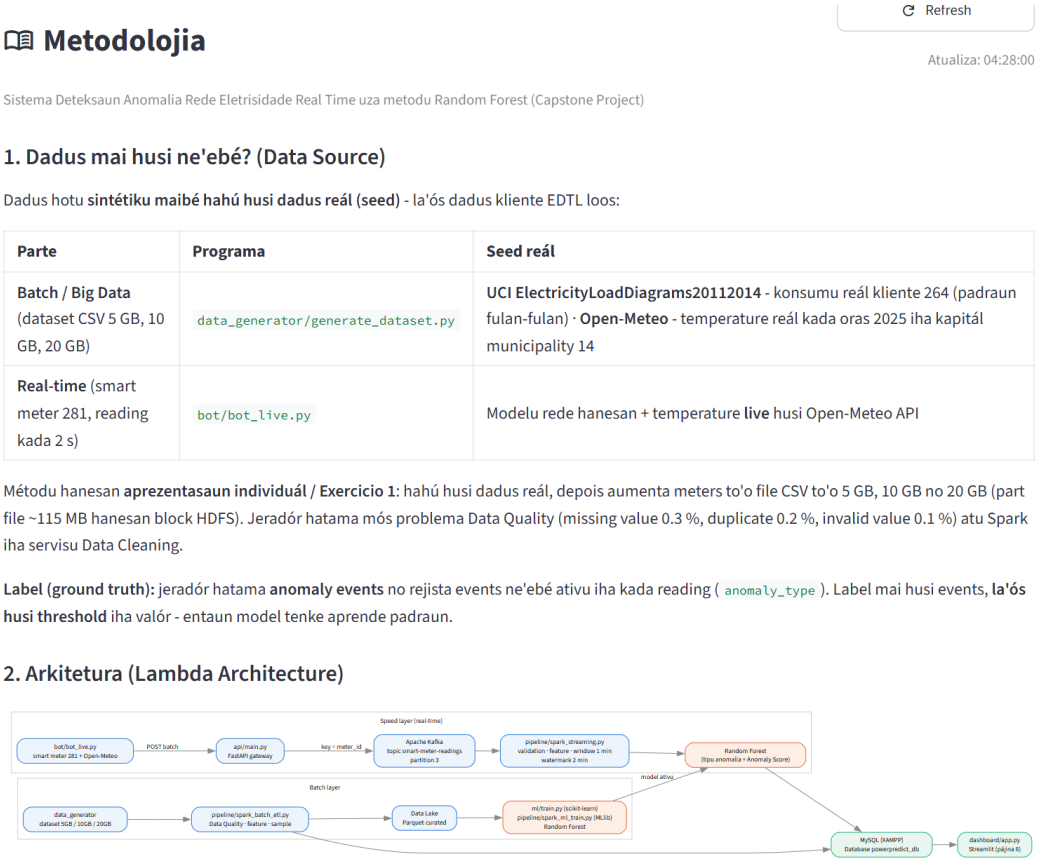
Admin kria konta foun (password karakter 8 ka liu), muda role, ativa ka dezativa konta, reset password no hamoos konta. Sistema la permite Admin hamoos ka dezativa nia konta rasik ka Admin ativu ikus. (Konta admin_relatoriu no user_relatoriu iha imajen mak konta temporáriu ba teste automátiku; hamoos ona depois teste.)



Imajen 4.22. Jestaun Uzuáriu

4.7.11. Metodolojia

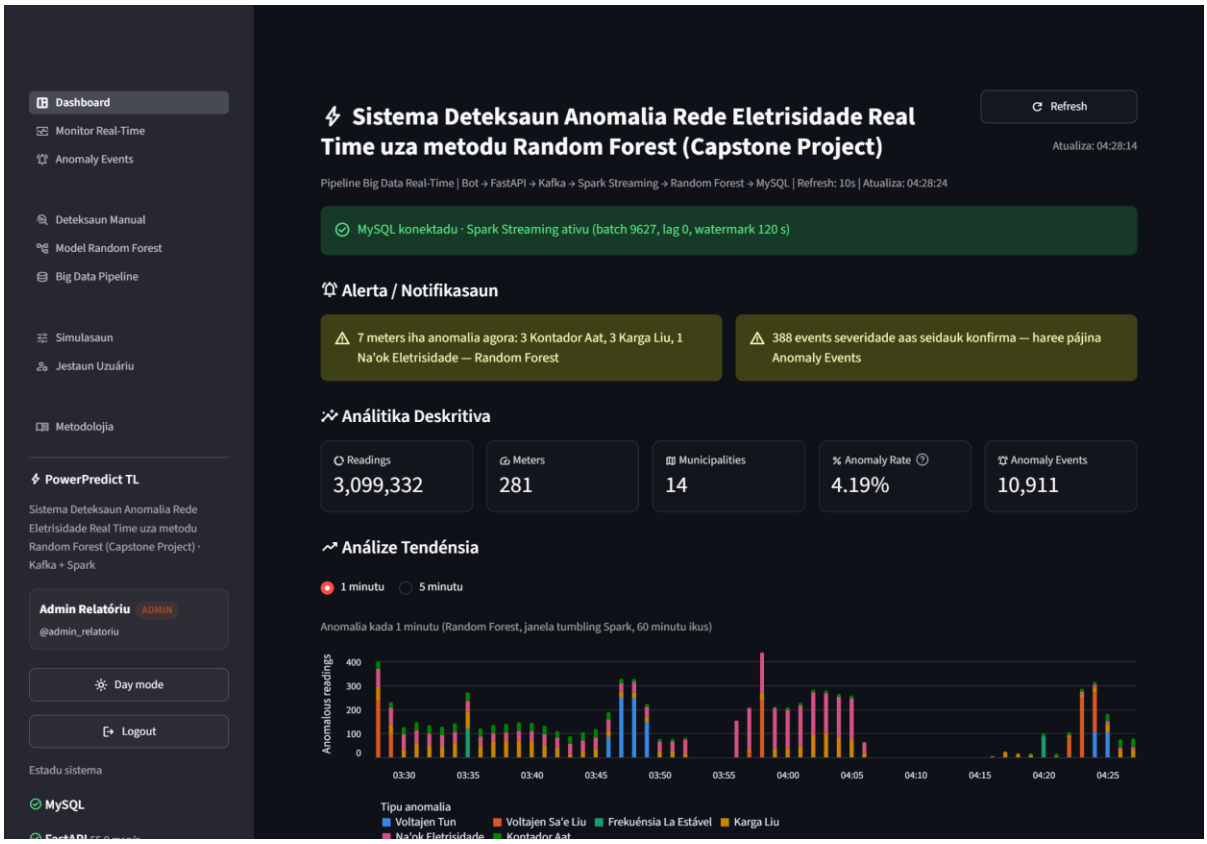
Pájina Metodolojia esplika data source, arkitetura Lambda, tipu anomalia, Random Forest no teknolojia.



Imajen 4.23. Metodolojia: data source no arkitetura

4.7.12. Night Mode

Butaun Night mode iha sidebar (ka iha pájina login) muda tema website tomak ba dark; gráfiku no kór tipu anomalia hela hanesan. Butaun Day mode fila fali ba tema light.



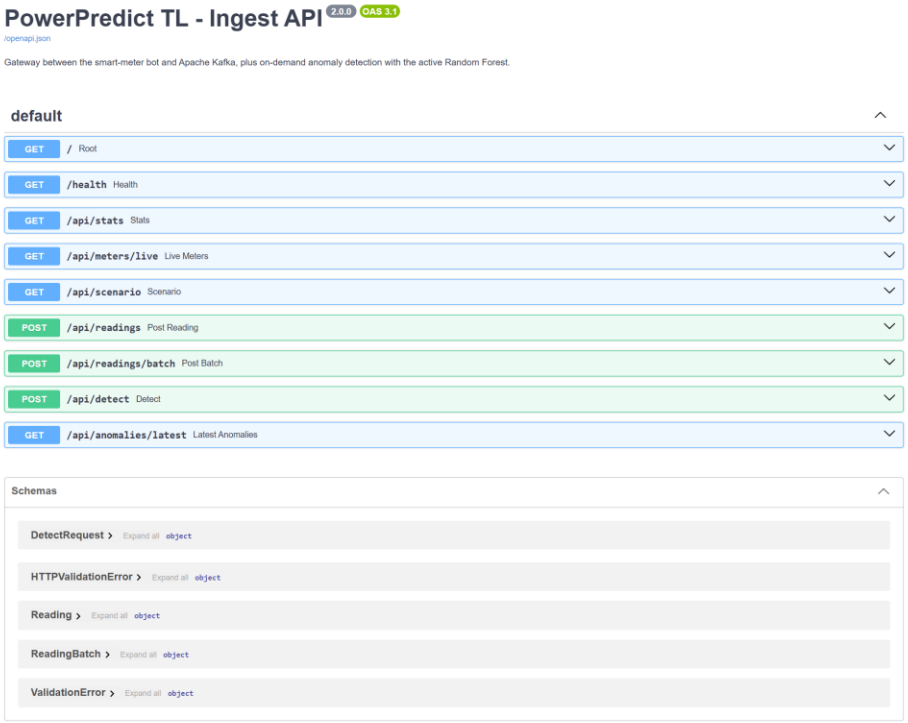
Imajen 4.24. Dashboard iha Night mode

4.8. API Gateway (FastAPI)

FastAPI (`api/main.py`) mak dalan tama ba dadus real-time no fornese dokumentasaun OpenAPI automátiku iha `/docs`:

Method	Endpoint	Funsaun
POST	/api/readings/batch	Simu readings barak husi bot, valida no publish ba Kafka
POST	/api/readings	Simu reading ida
GET	/api/scenario	Senáriu simulasaun ne'ebé bot tenke tuir (Admin muda iha website)
GET	/api/meters/live	Lista meters live 281 ba bot
POST	/api/detect	Deteksaun anomalia ba reading ida (Random Forest)
GET	/api/anomalies/latest	Anomaly events foun
GET	/health, /api/stats	Estadu MySQL no Kafka, númeru mensajen

Tabela 4.10. Endpoint FastAPI



Imajen 4.25. Dokumentasaun FastAPI (OpenAPI /docs) ba ingest gateway

4.9. Publikasaun Website ba Internet

Atu ema hotu (dosente, kolega, EDTL) bele haree sistema live la presiza instala buat ida, website publika ba internet ho solusaun gratuitu tolu ne'ebé la muda pipeline: (1) kópia website ida iha modu konvidadu read-only, (2) Cloudflare Tunnel ba HTTPS, no (3) link permanente iha GitHub Pages. Opsaun 24/7 iha pakote Docker (seksaun 4.9.3). Imajen 4.26 hatudu fluxu kompletu.

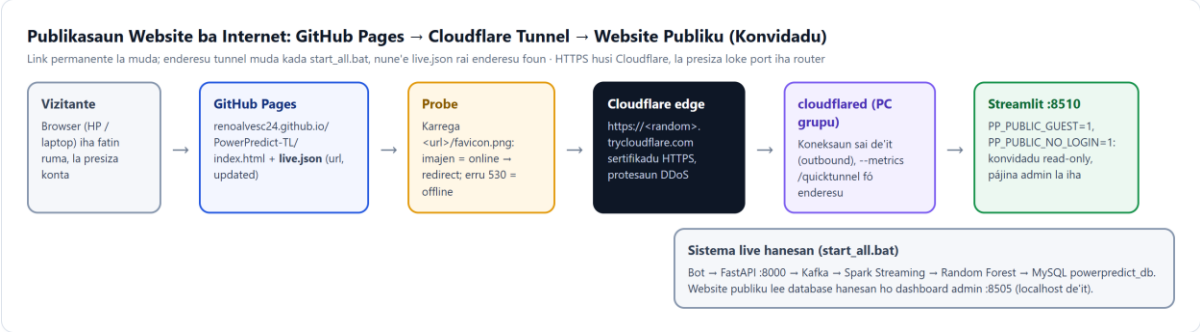
4.9.1. Modu Konvidadu (Read-only)

- **Kópia website rasik:** website publiku hala'o iha port 8510 ho `PP_PUBLIC_GUEST=1` no `PP_PUBLIC_NO_LOGIN=1`; website Admin iha port 8505 (localhost de'it) la muda.

- **Tama la ho konta:** butaun "Tama hanesan konvidadu (haree de'it)" kria sesaun role guest (KONVIDADU) ho cookie `pp_guest`; tab login User / Admin la hatudu no funsaun login taka iha kódigu (dashboard/auth.py).
- **Pájina:** konvidadu haree Dashboard, Monitor Real-Time, Anomaly Events, Deteksaun Manual, Model Random Forest, Big Data Pipeline no Metodolojia; pájina Simulasaun no Jestaun Uzuáriu la iha iha navigasaun, no URL direta hatudu "Page not found".
- **La rai dadus:** Deteksaun Manual husi konvidadu kalkula rezultadu maibé la rai iha database; Logout hamoos sesaun konvidadu.

4.9.2. Cloudflare Tunnel no Link Permanente

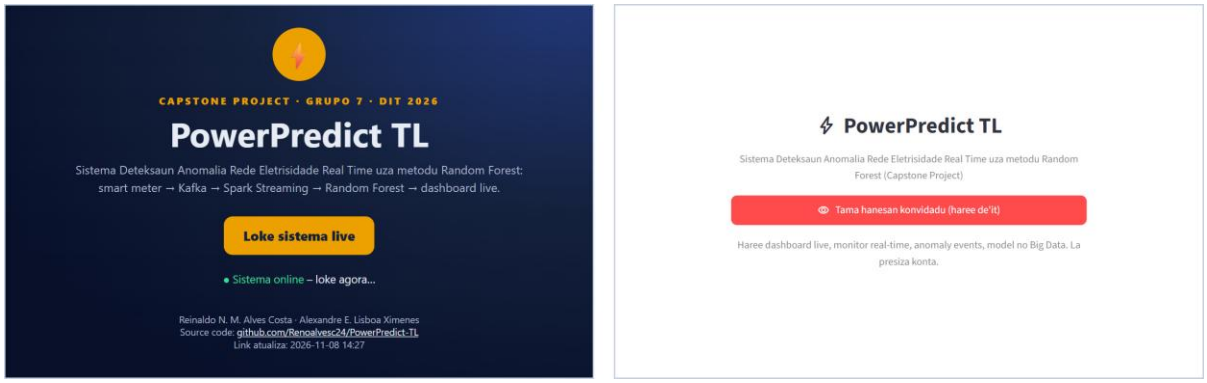
Programa cloudflared halo koneksaun sai husi komputadór grupu ba Cloudflare no simu enderesu HTTPS aleatóriu (https://<naran>.trycloudflare.com). Enderesu ne'e muda kada dala tunnel hahú, entaun setup/public_site.ps1 husu enderesu atuál husi cloudflared rasik (metrics /quicktunnel), verifika katak website responde (favicon.png fila imajen), no hakerek enderesu iha live.json iha branch gh-pages repozitóriu GitHub. Link permanente https://renoalvesc24.github.io/PowerPredict-TL/ lee live.json (la uza cache), koko karrega favicon husi tunnel, no se susesu hatudu "Sistema online" no loke website; se la susesu hatudu "Sistema offline agora" (Cloudflare, 2026; GitHub, 2026).



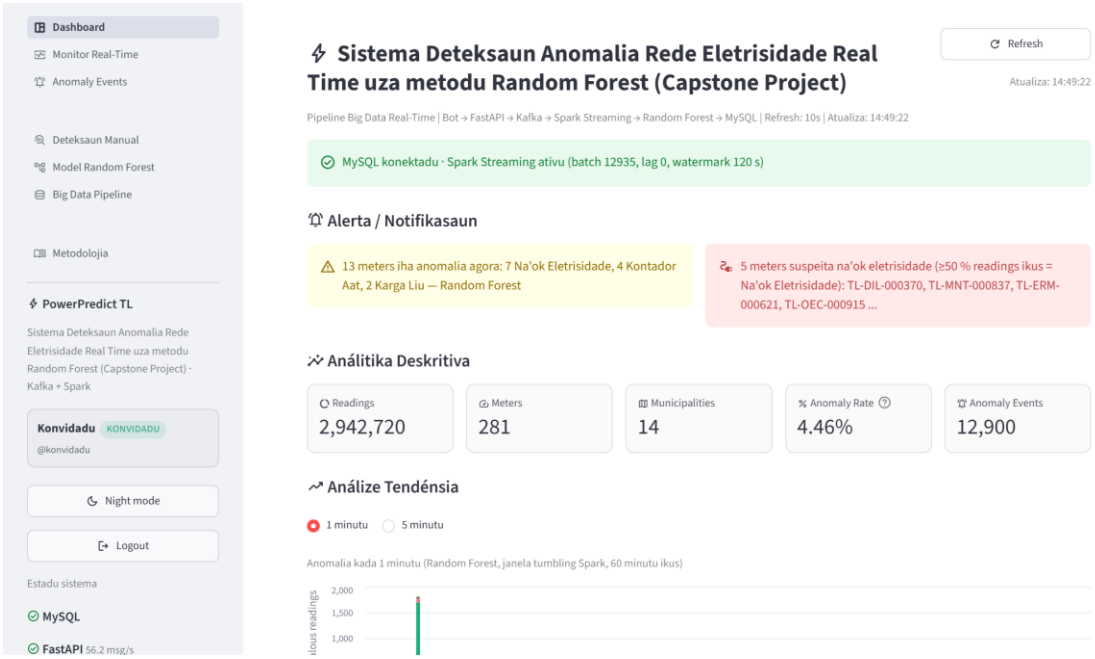
Imajen 4.26. Fluxu publikasaun website: GitHub Pages → Cloudflare Tunnel → website konvidadu

Komponente	File / komandu	Funsaun
start_all.bat pasu 6	setup/public_site.ps1	Hahú website publiku, tunnel no atualiza link permanente; seguru atu hala'o dala rua (reuza parte ne'ebé ativu)
Website publiku	streamlit … --server.port 8510	Kópia konvidadu read-only (PP_PUBLIC_GUEST=1, PP_PUBLIC_NO_LOGIN=1)
Cloudflare Tunnel	cloudflared tunnel --url http://localhost:8510	HTTPS públiku, koneksaun outbound de'it; metrics 127.0.0.1:20241
Link permanente	gh-pages: index.html + live.json	Landing page: lee enderesu tunnel foun, probe favicon, redirect ka "offline"
Start manuál	start_public.bat	Hahú parte publiku de'it bainhira start_all.bat ativu ona

Tabela 4.11. Komponente publikasaun website



Imajen 4.27. Link permanente GitHub Pages ("Sistema online") no pájina tama website publiku (konvidadu)



Imajen 4.28. Dashboard live iha website publiku ho role KONVIDADU (liuhusi Cloudflare Tunnel)

4.9.3. Pakote Deployment 24/7 (Docker)

Website liuhusi tunnel online de'it bainhira komputadór grupu moris. Ba operasaun 24/7, folder deploy/ iha pakote Docker Compose ba servidór Oracle Cloud Always Free (ARM, gratuitu): image aplikasaun ida (Python 3.11 + OpenJDK 17, versaun pakote hanesan ho komputadór dezenvolvimentu) ba servisu Python hotu, Kafka KRaft, MariaDB no Caddy ba HTTPS automátiku. Spark hala'o ho local[2] no memória driver 3 GB tuir kapasidade servidór, no database hahú ho readings 20,000 ikus. Pakote ne'e dokumenta iha deploy/ORACLE_SETUP.md maibé seidauk hala'o tanba presiza konta Oracle Cloud (Docker Inc., 2026).

Servisu	Image	Funsaun	Port públiku
kafka	apache/kafka:3.9.1	Broker KRaft, topic smart-meter-readings	–
db	mariadb:11.4	Database powerpredict_db (seed)	–
api	powerpredict-app	FastAPI ingest gateway	–
stream	powerpredict-app	Spark Streaming + Random Forest	–
bot	powerpredict-app	Smart meter live 281	–
dashboard	powerpredict-app	Website konvidadu (Streamlit)	–
caddy	caddy:2	Reverse proxy HTTPS (sertifikadu automátiku)	80, 443

Tabela 4.12. Servisu Docker Compose ba deployment 24/7

KAPITULU V
REZULTADU NO DISKUSAUN

5.1. Rezultadu Data Collection no Data Quality

5.1.1. Dataset ne'ebé Rekolla

Jeradór produz dataset tolu ho records totál 319,133,070 (35.8 GB CSV) — liu rekezitu mínimu records 10,000 iha manual. Proporsaun anomalia hanesan iha dataset tolu (maizumenus 3.8%), entaun rezultadu sira bele kompara.

Dataset	Tamañu	Records	Meters	Part-file	% anomalia	Tempu jera
5GB	5.09 GB	45,590,438	5,194	52	3.80%	20.5 s
10GB	10.20 GB	91,180,877	10,388	104	3.79%	19.3 s
20GB	20.49 GB	182,361,755	20,776	208	3.80%	39 s

Tabela 5.1. Dataset big data ne'ebé jera

5.1.2. Data Quality no Data Cleaning

Spark verifika dataset 20 GB (records 182,361,755) no hetan problema Data Quality tuir mai:

Check	Column	Records	%	Asaun
duplicate	meter_id+timestamp	363,995	0.200%	Hamoos
invalid	energy_kwh	3	0.000%	Hamoos
invalid	power_factor	45,586	0.025%	Hamoos
invalid	power_kw	45,605	0.025%	Hamoos
invalid	voltage_v	91,787	0.050%	Hamoos
missing	current_a	136,784	0.075%	Hamoos
missing	power_factor	136,722	0.075%	Hamoos
missing	temperature_c	136,771	0.075%	Hamoos
missing	voltage_v	136,720	0.075%	Hamoos

Tabela 5.2. Rezultadu verifikasaun Data Quality (dataset 20 GB)

Missing value 546,997, duplicate 363,995 no invalid value 182,981 (readings ida bele iha problema liu ida). Rezultadu cleaning ba dataset tolu:

Dataset	Records input	Records moos	Hamoos	%
5GB	45,590,438	45,317,415	273,023	0.60%
10GB	91,180,877	90,634,831	546,046	0.60%
20GB	182,361,755	181,269,709	1,092,046	0.60%

Tabela 5.3. Rezultadu Data Cleaning ba dataset tolu

5.1.3. Outliers

Manual seksaun 14.C husu verifika outliers. Regra IQR (valór < Q1 − 1.5·IQR ka > Q3 + 1.5·IQR) aplika ba sample test natural (readings 999,536, dataset 20 GB):

Feature	Limite ki'ik	Limite aas	Outliers	% readings	% ne'ebé anomalia loos
Voltage (V)	209.56	228.81	34,667	3.47%	43.5%
Current (A)	-3.52	10.37	74,343	7.44%	14.0%
Power (kW)	-0.99	2.68	112,696	11.27%	8.3%
Power factor	0.83	1.03	69,738	6.98%	4.3%
Frequency (Hz)	49.76	50.24	23,069	2.31%	36.8%
Temperature (°C)	16.00	34.40	2,378	0.24%	4.0%

Tabela 5.4. Outliers tuir regra IQR (test sample natural)

Readings 22.7% iha outlier iha feature ida ka liu, maibé husi outliers sira ne'e 13.9% de'it mak anomalia loos — barak liu mak readings normál husi kliente boot (komersiál, indústria) ka oras pico, tanba distribuisaun konsumu la simétriku. Tan ne'e outliers **la hamoos**: sira la'ós erru dadus (erru loos — valór imposível — hamoos ona nu'udar invalid). Regra IQR nu'udar detetór sei fó alarme falsu barak (precision ~0.14 kompara ho Random Forest 0.94) no hetan 80.6% husi anomalia hotu (Voltajen Tun 99%, Voltajen Sa'e Liu 100%, Frekuénsia La Estável 100%, Karga Liu 83%, Na'ok Eletrisidade 15%, Kontador Aat 30%): Na'ok Eletrisidade no Kontador Aat la mosu nu'udar outlier tanba sira-nia valór medidu parese normál. Ida ne'e hatudu tanba sa presiza model ne'ebé haree feature barak hamutuk (Random Forest), la'ós threshold ba feature ida.

5.2. Rezultadu Data Analytics (Dataset 20 GB)

5.2.1. Analítika Deskritiva

Husi readings moos 181,269,709, 3.80% mak anomalia. Tabela tuir mai hatudu readings no Anomaly Rate kada municipality (orden tuir Anomaly Rate):

Municipality	Readings	Anomalia	Anomaly Rate	Tipu barak liu
Atauro	1,352,354	74,666	5.52%	Voltajen Tun
Oecusse	10,941,142	590,839	5.40%	Voltajen Tun
Bobonaro	14,457,177	637,343	4.41%	Voltajen Tun
Manatuto	6,892,681	283,934	4.12%	Voltajen Tun
Liquiçá	11,342,511	464,204	4.09%	Karga Liu
Cova Lima	9,998,878	402,687	4.03%	Voltajen Tun
Lautém	9,457,876	375,732	3.97%	Voltajen Tun

Municipality	Readings	Anomalia	Anomaly Rate	Tipu barak liu
Manufahi	8,236,411	325,375	3.95%	Voltajen Tun
Aileu	7,293,546	279,475	3.83%	Voltajen Tun
Viqueque	10,810,246	410,480	3.80%	Voltajen Tun
Baucau	18,095,607	677,390	3.74%	Voltajen Tun
Ermera	18,636,553	685,689	3.68%	Voltajen Tun
Ainaro	9,859,269	362,749	3.68%	Voltajen Tun
Dili	43,895,458	1,313,991	2.99%	Karga Liu

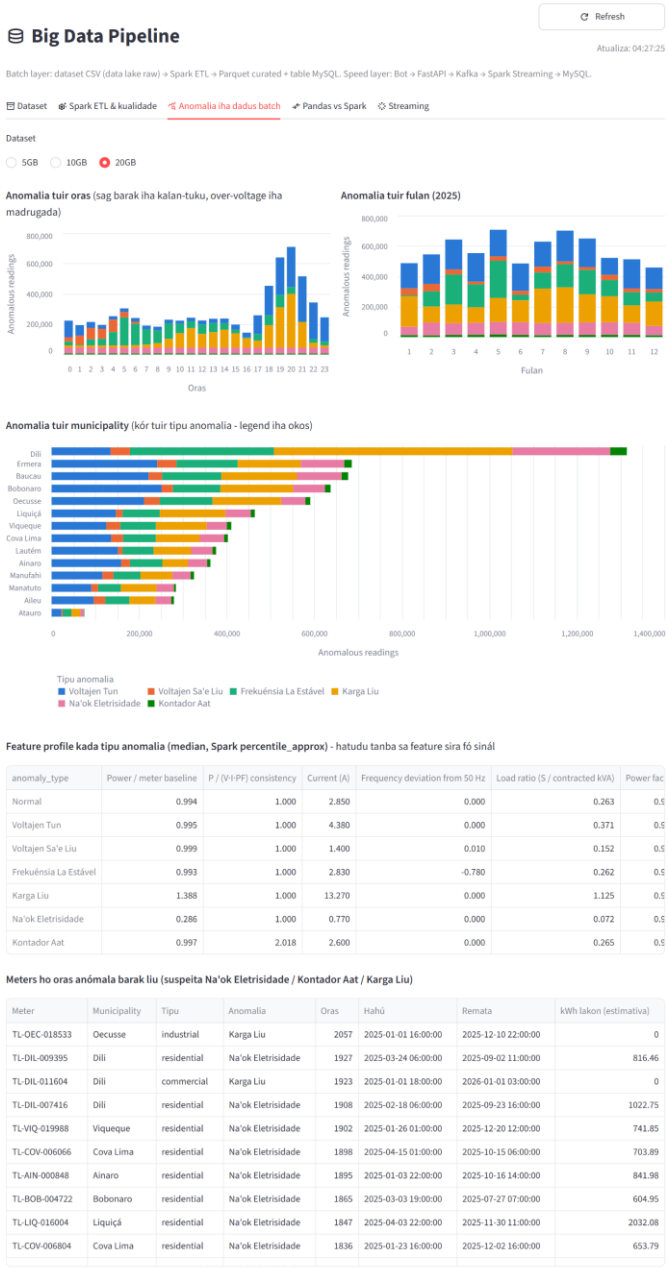
Tabela 5.5. Analítika deskritiva kada municipality

Atauro iha Anomaly Rate aas liu (5.52%) no Dili ki'ik liu (2.99%). Dili iha readings barak liu tanba meters barak, maibé nia Anomaly Rate ki'ik tanba liña distribuisaun badak no voltage drop ki'ik; municipality ho rede naruk ka sistema rasik (Atauro, Oecusse) iha Voltajen Tun no Frekuénsia La Estável barak liu.

5.2.2. Analítika Tendénsia

Tipu anomalia	Oras pico	Esplikasaun
Voltajen Tun	21:00	Karga aas iha kalan (oras pico) → voltage drop iha feeder naruk
Voltajen Sa'e Liu	4:00	Madrugada, karga ki'ik → voltage sa'e
Frekuénsia La Estável	5:00	Events generator trip (la depende maka'as ba oras)
Karga Liu	20:00	Kalan, uma-kain uza ekipamentu barak
Na'ok Eletrisidade	16:00	Episódiu naruk (loron to'o semana) → mosu iha oras hotu
Kontador Aat	3:00	Episódiu loron 1–7 → mosu iha oras hotu

Tabela 5.6. Tendénsia: oras ho anomalia barak liu kada tipu



Imajen 5.1. Anomalia tuir oras, fulan no municipality, feature profile no top meters (Spark, 20 GB)

5.2.3. Analítika Estatístika

Mediana (p50, Spark `percentile_approx`) feature prinsipál kada tipu anomalia hatudu tanba sa feature engineering fó sinál:

Tipu	Baseline ratio	Load ratio	Voltage deviation (%)	Frequency deviation (Hz)	Consistency ratio
Normal	0.994	0.263	-0.259	0.000	1.000
Voltajen Tun	0.995	0.371	-12.659	0.000	1.000
Voltajen Sa'e Liu	0.999	0.152	13.323	0.010	1.000
Frekuénsia La Estável	0.993	0.262	-0.318	-0.780	1.000
Karga Liu	1.388	1.125	-1.982	0.000	1.000
Na'ok Eletrisidade	0.286	0.072	-0.291	0.000	1.000
Kontador Aat	0.997	0.265	-0.291	0.000	2.018

Tabela 5.7. Mediana feature kada tipu anomalia

Na'ok Eletrisidade iha baseline ratio 0.29 (meter rejista ~29% husi konsumu normál), Karga Liu iha load ratio 1.12 (> 1.0), Kontador Aat iha consistency ratio 2.02 (current la tuir P/(V·PF)), no Frekuénsia La Estável iha frequency deviation -0.78 Hz. Ba Normal, feature hotu besik 1.0 ka 0.

5.2.4. Meters Suspeitu (Top Meters)

Meter	Municipality	Customer type	Anomalia	Oras anómala	kWh lakon (estimativa)
TL-OEC-018533	Oecusse	industrial	Karga Liu	2,057	–
TL-DIL-009395	Dili	residential	Na'ok Eletrisidade	1,927	816.5
TL-DIL-011604	Dili	commercial	Karga Liu	1,923	–
TL-DIL-007416	Dili	residential	Na'ok Eletrisidade	1,908	1,022.8
TL-VIQ-019988	Viqueque	residential	Na'ok Eletrisidade	1,902	741.9
TL-COV-006066	Cova Lima	residential	Na'ok Eletrisidade	1,898	703.9
TL-AIN-000848	Ainaro	residential	Na'ok Eletrisidade	1,895	842.0
TL-BOB-004722	Bobonaro	residential	Na'ok Eletrisidade	1,865	605.0

Tabela 5.8. Meters ho oras anómala barak liu (dataset 20 GB)

Lista ne'e mak output prátiku ba ekipa inspesaun EDTL: meters ho Na'ok Eletrisidade durante oras rihun ba rihun no estimativa kWh ne'ebé lakon.

5.3. Rezultadu Model Random Forest

5.3.1. Komparasaun ho Métodu Seluk

Métodu haat treina iha training set hanesan no test iha test set hanesan (readings 999,536 husi meters separadu, dataset 20 GB):

Métodu	Tipu	Precision	Recall	F1	PR-AUC	Desizaun
Random Forest	Supervised (ensemble)	**0.942**	**0.968**	**0.954**	**0.990**	**Hili – uza iha sistema**
Decision Tree	Supervised	0.881	0.973	0.925	0.880	La uza
Logistic Regression	Supervised (lineár)	0.903	0.888	0.896	0.934	La uza
Isolation Forest (Liu et al., 2008)	Unsupervised	0.265	0.099	0.144	0.251	La uza

Tabela 5.9. Komparasaun métodu machine learning iha test set hanesan (dataset 20 GB)

Decision Tree ho Logistic Regression hetan peza klase hanesan ho Random Forest (Normal = 4). Decision Tree iha recall aas maibé PR-AUC ki'ik (0.880) tanba tree ida fó probability 0 ka 1 de'it; Logistic Regression la kaptura relasaun non-linear; Isolation Forest (la uza label) la bele fahe anomalia husi variasaun normál konsumu.

5.3.2. Rezultadu Test Random Forest

Random Forest treina ho training set 1,079,003 readings husi dataset 20 GB no test iha 999,536 readings husi meters separadu (distribuisaun natural, 3.92% anomalia). Spark MLlib Random Forest treina distribuídu iha 2,298,650 readings no test iha readings hotu husi meters test (36,958,516).

Model Random Forest	Training	Test	Precision	Recall	F1	PR-AUC	Macro F1	Tempu
scikit-learn 20 GB (#13, ativu)	1,079,003	999,536	0.9416	0.9677	0.9545	0.9899	0.9626	49.5 s
Spark MLlib 20 GB (#12)	2,298,650	36,958,516	0.7855	0.9677	0.8671	–	0.8147	402.4 s

Tabela 5.10. Rezultadu test Random Forest (anomalia vs normal)

Random Forest scikit-learn atinje Precision 0.942, Recall 0.968, F1 0.954, PR-AUC 0.990 no ROC-AUC 0.9988, ho False Positive Rate 0.25% de'it. Cross-validation 3-fold (group = meter) fó F1 anomalia 0.981 ± 0.001, katak model estável. Spark MLlib Random Forest atinje F1 0.867 (Precision 0.785, Recall 0.968) iha readings 37.0 millaun: recall hanesan, maibé precision ki'ik liu tanba MLlib uza split ho bin 64 ba feature kontínuu, tree menus no la uza class weight (teste ho weight Normal = 2 fó F1 0.836, entaun model MLlib la ho weight mak rai).

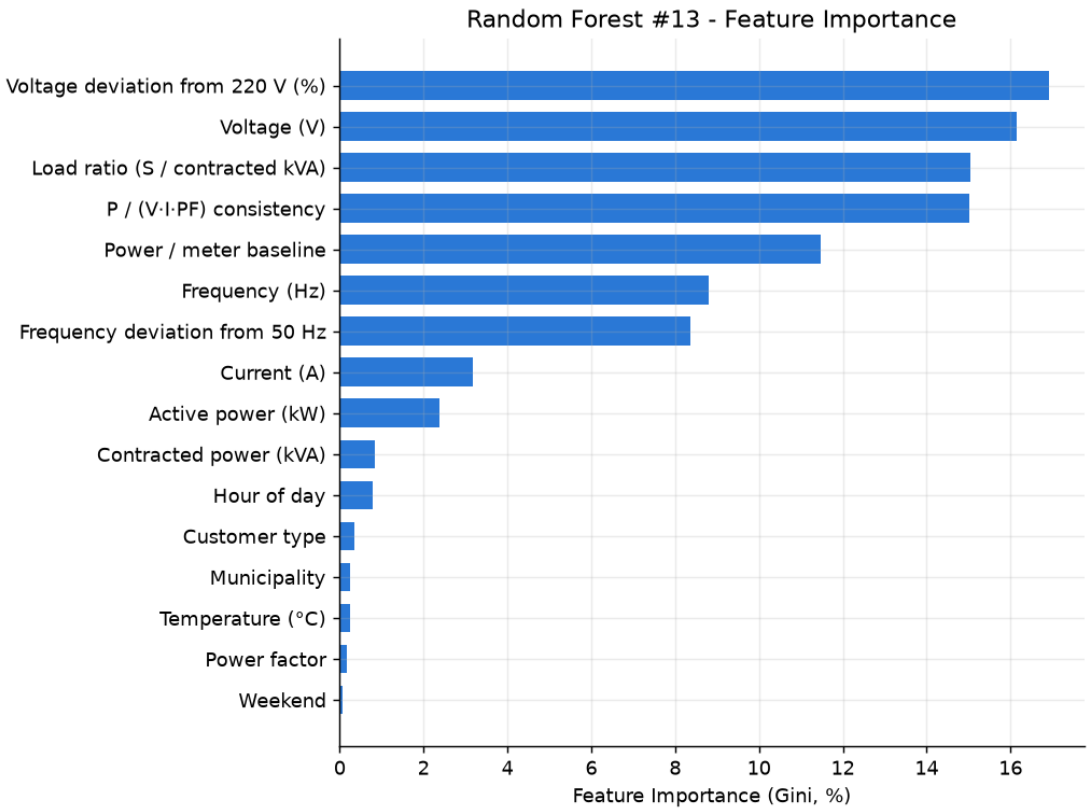
Tuning class weight. Random Forest ne'ebé treina la ho class weight (#11, dataset hanesan) atinje Precision 0.850, Recall 0.981 no F1 0.911 iha test set hanesan: training set undersample Normal, entaun model fó alarme falsu barak. Ho class_weight Normal = 4 (#13),

Precision sa'e ba 0.942 — alarme falsu menus maizumenus 65% — enkuantu Recall tun uitoan de'it (0.968), no F1 sa'e ba 0.954.

5.3.3. Feature Importance

No	Feature	Data type	Feature Importance (%)
1	Voltage deviation (%)	Numeric	16.91%
2	Voltage (V)	Numeric	16.16%
3	Load ratio	Numeric	15.05%
4	Consistency ratio	Numeric	15.01%
5	Baseline ratio	Numeric	11.47%
6	Frequency (Hz)	Numeric	8.81%
7	Frequency deviation (Hz)	Numeric	8.35%
8	Current (A)	Numeric	3.18%
9	Power (kW)	Numeric	2.38%
10	Contracted kVA	Numeric	0.83%
11	Hour	Numeric	0.79%
12	Customer type	Categorical	0.35%
13	Municipality	Categorical	0.25%
14	Temperature (°C)	Numeric	0.24%
15	Power factor	Numeric	0.17%
16	Weekend	Numeric	0.06%

Tabela 5.11. Feature Importance Random Forest (feature 16)



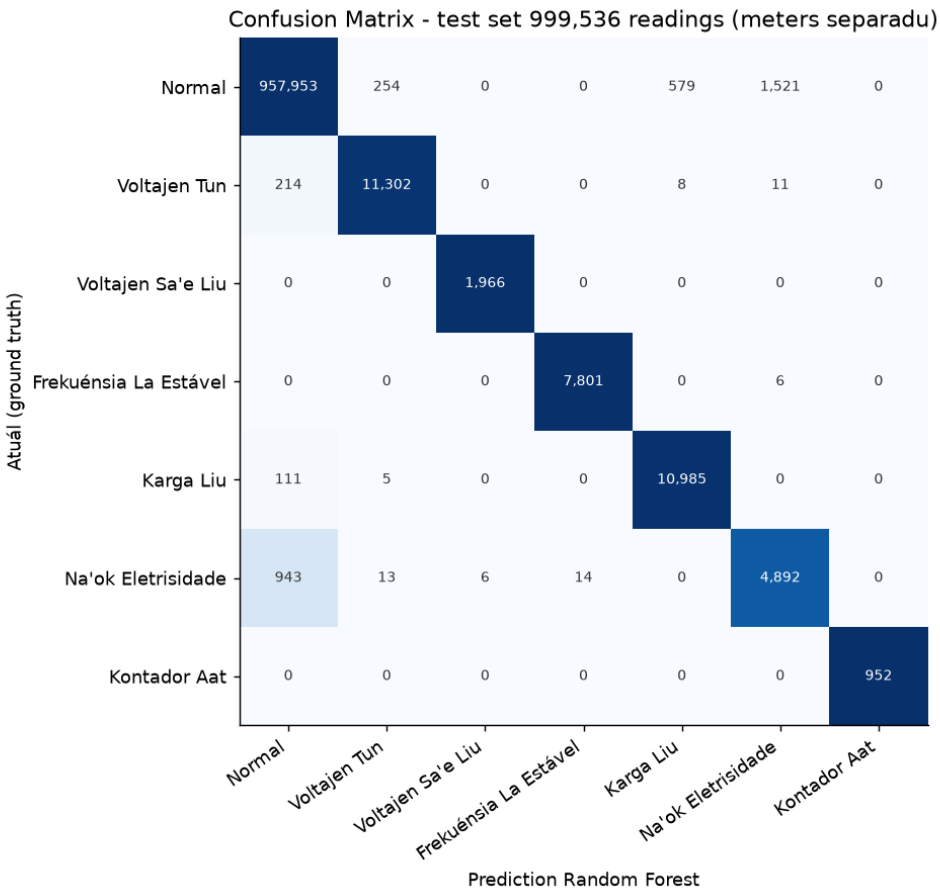
Imajen 5.2. Feature Importance Random Forest (Gini)

Feature **Voltage deviation (%)** (16.91%) no **Voltage (V)** (16.16%) mak importante liu, tuir husi Load ratio (15.05%) no Consistency ratio (15.01%). Feature engineering ho koñesimentu domíniu (load ratio, consistency ratio, baseline ratio) iha importánsia aas liu valór medidu barak, konsistente ho analítika estatística iha seksaun 5.2.3.

5.3.4. Métrika Kada Tipu Anomalia

Tipu	Precision	Recall	F1-Score	Readings (test)
Normal	0.9987	0.9975	0.9981	960,307
Voltajen Tun	0.9765	0.9798	0.9781	11,535
Voltajen Sa'e Liu	0.9970	1.0000	0.9985	1,966
Frekuénsia La Estável	0.9982	0.9992	0.9987	7,807
Karga Liu	0.9493	0.9896	0.9690	11,101
Na'ok Eletrisidade	0.7608	0.8337	0.7956	5,868
Kontador Aat	1.0000	1.0000	1.0000	952

Tabela 5.12. Métrika kada tipu anomalia (Random Forest, test set)



Imajen 5.3. Confusion Matrix Random Forest iha test set

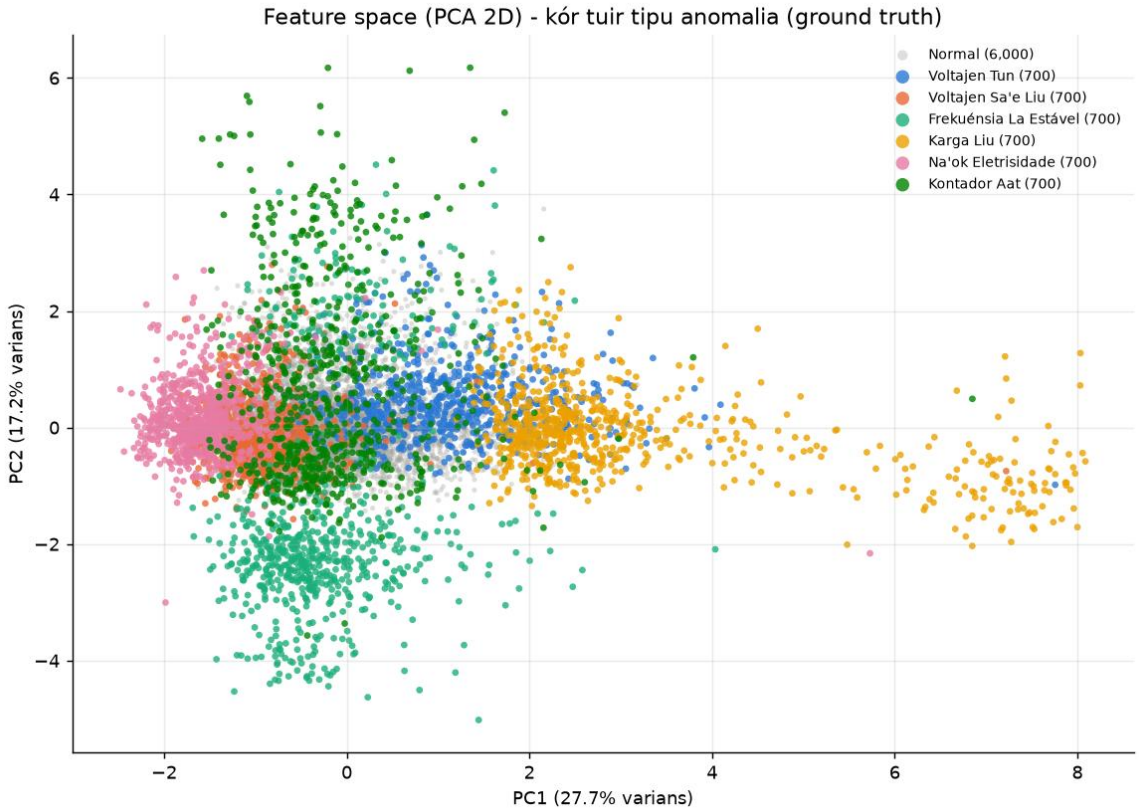
Voltajen Sa'e Liu, Frekuénsia La Estável, Kontador Aat deteta kuaze perfeitu (F1 > 0.99). Na'ok Eletrisidade mak susar liu (F1 0.80): uma-kain ne'ebé uza eletrisidade uitoan (ezemplu fulan ho konsumu ki'ik) parese ho na'ok. Tan ne'e sistema hatudu **suspeita** de'it bainhira meter ida repete Na'ok Eletrisidade iha ≥50% readings ikus.

5.3.5. Threshold Anomaly Score

Threshold Anomaly Score	Precision	Recall	F1-Score	Readings flag
0.2	0.8201	0.9836	0.8944	47,050
0.3	0.8867	0.9780	0.9301	43,268
0.4	0.9213	0.9724	0.9462	41,405
0.5	0.9409	0.9680	0.9543	40,360
0.6	0.9538	0.9634	0.9586	39,623
0.7	0.9666	0.9581	0.9623	38,884
0.8	0.9802	0.9481	0.9639	37,947
0.9	0.9915	0.9284	0.9589	36,733

Tabela 5.13. Precision no Recall tuir threshold Anomaly Score (test set)

Threshold aas hamenus false positive maibé lakon anomalia balun. Sistema uza threshold 0.6 ba events (Precision 0.954, Recall 0.963); Admin bele muda iha pájina Simulasaun.



Imajen 5.4. Feature space (PCA 2D) kór tuir tipu anomalia

5.4. Rezultadu Real-Time (Stream)

Iha stream, readings 3,135,118 husi meters 281 kompara ho ground truth bot. Random Forest atinje Accuracy 99.11%, Precision 0.9828, Recall 0.9412, F1 0.9615, ROC-AUC 0.9973 no PR-AUC 0.9899. Bot la iha variasaun fulan-fulan UCI, entaun readings normál estável liu; rezultadu oficiál mak husi test set (seksaun 5.3).

Tipu anomalia	Readings (live)	Recall Random Forest
Voltajen Tun	44,136	97.46%
Voltajen Sa'e Liu	85,293	99.84%
Frekuénsia La Estável	14,505	99.88%
Karga Liu	55,875	99.27%
Na'ok Eletrisidade	106,403	80.81%
Kontador Aat	20,485	98.51%

Tabela 5.14. Recall kada tipu anomalia iha stream real-time

Demonstrasaun live ba vídeo (Anexu C) hatudu tempu husi asaun to'o rezultadu: klik "Hahú na'ok eletrisidade" iha pájina Simulasaun iha 10:22:03 no anomaly events Na'ok Eletrisidade 6 mosu iha 10:22:04 (8 to'o 10:22:14); iha take daruak klik iha 10:29:36 no events 6 mosu iha 10:29:40. Website publiku lee database hanesan, entaun events ne'e mosu mós ba konvidadu iha internet.

5.5. Functional Testing

No	Test	Rezultadu esperadu	Rezultadu
1	Dadus tama?	Bot → FastAPI → Kafka	Loos – 2,024,886 mensajen iha topic
2	Dadus prosesa ho loos?	Spark parse, validation, feature, Random Forest	Loos – 3,150,953 mensajen, 15,735 invalid rejeita
3	Dadus rai ho loos?	MySQL no data lake	Loos – 3,135,118 readings iha tbl_readings_live
4	Dashboard funsiona?	Pájina 9 loke, KPI no gráfiku atualiza	Loos – screenshot pájina hotu (seksaun 4.7)
5	Anomaly events	Events ho tipu loos	Loos – 10,955 events, 91.4% tipu loos
6	Simulasaun Voltajen Tun	Meters Oecusse sai Voltajen Tun	Loos – readings 202 hotu deteta
7	Simulasaun Na'ok Eletrisidade	Meters 8 sai suspeita	Loos – 105 husi 117 readings deteta
8	Deteksaun Manual / API /api/detect	Tipu + votu tree	Loos – Na'ok Eletrisidade 150/150
9	Automated test (pytest)	Domain, feature, model, API	Loos – test 10 passa
10	Website: Login page	Pájina login mosu antes asesu website	Loos – tab User + tab Admin
11	Website: Password sala	Login rejeita	Loos – mensajen erru mosu
12	Website: Konta User iha tab Admin	Login rejeita (role la hanesan)	Loos – role tenke tuir tab
13	Website: Login User	Login susesu; menu Admin la mosu	Loos – seksaun Monitorizasaun, Análize, Informasaun de'it
14	Website: User loke /simulation	Pájina Admin la loke ba User	Loos – URL admin bloka
15	Website: User iha pájina Model	Retrain no ativa model: Admin de'it	Loos – butaun retrain la mosu
16	Website: Logout	Fila ba pájina login	Loos – sesaun hamoos
17	Website: Login Admin	Login susesu; menu Admin mosu	Loos – Simulasaun + Jestaun Uzuáriu
18	Website: Sesaun hela	Loke URL foun la presiza login fali	Loos – cookie sesaun pp_session
19	Website: Butaun Refresh	Dadus foti fali; oras 'Atualiza' muda	Loos – Atualiza: 04:26:59 -> Atualiza: 04:27:06

No	Test	Rezultadu esperadu	Rezultadu
20	Website: Night mode	Tema muda ba dark; login la lakon	Loos – rgb(255, 255, 255) -> rgb(14, 17, 23)
21	Website publiku husi internet (HTTPS)	Link permanente → tunnel → pájina tama	Loos – HTTP 200, pájina loke iha 11.3 s
22	Modu konvidadu	Butaun konvidadu → Dashboard ho role KONVIDADU	Loos – pájina 7 iha navigasaun
23	Pájina Admin taka ba konvidadu	URL /simulation no /users la loke	Loos – "Page not found"
24	Login taka iha versaun publiku	Tab login User / Admin la iha	Loos – tab login 0
25	Logout konvidadu	Fila ba pájina tama, maske reload	Loos – fila ba butaun konvidadu
26	Deteksaun offline	Landing page hatene tunnel mate	Loos – 530 (mate) vs 200 image/png (ativu)
27	Dadus live iha website publiku	Dashboard konvidadu hatudu batch Spark foun	Loos – Spark ativu, lag 0, MySQL konektadu

Tabela 5.15. Rezultadu functional testing

Pergunta haat iha manual (dadus tama, prosesa, rai, dashboard funsiona) hotu hetan resposta "loos". Teste website (No 10 to'o 20) hala'o automatikamente ho Playwright iha Microsoft Edge (`docs/report_build/shoot.py`), ho konta temporáriu Admin no User ne'ebé kria ho password aleatóriu no hamoos depois teste: 11 husi 11 passa. Teste website publiku (No 21 to'o 27) hala'o ho Playwright liuhusi link permanente iha internet, la ho konta: 7 husi 7 passa.

5.6. Performance Testing

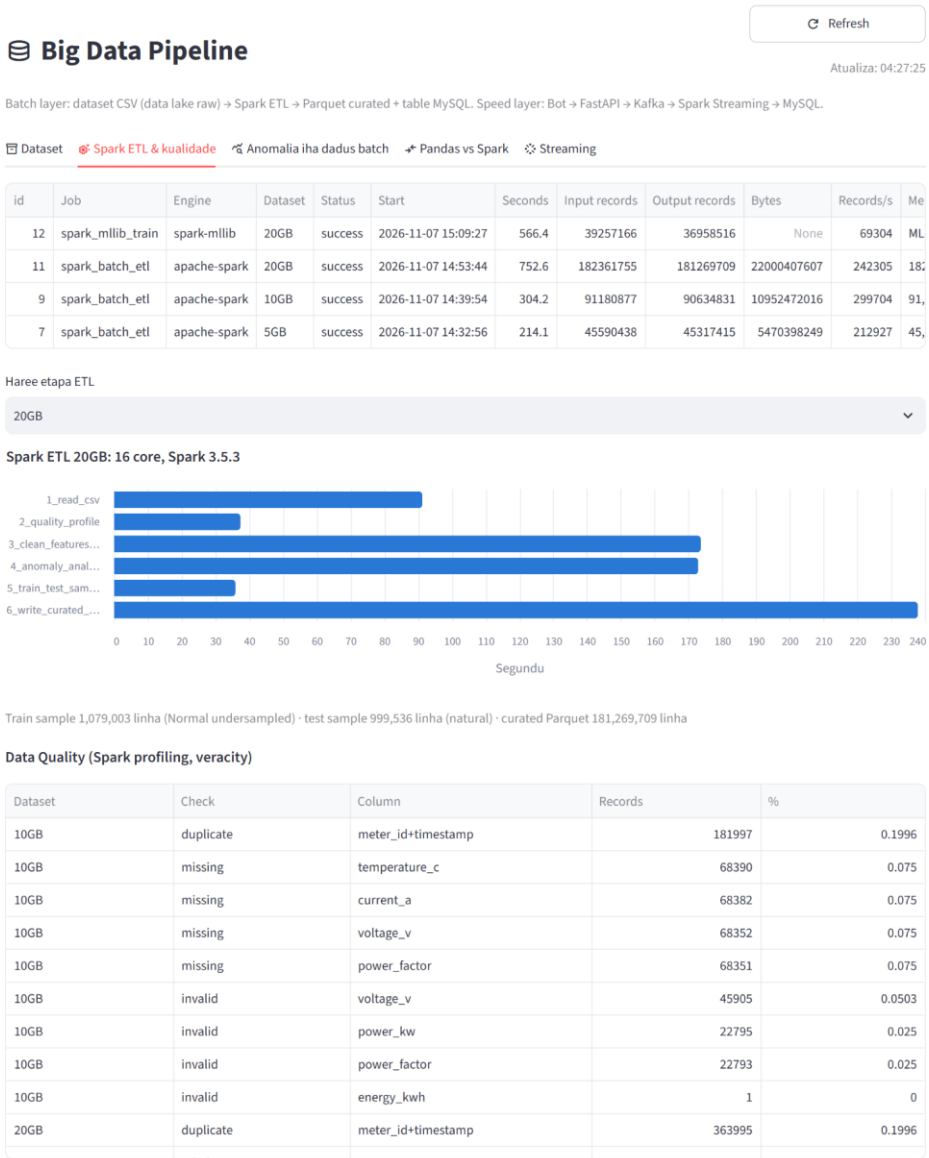
Manual seksaun 14.B husu sukat Processing Time, Throughput, Latency no CPU/Memory Usage.

5.6.1. Processing Time no Throughput (Batch)

Etapa Spark ETL	5 GB	10 GB	20 GB
Core Spark	local[32]	local[16]	local[16]
Read CSV	29.0 s	53.3 s	91.2 s
Data Quality profile	11.2 s	24.7 s	37.4 s
Cleaning + feature + join	68.7 s	98.1 s	173.7 s
Anomaly analytics	36.9 s	56.5 s	172.9 s
Training / test sample	18.2 s	20.6 s	35.9 s
Write curated Parquet	46.5 s	48.0 s	237.9 s
Totál	**214.1 s**	**304.2 s**	**752.6 s**
Records input → output	45.6 M → 45.3 M	91.2 M → 90.6 M	182.4 M → 181.3 M
Throughput	213 k records/s	300 k records/s	242 k records/s

Tabela 5.16. Tempu prosesamentu Spark batch ETL

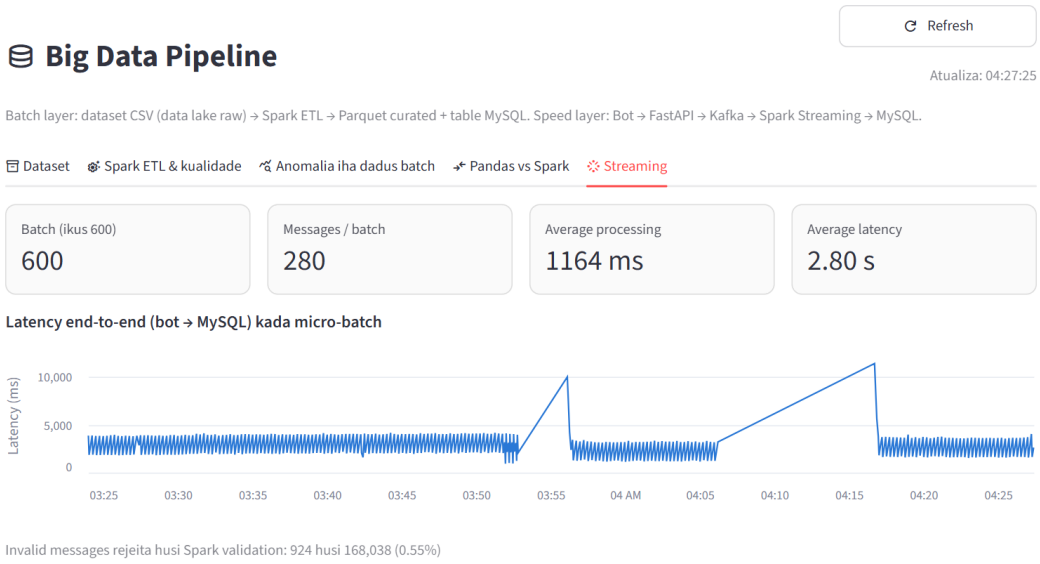
Dadus dala 4 boot liu (5 GB → 20 GB) presiza tempu dala 3.5. Ba 20 GB, Spark uza core 16 (la'ós 32) no shuffle partitions = 2 × partition input hodi Java heap la to'o limite (iha teste dahuluk, ho konfigurasaun default, job 20 GB para ho *Java heap space*). Spark MLlib Random Forest presiza 566 segundu (inklui lee Parquet no test iha readings 37.0 millaun).



Imajen 5.5. Spark ETL: istória job, tempu kada etapa no Data Quality

5.6.2. Latency no Throughput (Stream)

Ba speed layer, husi micro-batch 9,787: média mensajen 322 kada batch (throughput ~161 readings/s husi bot), prosesamentu 1,055 ms kada batch, no latency média bot → MySQL 2.81 s (latency inklui intervalu trigger 3 s). Random Forest klasifika readings 1,000 iha 2.3 ms.



Imajen 5.6. Estatística streaming: latency end-to-end kada micro-batch

5.6.3. CPU no Memory Usage

CPU no memória sukat ho psutil kada segundu 0.5 bainhira servisu benchmark hanesan hala'o (`docs/report_build/measure_cpu.py`, komputadór ho thread 32):

Engine	Volume	Tempu	CPU média	CPU pico	Memory pico
Pandas	1M	3.6 s	6.0%	10%	0.61 GB
Apache Spark	1M	16.0 s	42.6%	100%	3.79 GB
Apache Spark	5GB	61.3 s	73.2%	100%	12.89 GB
Apache Spark	20GB	200.4 s	78.0%	100%	12.94 GB

Tabela 5.17. CPU no memory usage (Pandas vs Spark)

Pandas uza CPU 6% de'it (core ida husi 32) — lalais ba dadus ki'ik maibé la uza komputadór tomak. Spark uza core hotu (CPU média 78%, pico 100% iha 20 GB) no memória hela ~13 GB maske dadus aumenta dala haat, tanba Spark prosesa partition-partition.

5.7. Data Quality Testing

Manual seksaun 14.C husu verifika Missing Values, Duplicate Records, Invalid Records no Outliers. Batch: Spark verifika kada dataset (seksaun 5.1.2) no outliers analiza ho regra IQR (seksaun 5.1.3). Stream: kada mensajen valida (schema, limite valór, duplicate) antes tama ba model — mensajen invalid 15,735 husi 3,150,953 (0.50%) rejeita, hanesan taxa 0.5% ne'ebé bot hatama.

5.8. Scalability Testing

Tuir manual seksaun 14.D (volume 10,000; 50,000; 100,000; 500,000; 1,000,000) no aumenta ho dataset 5 GB, 10 GB no 20 GB, scalability testing sukat servisu hanesan (lee CSV, hamoos, totál kWh kada municipality, top-10 meters, distribuisaun tipu anomalia, média voltage kada customer type) ho Pandas no Spark, run 3 kada engine, process foun kada run. Volume ki'ik mak subset husi dataset 5 GB. Engine rua fó rezultadu hanesan (records no totál kWh hanesan). Benchmark para Pandas bainhira memória komputadór to'o 85% (ka livre < 0.5 GB), hodi Windows la hahú swap, no rai rezultadu "FAILED: out of memory".

Volume	Records	Pandas (s)	Pandas RAM	Spark (s)	Spark RAM	Manán
10k	9,958	0.72	0.08 GB	11.69	1.67 GB	Pandas 16.2×
50k	49,808	0.79	0.09 GB	11.86	1.49 GB	Pandas 15.0×
100k	99,600	0.87	0.11 GB	11.39	1.66 GB	Pandas 13.2×
500k	497,994	1.29	0.24 GB	13.95	2.41 GB	Pandas 10.8×
1M	994,189	2.04	0.54 GB	15.18	2.79 GB	Pandas 7.4×
5GB	45,317,416	× out of memory	≥ 20.3 GB	50.46	12.83 GB	**Spark de'it**
10GB	90,634,832	× out of memory	≥ 20.6 GB	81.16	12.90 GB	**Spark de'it**
20GB	181,269,712	× out of memory	≥ 21.4 GB	189.38	12.95 GB	**Spark de'it**

Tabela 5.18. Rezultadu scalability testing Pandas vs Spark (média run 3)



Imajen 5.7. Dataset size vs execution time no memory: Pandas vs Spark

Ba dadus ki'ik (10k to'o 1M records) Pandas lalais liu tanba Spark iha kustu fixu ~11 segundu (JVM no planu job): iha 1M, Pandas 2.0 s vs Spark 15.2 s. Maibé Pandas tenke karrega dataset tomak iha RAM (maizumenus dala 4 tamañu CSV), entaun iha dataset 5 GB, 10 GB no 20 GB **Pandas la konsege remata** — memória to'o 20.3 GB–21.4 GB no benchmark para. Spark prosesa partition-partition no bele *spill* ba disk: 5 GB iha 50.5 s, 10 GB iha 81.2 s no 20 GB iha 189.4 s, ho memory maizumenus 13 GB de'it ba volume hotu. Husi 5 GB ba 20 GB, eksponente eskala Spark k = 0.95 (kuaze lineár). **Konkluzaun: ba dataset 5 GB ba leten, Apache Spark de'it mak bele uza** iha komputadór ne'e.

5.9. Kumprimentu ho Rekezitu Mínimu Manual

Tabela tuir mai kompara rekezitu tékniku mínimu iha manual Capstone Project (seksaun 12 no 13) ho saida mak PowerPredict TL implementa:

Rekezitu	Mínimu (manual)	PowerPredict TL	Estadu
Data Source	Fonte online 1	UCI Open Data, Open-Meteo REST API (arkivu + live), bot simulated real-time	Kumpre
Data Records	≥ 10,000	319.1 M records batch + 3.14 M readings live	Kumpre
Data Type	≥ tipu 2	Numeric, categorical, datetime; CSV, JSON, Parquet, MySQL	Kumpre
Streaming / Real-Time	Obrigatóriu	Bot → FastAPI → Kafka → Spark Streaming kada 3 s; dashboard refresh 5–10 s	Kumpre
Apache Kafka	Obrigatóriu	Kafka 3.9.1 KRaft, topic smart-meter-readings, partition 3	Kumpre
Apache Spark	Obrigatóriu	Batch ETL 5 / 10 / 20 GB, Spark Streaming, Spark MLlib	Kumpre
Data Storage	Obrigatóriu	Data Lake (CSV + Parquet) + MySQL (table 19)	Kumpre
Data Analytics	Obrigatóriu	Descriptive, trend, statistical + Random Forest (anomaly detection)	Kumpre
Visualization	Obrigatóriu	KPI, gráfiku Altair, mapa, confusion matrix, tendénsia	Kumpre
Dashboard	Obrigatóriu	Streamlit: login Admin / User, pájina 9, Refresh, Night mode; versaun publiku konvidadu iha internet	Kumpre
Documentation	Obrigatóriu	Relatóriu ida ne'e + README + kódigu ho komentáriu; source code públiku iha GitHub	Kumpre
Live Demonstration	Obrigatóriu	start_all.bat + pájina Simulasaun (Admin); vídeo YouTube 1080p (17:47 + closing 1:07); website publiku live (https://renoalvesc24.github.io/PowerPredict-TL/)	Kumpre

Tabela 5.19. Kumprimentu ho rekezitu tékniku mínimu manual

5.10. Diskusaun

Seksaun ida ne'e hatán pergunta peskiza (seksaun 1.3) bazeia ba rezultadu:

1. **Dezempeñu Random Forest.** Iha meters ne'ebé model nunka haree, Random Forest atinje F1 0.954, PR-AUC 0.990 no ROC-AUC 0.999; iha stream real-time F1 0.962. Tipu kualidade enerjia (Voltajen Tun, Voltajen Sa'e Liu, Frekuénsia La Estável) no Kontador Aat kuaze perfeitu; Na'ok Eletrisidade mak dezafiu (F1 0.80).
2. **Tanba sa Random Forest fit.** Dadus iha label, klase la balansu no anomalia depende ba kombinasaun feature. Random Forest manán métodu tolu seluk iha test set hanesan, estável iha cross-validation (± 0.001), no class weight bele kontrola balansu Precision–Recall. Ida ne'e konsistente ho Fernández-Delgado et al. (2014). Regra outlier (IQR) hetan 81% husi anomalia maibé ho precision 0.14 de'it no Na'ok Eletrisidade 15% de'it, entaun threshold simples la to'o.

3. **Big Data no Pandas vs Spark.** Pipeline FastAPI–Kafka–Spark–MySQL prosesa dataset to'o 20 GB no stream kontínuu ho latency ~2.8 s. Konfirma aprezentasaun individuál (Costa, 2026a): Pandas manán ba dadus ki'ik, maibé iha komputadór RAM 32 GB Pandas la konsege ona iha 5 GB — Spark mak solusaun ba dataset 5 GB ba leten.
4. **Feature importante.** Voltage deviation (%), Voltage (V), Load ratio no Consistency ratio mak importante liu. Hanesan Jokar et al. (2016), konsumu kompara ho padraun normál (baseline ratio) mak sinál ba na'ok; consistency ratio mak sinál ba Kontador Aat.
5. **Website.** Website Streamlit ho login no role fasil uza no seguru: teste automátiku hotu passa, User la bele loke pájina Admin, no funsaun Refresh no Night mode servisu. Versaun publiku iha internet (seksaun 4.9) hatudu katak arkitetura ne'e bele fó asesu real-time ba ema hotu ho kustu zero: HTTPS husi Cloudflare Tunnel, modu konvidadu read-only, no teste publiku 7 husi 7 passa.

Limitasaun rezultadu. (1) Dadus sintétiku: maske seed mai husi dadus reál, padraun anomalia mai husi jeradór, entaun rezultadu iha dadus EDTL reál bele ki'ik liu. (2) Rezultadu stream aas liu test set tanba bot la iha variasaun fulan-fulan. (3) Sistema hala'o iha komputadór ida ho Kafka broker ida; iha produsaun presiza cluster no replication. (4) Na'ok Eletrisidade ho konsumu ki'ik loos sei bele sai alarme falsu — tan ne'e sistema uza "suspeita" no inspesaun manuál. (5) Website publiku online de'it bainhira komputadór grupu ativu; ba 24/7 presiza servidór (pakote Docker iha seksaun 4.9.3).

KAPITULU VI
KONKLUZAUN NO REKOMENDASAUN

6.1. Konkluzaun

Bazeia ba rezultadu peskiza no dezenvolvimentu Sistema Deteksaun Anomalia Rede Eletrisidade Real Time uza metodu Random Forest, hakerek-na'in sira foti konkluzaun tuir mai:

1. Sistema Big Data implementa ho susesu pipeline kompletu tuir manual Capstone Project: Data Collection (bot smart meter, Open-Meteo, jeradór dataset) → Data Ingestion (FastAPI + Apache Kafka) → Data Storage (Data Lake Parquet + MySQL) → Data Processing (Spark batch ETL no Spark Streaming ho window no watermark) → Machine Learning (Random Forest) → Visualization (website Streamlit ho login no pájina 9) → Real-Time Application (Anomaly Score, anomaly events no simulasaun).
2. Métodu eskala dadus husi aprezentasaun individuál aplika ho susesu: seed reál (UCI no Open-Meteo) eskala ba dataset 5 GB, 10 GB no 20 GB ho records totál 319,133,070 ho events anomalia neen no ground truth label. Spark ETL prosesa dataset 20 GB (records 182.4 millaun) iha 753 segundu.
3. Random Forest — métodu machine learning ida de'it iha sistema — deteta anomalia ho Precision 0.942, Recall 0.968, F1-score 0.954 no PR-AUC 0.990 iha meters ne'ebé model nunka haree, no klasifika tipu anomalia ho Macro F1 0.963. Iha test set hanesan, nia accurate liu Decision Tree (F1 0.925), Logistic Regression (F1 0.896) no Isolation Forest (F1 0.144). Spark MLlib Random Forest ne'ebé treina iha records 2,298,650 no test iha records 36,958,516 atinje F1 0.867, hatudu katak métodu ne'e bele hala'o iha eskala Big Data.
4. Random Forest fit ho dadus tanba dadus iha ground truth label, anomalia depende ba kombinasaun feature (non-linear), no votu tree 150 fó Anomaly Score ne'ebé estável (cross-validation F1 0.981 ± 0.001). Class weight Normal = 4 hamenus alarme falsu: Precision sa'e husi 0.850 ba 0.942.
5. Feature **Voltage deviation (%)** (16.91%) no **Voltage (V)** (16.16%) mak importante liu. Feature engineering ho koñesimentu domíniu (baseline ratio no consistency ratio)

mak xave ba deteksaun Na'ok Eletrisidade no Kontador Aat. Na'ok Eletrisidade mak tipu susar liu (F1 0.80).

6. Scalability testing hatudu katak Pandas lalais liu ba dadus ki'ik (10k–1M records), maibé iha komputadór RAM 32 GB Pandas la konsege prosesa dataset 5 GB, 10 GB no 20 GB (out of memory). Apache Spark prosesa 20 GB iha 189 segundu ho memory 13 GB de'it; tan ne'e ba dataset 5 GB ba leten, Spark mak tools ida de'it ne'ebé bele uza.
7. Sistema real-time prosesa mensajen 3,150,953 husi meters 281 ho latency média 2.8 segundu no F1 0.962 iha stream. Website Streamlit ho login Admin / User (password hash PBKDF2, cookie sesaun), butaun Refresh no Night mode — Dashboard, Monitor Real-Time, Anomaly Events, Deteksaun Manual, Model Random Forest, Big Data Pipeline, Simulasaun, Jestaun Uzuáriu no Metodolojia — passa teste website 11 husi 11, no bele ajuda EDTL deteta problema rede no suspeita na'ok eletrisidade sedu.
8. Sistema publika ona ba internet ho kustu zero: website konvidadu read-only ho HTTPS liuhusi Cloudflare Tunnel, link permanente https://renoalvesc24.github.io/PowerPredict-TL/ (GitHub Pages) ne'ebé atualiza automátiku husi start_all.bat, source code públiku iha https://github.com/Renoalvesc24/PowerPredict-TL, no pakote Docker Compose prontu ba deployment 24/7.

6.2. Rekomendasaun

Rekomendasaun ba futuru:

1. **Integrasaun Dadus Reál EDTL:** kolabora ho EDTL hodi hetan dadus smart meter reál ne'ebé anonimizadu no rezultadu inspesaun (label reál ba Na'ok Eletrisidade no Kontador Aat).
2. **Feature Tempu Naruk:** aumenta feature istóriku meter (ezemplu média semana 4 ikus, tendénsia konsumu) hodi hadi'a deteksaun Na'ok Eletrisidade.
3. **Cluster no Structured Streaming:** implementa iha cluster (Spark node barak, Kafka broker 3 ho replication) ka iha Linux/cloud hodi uza Structured Streaming ho checkpoint.

4. **Retraining Automátiku:** treina fali Random Forest kada fulan ho dadus foun no monitoriza *model drift*.
5. **Notifikasaun:** haruka events severity aas liuhusi SMS ka WhatsApp ba tékniku EDTL.
6. **Edge Computing:** hala'o model iha subestasaun ka gateway lokál hodi kontinua funsiona maske internet mate.
7. **Seguransa Website:** HTTPS no modu konvidadu read-only implementa ona iha website publiku (seksaun 4.9); futuru aumenta autentikasaun faktór rua, log auditoria ba asaun Admin no rate limiting bainhira sistema hala'o iha rede EDTL.
8. **Deployment 24/7:** hala'o pakote deploy/ (Docker Compose) iha Oracle Cloud Always Free ka servidór EDTL, ho domíniu rasik no monitorizasaun uptime, hodi website publiku la depende ba komputadór grupu.

ANEXU

Anexu A. Estrutura Kódigu Sistema

Folder / file	Funsaun
config.py, .env, .env.example	Konfigurasaun (MySQL, Kafka, Spark, data lake iha D:, konta website dahuluk)
common/domain.py	Tipu anomalia, load shape, baseline, feature engineering, evidénsia
common/db.py, common/spark.py	Schema table 19 (SQLAlchemy); SparkSession, feature Spark, Parquet writer
data_generator/, data/seed/uci_real/	Download temperature Open-Meteo; seed UCI; jeradór dataset 5 GB / 10 GB / 20 GB ho anomalia
bot/bot_live.py, bot/bot_replay.py	Data source real-time (smart meter 281) no replay dataset
api/main.py, api/schemas.py	FastAPI gateway: ingest, health, stats, detect
pipeline/spark_streaming.py	Kafka → Spark Streaming → Random Forest → MySQL
pipeline/spark_batch_etl.py	Spark ETL: Data Quality, cleaning, feature, analytics, training/test split
pipeline/spark_ml_train.py, pipeline/benchmark.py	Spark MLlib Random Forest; benchmark Pandas vs Spark (ho memory guard)
ml/train.py, ml/predict.py	Training (class weight) no serving Random Forest (scikit-learn)
dashboard/app.py, auth.py, ui.py	Website: navigation, login no sesaun (PBKDF2, cookie), Refresh, Night mode, ikon, modu konvidadu
dashboard/views/ (file 10)	Pájina login + pájina 9 (inklui users.py ba Jestaun Uzuáriu)
setup/, setup_once.bat, start_all.bat	Kria database, hala'o Kafka (check_running.ps1 hodi la hahú dala rua), instalasaun no hahú sistema; start_all.bat pasu 6 hahú website publiku
start_public.bat, setup/public_site.ps1, setup/public_landing.html	Website publiku: kópia konvidadu port 8510, Cloudflare Tunnel, live.json + landing page iha GitHub Pages
deploy/ (Dockerfile, docker-compose.yml, Caddyfile, ORACLE_SETUP.md)	Pakote deployment 24/7 ba Oracle Cloud Always Free (Kafka, MariaDB, app, Caddy HTTPS)
tests/test_anomaly.py, data/samples/	Automated test no sample CSV 100 records
docs/report_build/	Script relatóriu: collect_facts.py, shoot.py (screenshot + teste website), build_report.js

Tabela 6.1. Estrutura kódigu sistema

Anexu B. Oinsá Hala'o Sistema

1. Loke XAMPP Control Panel no hahú **MySQL**.
2. Kopia `.env.example` ba `.env`; hakerek `PP_DATA_DIR` (data lake) no username / password ba konta Admin no User dahuluk (`PP_ADMIN_*`, `PP_USER_*`, password karakter 8 ka liu).
3. Dala uluk de'it: hala'o `setup_once.bat` (instala pakote, download temperature, jera dataset 5 GB / 10 GB / 20 GB, kria database, Spark ETL no training Random Forest).

4. Hala'o `start_all.bat`: hahú Kafka, FastAPI, Spark Streaming, bot no website. Script ne'e seguru atu hala'o dala rua — komponente ne'ebé ativu ona la hahú fali. Pasu 6 hahú website publiku (konvidadu) ba internet no atualiza link permanente.
5. Loke browser iha `http://localhost:8505`, login iha tab Admin ka User, no `http://127.0.0.1:8000/docs` (API).
6. Website publiku: loke https://renoalvesc24.github.io/PowerPredict-TL/ husi HP ka komputadór seluk (konvidadu, la presiza konta). Bainhira start_all.bat la ativu, pájina ne'e hatudu "Sistema offline agora". Atu hahú parte publiku de'it, hala'o start_public.bat.
7. Demonstrasaun live (Admin): loke pájina Simulasaun, klik "Voltajen Tun iha municipality" ka "Hahú na'ok eletrisidade", depois haree Dashboard, Monitor Real-Time no Anomaly Events.
8. Se janela Kafka hatudu `CorruptRecordException`: taka janela Kafka, hala'o `setup\start_kafka.bat reset` (hamoos dadus Kafka de'it), depois hala'o fali `start_all.bat`.
9. Opsionál 24/7: tuir deploy/ORACLE_SETUP.md hodi hala'o sistema iha servidór Oracle Cloud Always Free ho Docker Compose.

Anexu C. Vídeo Demonstrasaun (YouTube)

Link vídeo demonstrasaun YouTube: _____

Vídeo prontu ona: parte prinsipál 17:47 no closing 1:07 (MP4 1080p, 30 fps), ho demonstrasaun live (Simulasaun na'ok eletrisidade → events mosu iha segundu 1–4). Source code públiku: https://github.com/Renoalvesc24/PowerPredict-TL. Website publiku live: https://renoalvesc24.github.io/PowerPredict-TL/.

Vídeo prepara tuir manual Capstone Project seksaun 17 no 18:

Komponente	Rekezitu manual	PowerPredict TL
Plataforma / Visibility	YouTube, Unlisted ka Public	YouTube (Unlisted ka tuir instrusaun dosente)
Durasaun / Formatu	10–20 minutu, MP4, 1080p rekomendadu	17:47 + closing 1:07 (~18:54), MP4 1080p 30 fps
Screen Recording	Obrigatóriu	Grava ekran tomak (website, janela start_all, phpMyAdmin)
Live System / Dashboard Demo	Obrigatóriu	start_all.bat → login Admin → Simulasaun → Dashboard / Monitor atualiza; website publiku iha internet
Source Code Demo	Rekomendadu	Repozitóriu GitHub públiku (github.com/Renoalvesc24/PowerPredict-TL): folder bot / api / pipeline / dashboard
YouTube Link	Entrega ho relatóriu	Hakerek iha leten

Tabela 6.2. Rekezitu vídeo demonstrasaun

Parte	Durasaun	Konteúdu PowerPredict TL
1 – Introdusaun	1–2 min	Naran no NIM membru grupu na'in rua, títulu projetu, unidade kurikuler Big Data, problema: anomalia rede no na'ok eletrisidade
2 – Problem & Objectives	1–2 min	Problem statement (seksaun 1.2), objetivu (1.4), benefísiu ba EDTL (1.5)
3 – Big Data Architecture	2–3 min	Bot / Open-Meteo / UCI → FastAPI → Kafka → Spark → Data Lake + MySQL → Random Forest → Dashboard (Imajen 4.1, 4.3)
4 – Implementasaun	3–5 min	Dataset 5 / 10 / 20 GB, Kafka topic, Spark Streaming, Spark ETL, Data Lake, Random Forest (scikit-learn + MLlib)
5 – Live Demonstration	3–5 min	Simulasaun Voltajen Tun ka Na'ok Eletrisidade → dadus tama → Spark prosesa → Dashboard no Monitor atualiza; Pandas vs Spark
6 – Conclusion	1–2 min	Rezultadu prinsipál (F1, Spark vs Pandas), limitasaun no future improvement (Kapitulu VI)

Tabela 6.3. Estrutura vídeo demonstrasaun

Anexu D. Mapa Relatóriu ba Avaliasaun no Learning Outcomes

Tabela tuir mai hatudu iha ne'ebé mak relatóriu ida ne'e hatán kada komponente avaliasaun (manual seksaun 15) no kada Learning Outcome (manual seksaun 4):

Komponente avaliasaun	Peso	Seksaun relatóriu
Problem & Project Proposal	10%	1.1–1.6
Literature Review	5%	Kapitulu II
Data Collection	10%	3.4, 3.5, 5.1
Data Engineering	15%	3.7, 3.8, 4.3, 4.4, 5.1.2
Kafka & Streaming	15%	3.6, 4.2, 4.3.2, 5.4, 5.6.2
Spark Processing	15%	3.8, 4.3.1, 5.6, 5.8
Analytics / Machine Learning	10%	3.9, 4.5, 5.2, 5.3
Dashboard	10%	3.10, 4.6, 4.7
Final Report	5%	Relatóriu ida ne'e
Presentation & Live Demo	5%	Anexu B, Anexu C

Tabela 6.4. Komponente avaliasaun Capstone Project no seksaun relatóriu

LO	Learning Outcome (manual)	Evidénsia iha relatóriu
LO1	Esplika konseitu no karakteristika Big Data	2.1.3 (5V), 2.1.5–2.1.7
LO2	Identifika fonte no karakteristika dadus	3.4, 3.5.2
LO3	Dezenvolve prosesu data ingestion	3.6, 4.3.2, 4.8
LO4	Implementa distributed data processing	3.8, 4.3.1, 5.6.3
LO5	Prosesu batch no streaming data	4.3.1, 4.3.2, 5.4
LO6	Halo Big Data analytics	3.9, 5.2, 5.3
LO7	Dezenvolve visualization no dashboard	3.10, 4.6, 4.7
LO8	Integra teknolojia Big Data	4.1, 4.2, 4.3.3, 4.9
LO9	Avalia performance sistema	5.5–5.8
LO10	Komunika rezultadu projetu ho profisionál	Relatóriu, aprezentasaun, vídeo (Anexu C) no website publiku (4.9)

Tabela 6.5. Learning Outcomes no evidénsia

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